# PT3S Documentation

*Release 90.11.01.20.dev1*

**3S**

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Use SIR 3S Modeldata and SIR 3S Calculationresults in pure Python. With pandas, matplotlib and others. For documentation, test, verification, analysis, reporting, prototyping, play.

Use SIR 3S Modeldata and SIR 3S Results in pure Python.

With pandas, matplotlib and others.

For documentation, test, verification, analysis, reporting, prototyping, play.

CHAPTER

ONE

# MX

SIR 3S MX-Interface (short: MX)

MX is a file based, channel-oriented interface for SIR 3S' calculation results.

This module contains stuff to utilize SIR 3S' MX calculation results in pure Python.

SIR 3S MX calculation results overview:

- Binary .MXS-Files contain the SIR 3S calculations results.
- A Model calculation run creates at least one .MXS-File.
- There is one .MX1-File (an XML-File) for the corresponding .MXS-File(s).
- This .MX1-File defines (in XML) a sequence of MX-Channels in the corresponding .MXS-File(s).
- And - as a result - the Byte-Layout of a single Record in the corresponding .MXS-File(s).
- A single Record is called a MX3-Record. .MXS-File(s) contain a sequence of MX3-Records.
- A MX3-Record contains calculation results for one TIMESTAMP. TIMESTAMP ist Scenariotime.
- A corresponding .MX3-File contains one MX3-Record - the last Scenariotime calculated.
- Summary:
- .MXS-File(s): MX3-Records: MX-Channels
- MX3-Record Byte-Layout (the MX-Channels) defined in corresponding .MX1-File.

A MX-Channel can be:

- a single Value
- or a Vector: Sequence of Values of the same Type:
  - for all Objects of a certain Type or (called Vectorchannels)
  - number of interior Points for all Pipes (special Vectorchannels: Pipevectorchannels)
  - Vectors with ATTRTYPE in: {'SVEC', 'PVECMIN_INST', 'PVECMAX_INST'}

For Vectorchannels (including Pipevectorchannels) the sequence of Objects is defined in the .MX2-File.

```
>>> # ---
>>> # SETUP
>>> # ---
>>> import os
>>> import time
>>> import logging
>>> logger = logging.getLogger('PT3S.Mx')
```

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```
>>> # ---
>>> # path
>>> # ---
>>> if __name__ == "__main__":
...     try:
...         dummy=__file__
...         logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ','path =␣
↪os.path.dirname(__file__)'," ."))
...         path = os.path.dirname(__file__)
...     except NameError:
...         logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ","path =
↪'.' because __file__ not defined and: "," from Mx import Mx"))
...         path = '.'
...         from Mx import Mx
... else:
...     logger.debug("{0:s}{1:s}{2:s}{3:s}".format('DOCTEST: Not __main__ Context: ','_
↪_name__: ',__name__,"path = '."))
...     path = '.'
>>> # ---
>>> # testDir
>>> # ---
>>> # globs={'testDir':'testdata'}
>>> try:
...     dummy= testDir
... except NameError:
...     testDir='testdata'
>>> # ---
>>> # dotResolution
>>> # ---
>>> # globs={'dotResolution':''}
>>> try:
...     dummy= dotResolution
... except NameError:
...     dotResolution=''
>>> import zipfile
>>> import pandas as pd
>>> # ---
>>> # Init
>>> # ---
>>> h5File=os.path.join(path,os.path.join(testDir,'OneLPipe.h5'))
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDOneLPipe\B1\V0\BZ1\M-1-0-1
↪'+dotResolution+'.MX1'))
>>> mx=Mx(mx1File=mx1File,NoH5Read=True,NoMxsRead=True)
>>> isinstance(mx.mx1Df,pd.core.frame.DataFrame) # MX1-Content
True
>>> isinstance(mx.df,type(None)) # MXS-Content
True
>>> # ---
>>> # Clean Up
>>> # ---
>>> mx.delFiles()
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> mxsDumpFile=mx.mxsFile+'.dump'
>>> if os.path.exists(mxsDumpFile):
...     os.remove(mxsDumpFile)
>>> # ---
```

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```
>>> # 1st Read MXS
>>> # ---
>>> logger.debug("{0:s}: 1st Read MXS".format('DOCTEST'))
>>> mx.setResultsToMxsFile() # looks for M-1-0-1.MXS in same Dir
4
>>> isinstance(mx.df,pd.core.frame.DataFrame) # MXS-Content
True
>>> rowsDf,colsDf = mx.df.shape
>>> (firstTime,lastTime,rows)=mx._checkMxsVecsFile()
>>> rowsDf==rows
True
>>> mx.df.index[0]==firstTime
True
>>> # ---
>>> # Write H5
>>> # ---
>>> if os.path.exists(mx.h5File):
...     os.remove(mx.h5File)
>>> mx.ToH5() # M-1-0-1.h5 in same Dir
>>> os.path.exists(mx.h5File)
True
>>> # ---
>>> # Init with H5
>>> # ---
>>> mx=Mx(mx1File=mx1File) # looks for M-1-0-1.h5 in same Dir
>>> # and reads the .h5 if newer than .MX1 and newer than an existing .MXS
>>> isinstance(mx.mx1Df,pd.core.frame.DataFrame) # MX1-Content
True
>>> isinstance(mx.df,pd.core.frame.DataFrame) # MXS-Content
True
>>> # ---
>>> # 1st Read MXS Zip
>>> # ---
>>> # create the Zip first
>>> with zipfile.ZipFile(mx.mxsZipFile,'w') as myzip:
...     myzip.write(mx.mxsFile)
>>> logger.debug("{0:s}: 1st Read MXS Zip".format('DOCTEST'))
>>> mx.setResultsToMxsZipFile() # looks for M-1-0-1.ZIP in same Dir
4
>>> isinstance(mx.df,pd.core.frame.DataFrame) # MXS-Content
True
>>> rowsMxs,colsMxs = mx.df.shape
>>> mx.df.index.is_unique # all setResultsTo... will ensure this uniqueness under all␣
↪circumstances
True
>>> # uniqueness under all circumstances: also when add=True (setResultsTo... shall␣
↪add the MXS-Content) is used
>>> # ---
>>> # 1st Add same MXS (for testing ensuring uniqueness)
>>> # ---
>>> oldShape=mx.df.shape
>>> logger.debug("{0:s}: 1st Add same MXS (for testing ensuring uniqueness)".format(
↪'DOCTEST'))
>>> mx.setResultsToMxsFile(add=True) # looks for M-1-0-1.MXS in same Dir
0
>>> newShape=mx.df.shape
>>> newShape==oldShape
```

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```
True
>>> # ---
>>> # 1st Add same Zip (for testing ensuring uniqueness)
>>> # ---
>>> logger.debug("{0:s}: 1st Add same Zip (for testing ensuring uniqueness)".format(
↪'DOCTEST'))
>>> mx.setResultsToMxsZipFile(add=True) # looks for M-1-0-1.ZIP in same Dir
0
>>> newShape=mx.df.shape
>>> newShape==oldShape
True
>>> # ---
>>> # 1st Read MXS Zip with overlapping Timestamps (for testing ensuring uniqueness)
>>> # ---
>>> with zipfile.ZipFile(mx.mxsZipFile,'w') as myzip:
...     myzip.write(mx.mxsFile)
...     myzip.write(mx.mxsFile,arcname=mx.mxsFile+'.2')
>>> logger.debug("{0:s}: 1st Read MXS Zip with overlapping Timestamps (for testing␣
↪ensuring uniqueness)".format('DOCTEST'))
>>> mx.setResultsToMxsZipFile() # looks for M-1-0-1.ZIP in same Dir
4
>>> newShape=mx.df.shape
>>> newShape==oldShape
True
>>> # ---
>>> # shift to younger Timestamps (for testing purposes)
>>> # ---
>>> lastTimestamp=mx.df.index[-1]
>>> firstTimestamp=mx.df.index[0]
>>> timeSpan=lastTimestamp-firstTimestamp
>>> if len(mx.df.index)>1:
...     timeStep=mx.df.index[-1]-mx.df.index[-2]
... else:
...     timeStep=pd.to_timedelta('1 second')
>>> mx.df.index=mx.df.index-(timeSpan+timeStep)
>>> # ---
>>> # 1st Read MXS (with the original Timestamps)
>>> # ---
>>> logger.debug("{0:s}: 1st Read MXS (with the original Timestamps)".format('DOCTEST
↪'))
>>> mx.setResultsToMxsFile(add=True) # looks for M-1-0-1.MXS in same Dir
4
>>> rowsNew,colsNew=mx.df.shape
>>> rowsOld,colsOld=oldShape
>>> rowsNew==2*rowsOld
True
>>> colsNew==colsOld
True
>>> # ---
>>> # shift to older Timestamps (for testing purposes)
>>> # ---
>>> lastTimestamp=mx.df.index[-1]
>>> firstTimestamp=mx.df.index[0]
>>> timeSpan=lastTimestamp-firstTimestamp
>>> if len(mx.df.index)>1:
...     timeStep=mx.df.index[-1]-mx.df.index[-2]
... else:
```

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```
...       timeStep=pd.to_timedelta('1 second')
>>> mx.df.index=mx.df.index+(timeSpan+timeStep)
>>> # ---
>>> # 2nd Read MXS (with the original Timestamps)
>>> # ---
>>> logger.debug("{0:s}: 2nd Read MXS (with the original Timestamps)".format('DOCTEST
↪'))
>>> mx.setResultsToMxsFile(add=True) # looks for M-1-0-1.MXS in same Dir
4
>>> rowsNew,colsNew=mx.df.shape
>>> rowsNew==3*rowsOld
True
>>> colsNew==colsOld
True
>>> # ---
>>> # Write Dump
>>> # ---
>>> mx.dumpInMxsFormat() # dumps to .MXS.dump-File in same Dir
(12, 8)
>>> # ---
>>> # Read Dump
>>> # ---
>>> logger.debug("{0:s}: Read Dump".format('DOCTEST'))
>>> mx.setResultsToMxsFile(mxsFile=mxsDumpFile)
12
>>> with zipfile.ZipFile(mx.mxsZipFile,'w') as myzip:
...     myzip.write(mx.mxsFile)
...     myzip.write(mxsDumpFile)
>>> # ---
>>> # Read Zip with Orig and Dump
>>> # ---
>>> logger.debug("{0:s}: Read Zip with Orig and Dump".format('DOCTEST'))
>>> mx.setResultsToMxsZipFile()
12
>>> (rows,cols)=mx.df.shape
>>> rows
12
>>> mx.ToH5()
>>> # ---
>>> # Without MX1, MXS
>>> # ---
>>> os.rename(mx.mx1File,mx.mx1File+'.blind')
>>> os.rename(mx.mxsFile,mx.mxsFile+'.blind')
>>> mx=Mx(mx1File=mx1File)
>>> os.rename(mx.mx1File+'.blind',mx.mx1File)
>>> os.rename(mx.mxsFile+'.blind',mx.mxsFile)
>>> # ---
>>> sir3sIdTimestamp=mx.mx1Df['Sir3sID'].iloc[mx.idxTIMESTAMP]
>>> mx.mx1Df['Sir3sID'][mx.mx1Df['Sir3sID']==sir3sIdTimestamp].index[0]
0
>>> mx.mx1Df['unpackIdx'][mx.mx1Df['Sir3sID']==sir3sIdTimestamp].iloc[0]
0
>>> (rows,cols)=mx.df.shape
>>> rows
12
>>> isinstance(mx.df.index[0],pd.tslib.Timestamp)
True
```

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```
>>> str(mx.df.index[0])
'2018-03-03 00:00:00+00:00'
>>> sir3sId=mx.mx1Df['Sir3sID'][mx.mx1Df['Sir3sID'].str.contains('KNOT~\S*~~
↪5642914844465475844~QM')].iloc[0] #KNOT~I~~5642914844465475844~QM
>>> ts=mx.df[sir3sId]
>>> isinstance(ts,pd.core.series.Series)
True
>>> "{:06.2f}".format(round(ts.iloc[0],2))
'176.71'
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.31',1,'New','_checkMxsVecsFile: (...,fullCheck=False,...)'))
>>> mx._checkMxsVecsFile()
(Timestamp('2018-03-03 00:00:00+0000', tz='UTC'), Timestamp('2018-03-03 00:00:11+0000
↪', tz='UTC'), 12)
>>> mx._checkMxsVecsFile(fullCheck=True)
(Timestamp('2018-03-03 00:00:00+0000', tz='UTC'), Timestamp('2018-03-03 00:00:11+0000
↪', tz='UTC'), 12)
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',1,'Change','setResultsToMxsFile: finally: h5.close()'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',2,'Change','setResultsToMxsZipFile: finally: h5.close()'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',3,'Change','ToH5: finally: h5.close()'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',4,'Change','FromH5: finally: h5.close()'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',5,'Change','*: except Exception as e'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',6,'Change','setResultsToMxsFile: finally: NewH5Vec=False'))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',7,'Change','__init__(...,NoH5Read=True,...)'))
>>> if os.path.exists(mx.h5FileVecs):
...     os.remove(mx.h5FileVecs)
>>> mx=Mx(mx1File=mx1File,NoH5Read=True)
>>> os.path.exists(mx.h5FileVecs) # h5 was written (h5 did not exist)
True
>>> h5VecsFileTime=os.path.getmtime(mx.h5FileVecs)
>>> mx=Mx(mx1File=mx1File,NoH5Read=True) # h5 will be written again (h5 exists)
>>> h5VecsFileTimeNow=os.path.getmtime(mx.h5FileVecs)
>>> logger.debug("1 h5VecsFileTime:{:s} < h5VecsFileTimeNow:{:s}".format(time.
↪strftime("%Y-%m-%d %H:%M:%S %z",time.gmtime(h5VecsFileTime)),time.strftime("%Y-%m-
↪%d %H:%M:%S %z",time.gmtime(h5VecsFileTimeNow))))
>>> h5VecsFileTime<h5VecsFileTimeNow # 1 h5 was written again (h5 did exist before)
True
>>> h5VecsFileTime=os.path.getmtime(mx.h5FileVecs)
>>> mx=Mx(mx1File=mx1File) # h5 is read - not written
>>> h5VecsFileTimeNow=os.path.getmtime(mx.h5FileVecs)
>>> logger.debug("2 h5VecsFileTime:{:s} == h5VecsFileTimeNow:{:s}".format(time.
↪strftime("%Y-%m-%d %H:%M:%S %z",time.gmtime(h5VecsFileTime)),time.strftime("%Y-%m-
↪%d %H:%M:%S %z",time.gmtime(h5VecsFileTimeNow))))
>>> mx.setResultsToMxsFile() # h5 will not be updated
0
>>> h5VecsFileTimeNow=os.path.getmtime(mx.h5FileVecs)
>>> logger.debug("3 h5VecsFileTime:{:s} == h5VecsFileTimeNow:{:s}".format(time.
↪strftime("%Y-%m-%d %H:%M:%S %z",time.gmtime(h5VecsFileTime)),time.strftime("%Y-%m-
↪%d %H:%M:%S %z",time.gmtime(h5VecsFileTimeNow))))
>>> mx.setResultsToMxsFile(NewH5Vec=True) # h5 is written
```

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```
4
>>> h5VecsFileTimeNow=os.path.getmtime(mx.h5FileVecs)
>>> logger.debug("h5VecsFileTime:{:s} == h5VecsFileTimeNow:{:s}".format(time.strftime(
↪"%Y-%m-%d %H:%M:%S %z",time.gmtime(h5VecsFileTime)),time.strftime("%Y-%m-%d %H:%M:
↪%S %z",time.gmtime(h5VecsFileTimeNow))))
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.32',8,'Change','setResultsToMxsZipFile: finally: NewH5Vec=False'))
>>> mx.ToH5()
>>> mx=Mx(mx1File=mx1File,NoH5Read=True)
>>> os.path.exists(mx.h5File)
False
>>> pd.set_option('display.max_columns',None)
>>> pd.set_option('display.max_rows',None)
>>> pd.set_option('display.max_colwidth',666666)
>>> pd.set_option('display.width',666666666)
>>> print(mx._getMx1DfAsOneString().replace('\n','\n '))
ATTRTYPE  DATALENGTH DATATYPE  DATATYPELENGTH  FLAGS OBJTYPE           OBJTYPE_PK    ␣
↪           TITLE    UNIT
     PH           4     REAL               4   1265    KNOT  5289899964753656852    ␣
↪           Druck   [bar]
     QM           4     REAL               4   1265    KNOT  5289899964753656852 ␣
↪Externer Durchfluss  [m3/h]
     PH           4     REAL               4   1265    KNOT  5642914844465475844    ␣
↪           Druck   [bar]
     QM           4     REAL               4   1265    KNOT  5642914844465475844 ␣
↪Externer Durchfluss  [m3/h]
>>> print("'''{:s}'''".format(repr(mx.mx2Df).replace('\n','\n   ')))
'''       AttrType                                        Data  DataLength DataType ␣
↪DataTypeLength  NOfItems       ObjType
   0  tk            [5642914844465475844, 5289899964753656852]          40     CHAR  ␣
↪           20         2  KNOT
   1  pk                                  [5252810657060947333]          20     CHAR  ␣
↪           20         1  LFKT
   2  pk                                  [5502689500012692689]          20     CHAR  ␣
↪           20         1  PHI1
   3  pk                                  [5732781659713982525]          20     CHAR  ␣
↪           20         1  PUMD
   4  pk                                  [5163733225086798083]          20     CHAR  ␣
↪           20         1  PVAR
   5  pk                                  [4742976321174242828]          20     CHAR  ␣
↪           20         1  QVAR
   6  tk                                  [4737064599036143765]          20     CHAR  ␣
↪           20         1  ROHR
   7  N_OF_POINTS                                        (101,)           4     INT4  ␣
↪            4         1  ROHR
   8  pk                                  [5396761270498593493]          20     CHAR  ␣
↪           20         1  SWVT        '''
>>> print(mx._getDfAsOneString())
                         KNOT~~~5289899964753656852~PH KNOT~~~5289899964753656852~QM␣
↪KNOT~~~5642914844465475844~PH KNOT~~~5642914844465475844~QM
2018-03-03 00:00:00+00:00                          0.0                          -176.7␣
↪                        4.2                          176.7
2018-03-03 00:00:01+00:00                          0.0                          -176.7␣
↪                        4.2                          176.7
2018-03-03 00:00:02+00:00                          0.0                          -176.7␣
↪                        4.2                          176.7
2018-03-03 00:00:03+00:00                          0.0                          -176.7␣
↪                        4.2                          176.7
```

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```
>>> # ---
>>> # Clean Up OneLPipe
>>> # ---
>>> mx.delFiles()
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> mxsDumpFile=mx.mxsFile+'.dump'
>>> if os.path.exists(mxsDumpFile):
...     os.remove(mxsDumpFile)
>>> # ---
>>> # LocalHeatingNetwork
>>> # ---
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDLocalHeatingNetwork\B1\V0\BZ1\M-
↪1-0-1'+dotResolution+'.MX1'))
>>> mx=Mx(mx1File=mx1File,NoH5Read=True,NoMxsRead=True)
>>> # ---
>>> # Clean Up LocalHeatingNetwork
>>> # ---
>>> mx.delFiles()
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> mxsDumpFile=mx.mxsFile+'.dump'
>>> if os.path.exists(mxsDumpFile):
...     os.remove(mxsDumpFile)
>>> mx.setResultsToMxsFile(maxRecords=1)
1
>>> print(mx._getDfAsOneString())
                          KNOT~~~5356267303828212700~PH KNOT~~~5397990465339071638~QM␣
↪KNOT~~~5736262931552588702~PH
2004-09-22 08:30:00+00:00                          2.0                          0.0␣
↪                          4.1
>>> logger.debug("{:s}: CHANGEHISTORY: {:>10s}: {:>3d}: {:>6s}: {:s}".format('DOCTEST
↪','0.0.41',1,'New',"getMxsVecsFileData"))
>>> timesReq=[]
>>> timesReq.append(mx.df.index[0])
>>> plotTimeDfs=mx.getMxsVecsFileData(timesReq=timesReq)
>>> len(plotTimeDfs)
1
>>> isinstance(plotTimeDfs[0],pd.core.frame.DataFrame)
True
>>> print(mx._getDfVecAsOneString(df=plotTimeDfs[0],regex='KNOT~\S*~\S*~\S*~T$'))
                         KNOT~~~~T
2004-09-22 08:30:00+00:00      60.00
>>> timesReq[0]=timesReq[0]-pd.to_timedelta('1 second')
>>> plotTimeDfs=mx.getMxsVecsFileData(timesReq=timesReq)
>>> len(plotTimeDfs)
0
>>> # ---
>>> # Clean Up LocalHeatingNetwork
>>> # ---
>>> mx.delFiles()
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> mxsDumpFile=mx.mxsFile+'.dump'
>>> if os.path.exists(mxsDumpFile):
...     os.remove(mxsDumpFile)
>>> # ---
```

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```
>>> # TinyWDN
>>> # ---
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDTinyWDN\B1\V0\BZ1\M-1-0-1
↪'+dotResolution+'.MX1'))
>>> mx=Mx(mx1File=mx1File,NoH5Read=True,NoMxsRead=True)
>>> print(mx._getMx1DfAsOneString().replace('\n','\n '))
ATTRTYPE  DATALENGTH DATATYPE  DATATYPELENGTH  FLAGS OBJTYPE           OBJTYPE_PK␣
↪TITLE    UNIT
     PH           4     REAL               4      1    KNOT  4711309381204507891    ␣
↪    [bar]
     PH           4     REAL               4      1    KNOT  5179406559406617933    ␣
↪    [bar]
     QM           4     REAL               4      1    KNOT  5179406559406617933    ␣
↪   [m3/h]
>>> print("'''{:s}'''".format(repr(mx.mx2Df).replace('\n','\n   ')))
'''        AttrType                                                                ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                          Data  DataLength DataType ␣
↪DataTypeLength  NOfItems       ObjType
   0   tk                                                                          ␣
↪                                          [5179406559406617933, 5706345341889312301,␣
↪5028754475676510796, 4880261452311588026, 4711309381204507891, 5697271655044179265,␣
↪5165939128645634755, 4790715298669926433, 5629305599838467353, 5388350113283448399,␣
↪4702108116010765829, 5095105260880965618, 5113971272348625691, 5042575626021291052,␣
↪4780213881308610359, 4832703654265095420, 5498009282312522569]         340     CHAR␣
↪             20        17  KNOT
   1   pk                                                                          ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                          [4673068329861421062]          20     CHAR ␣
↪            20         1  LFKT
   2   tk                                                                          ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                          [4914542339545953765]          20     CHAR ␣
↪            20         1  OBEH
   3   pk                                                                          ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                          [5295290952622367198]          20     CHAR ␣
↪            20         1  PHI1
   4   pk                                                                          ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                                                                  ␣
↪                                          [5623478865255556069]          20     CHAR ␣
↪            20         1  PUMD
```

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```
  5   pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [4728046487162557851]          20     CHAR ␣
↪            20         1  PVAR
  6   pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [5377391575499483345]          20     CHAR ␣
↪            20         1  QVAR
  7   tk            [4820249423461369400, 5508607066407516090, 5095827458323717990,␣
↪5700891034663173564, 5729543164571807746, 5067763789937102564, 4799443678518131207,␣
↪5605831797210762998, 5126294008398890918, 5419686039699182683, 5312551747276023841,␣
↪5296152509037292880, 5670035893444309530, 5433880705192526755, 4648047345314768819,␣
↪4984438795139137900, 5644872080928983958, 5148090523913666712, 5461179577260327606,␣
↪4978978527327130204, 5497762617222653432, 5076321356874807093]         440     CHAR␣
↪             20        22  ROHR
  8   N_OF_POINTS                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                          (5,␣
↪6, 5, 3, 4, 3, 3, 3, 3, 3, 3, 3, 3, 2, 5, 4, 3, 3, 4, 3, 2, 2)          88     INT4␣
↪              4        22  ROHR
  9   pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [4955975670134047353]          20     CHAR ␣
↪            20         1  SWVT
  10  pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [4821755615165519990]          20     CHAR ␣
↪            20         1  TEVT
  11  tk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [5466655470152247657]          20     CHAR ␣
↪            20         1  VENT
  12  pk                                                                    ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                                                           ␣
↪                                        [5736734929574151957]          20     CHAR ␣
↪            20         1  WEVT        '''
>>> mx.setResultsToMxsFile(maxRecords=1)
```

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```
1
>>> print(mx._getDfAsOneString())
                        KNOT~~~4711309381204507891~PH KNOT~~~5179406559406617933~PH␣
↪KNOT~~~5179406559406617933~QM
2002-05-22 16:16:16+00:00                        1.0                          3.0␣
↪                       140.0
>>> # ---
>>> # Clean Up Tiny WDN
>>> # ---
>>> mx.delFiles()
>>> # ---
>>> # GPipe
>>> # ---
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDGPipe\B1\V0\BZ1\M-1-0-1
↪'+dotResolution+'.MX1'))
>>> mx=Mx(mx1File=mx1File)
>>> plotTimeDfs=mx.getMxsVecsFileData()
>>> len(plotTimeDfs[0]['ROHR~*~*~*~PVEC'].iloc[0]) == mx.mx2Df[mx.mx2Df['AttrType'].
↪str.contains('N_OF_POINTS')].iloc[0].Data[0]
True
>>> plotTimeDfs[0]['ROHR~*~*~*~PVEC'].iloc[0][0]
41.0
>>> # ---
>>> # Clean Up GPipe
>>> # ---
>>> mx.delFiles()
```

**class** Mx.**Mx**(*mx1File*, *NoH5Read=False*, *NoMxsRead=False*)

Reading SIR 3S' MX-Files.

**Summary:**

- (mx1File): use this to profit from previous reads finalized with ToH5()
- (mx1File,NoH5Read=True): use this for a fresh start with implicit .MXS-File read; finalize with ToH5()
- (mx1File,NoH5Read=True,NoMxsRead=True): use this for a fresh start; call setResultsTo...() explicit; finalize with ToH5()
- note that base.Y.h5-File has to be dunped explicitely with ToH5()
- and base.Y.vec.h5-File is written implicitely while (implicit or explicit calls to) setResultsTo...()
- and is deleted explicitely (mx1File,NoH5Read=True) or implicitely (because it is i.e. to old)

**Args:**

- mx1File (str): base.MX1-File (an XML-File) (base.Y.MX1-File from 90-10 on)
- **NoH5Read (bool):**

  **False (default - use this to profit from previous reads finalized with ToH5()):**

  - **If a base.Y.h5-File**
    - exists
    - and is newer (>) than an .MX1-File (base.Y.Mx1-File from 90-10 on)
    - and is newer (>) than an .MXS-File (base.Y.MXS-File from 90-10 on):
      - The base.Y.h5-File is read instead of the .MX1-File.

**True (use this for a fresh start):**

- An base.Y.h5-File is deleted if existing.
- The base.Y.Mx1-File is read.
- The base.Y.vec.h5-File is newly created in case of an .MXS-File read.

- **NoMxsRead (bool):**

  **True:**

  - a base.Y.MXS-File is not read
  - a base.Y.vec.h5-File is not touched

  **False (default):**

  - **If a base.Y.MXS-File**
    - exists
    - and is newer (>=) than base.Y.Mx1-File
    - and base.Y.h5-File is not read:
      - The base.Y.MXS-File is read.
      - NoH5Read=True will delete base.Y.vec.h5-File.

**Attributes:**

- **states**
  - h5Read: True, if read from H5
- **fileNames**
  - .mx1File: base.Y.MX1-File
  - **derived from mx1File**
    - .mx2File: base.MX2-File
    - .mxsFile: base.Y.MXS-File
    - .mxsZipFile base.ZIP
    - constructed from MX during Init and Usage:
    -
    - .h5File: base.Y.h5-File
    - .h5FileVecs: base.Y.vec.h5-File: MXS-H5Dump written implicitely
    - .h5FileMx1FmtString: base.Y.h5-File.metadata written implicitely
- **.mxRecordStructFmtString**
  - usage: struct.unpack(self.mxRecordStructFmtString,a_MXS_Record)
  - **.h5FileMx1FmtString:**
    - it was not possible to store mxRecordStructFmtString in H5-Format as Metadata
    - therefore mxRecordStructFmtString is stored in a file named .h5FileMx1FmtString
    - in .h5File the Link to this file is stored as Metadata

        * as pointed out with usage above mxRecordStructFmtString has nothing to do with writing to or reading from H5
        * mxRecordStructFmtString is only about reading from (and writing to for test purposes) MXS
        * the .mxRecordStructFmtString/.h5FileMx1FmtString stuff is only about performance:
        * if after reading from H5 only a(nother) MXS shall be read again ...
        * ... the stuff avoids the time-consuming reconstruction of mxRecordStructFmtString
    - **dataFrames**
      - .mx1Df
      - .mx2Df
      - **.df**
        * the base.Y.MXS-File(s) Content
        * non Vectordata only
        * index: TIMESTAMP (scenario time)
        * **columns: Values**
          · The following (String-)ID - called Sir3sID - is used as Columnlabel:
          · this Sir3sID consists of ~ (Mx.reSir3sIDSep) separated .MX1-File terms:
          · OBJTYPE~NAME1~NAME2~OBJTYPE_PK~ATTRTYPE
          · Sir3sID regExp Example: Mx.reSir3sID: (?P<OBJTYPE>S+)~(?P<NAME1>[S]*)~(?P<NAME1>S*)~(?P<OBJTYPE_PK>d+)~(?P<ATTRTYPE>S+)'
      - **.dfVecAggs**
        * some base.Y.vec.h5-File (Vectordata only) Aggregates as df
        * **MultiIndex:**
          · TYPE: SNAPSHOTTYPEs: TIME,TMIN,TMAX (from _readMxsFile) or Aggregates from 2 Times: MIN,MAX,... (from getVecAggs)
          · Sir3sID
          · TIMESTAMPL
          · TIMESTAMPR
        * Cols: mx2Idx
        * Value: Aggregate (i.e. TMIN) for mx2Idx-Col

**Raises:** MxError

**delFiles**()

Deletes Files constructed by MX during Init and Usage.

**_initWithMx1**()

(Re-)initialize .mx1Df, .mx2Df, .mxRecordStructFmtString and related stuff with .mx1File.

**Calls:**

- ._parseMx1()
- ._parseMx2()

- ._buildMxRecordStructUnpackFmtString()
- ._buildMxRecordStructUnpackFmtStringPost()

**Raises:** MxError

**_parseMx1**()

Parses .mx1File.

**Sets**

- .mx1Df

**Raises:** MxError

**_parseMx2**()

Parses .mx2File.

**Sets**

- .mx2Df

**Raises:** MxError

**_buildMxRecordStructUnpackFmtString**()

(Re-)builds .mxRecordStructFmtString and releated stuff.

**Sets**

- .mxRecordStructFmtString
- .mx1Df['unpackIdx']

**Raises:** MxError

**_buildMxRecordStructUnpackFmtStringPost**()

Stuff todo after buildMxRecordStructUnpackFmtString.

**Sets**

- .idxCVERSO (idx of CVERSO in MX1)
- .unpackIdxCVERSO (idx of CVERSO in recordData)
- .idxTIMESTAMP (idx of TIMESTAMP in MX1)
- .unpackIdxTIMESTAMP (idx of TIMESTAMP in recordData)
- .mxColumnNames=[] (of non Vectordata without TIMESTAMP in MX1-Sequence)
- .mxColumnNamesVecs=[] (of Vectordata without TIMESTAMP in MX1-Sequence)
- .idxUnpackNonVectorChannels[] (idx in recordData)
- .idxUnpackVectorChannels[] (idx in recordData of the 1st ([0]) Element of the Vector)
- .idxOfNonVectorChannels[] (idx in MX1 without TIMESTAMP)
- .idxVectorChannels[] (idx in MX1)

**Raises:** MxError

**_readMxsFile**(*mxsFilePtr*, *mxsVecsH5StorePtr*, *firstTime=None*, *maxRecords=None*)

**Args:**

- mxsFilePtr: .MXS-File
- mxsVecsH5StorePtr: .vec.h5-File (Vectordata only)

- **firstTime: used to calculate h5Key for mxsVecsH5Store**
  - None (default): firstTime is set to 1st TIMESTAMP in .MXS-File
  - **else:**
    - caller sets firstTime - in general the youngest TIMESTAMP in the df
- maxRecords: Anzahl der max. zu lesenden Records

**Returns:** df: the mxsFile (non Vectordata only) as DataFrame

- index: TIMESTAMP
- df.index.is_unique
  - might be False
  - because of SIR 3S'
  - 1st Time twice (SNAPSHOTTYPE: STAT+TIME) and Last Time triple (SNAPSHOTTYPE: TIME+TMIN/TMAX)

timesWrittenToMxsVecs: the times written to the mxsVecsH5Store (Vectordata only)

implicit Return: about mxsVecsH5StorePtr (Vectordata only):

- is used to update the mxsVecsH5Store-File with mxsFile-Content (Vectordata only):
  - **Key: microseconds from firstTime**
    - can be negative if firstTime (in general the youngest TIMESTAMP in the df) is older than the oldest TIMESTAMP in the .MXS-File
  - **dfVecs: a df containing the Vectordata**
    - index: TIMESTAMP
    - values: Vectordata for the TIMESTAMP
- the Vectordata for a TIMESTAMP is only written when the Key does _not already exist

**implicit Return: about .dfVecAggs:**

- some base.Y.vec.h5-File (Vectordata only) Aggregates as df
- **MultiIndex:**
  - TYPE: SNAPSHOTTYPEs: TIME,TMIN,TMAX (from _readMxsFile)
  - Sir3sID
  - TIMESTAMPL
  - TIMESTAMPR
- Cols: mx2Idx
- Value: Aggregate (i.e. TMIN) for mx2Idx-Col

**Raises:** MxError

```
>>> mx=mxs['LocalHeatingNetwork']
>>> mx.delFiles()
>>> mx.setResultsToMxsFile() # reads 5 TIMESTAMPS and constructs .vec.h5␣
↪while reading
5
```

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```
>>> mx.dfVecAggs.loc[(['TIME','TMIN','TMAX'],'KNOT~*~*~*~PH',slice(None),
↪slice(None)),0:22]
                                                                    0         1␣
↪        2         3         4         5         6         7         8         ␣
↪  9         10        11        12        13   14        15        16         ␣
↪ 17        18   19        20        21        22
TYPE Sir3sID       TIMESTAMPL          TIMESTAMPR                                ␣
↪                                                                                ␣
↪                                                                                ␣
↪
TIME KNOT~*~*~*~PH 2004-09-22 08:30:00 2004-09-22 08:30:00  2.302971  3.
↪985846  4.083384  4.121495  2.043288  2.283566  2.004937  4.311307  4.
↪126019  2.309655  4.291591  2.000133  2.141440  3.825970  2.0  3.819467  2.
↪314658  3.816599  2.312659  2.0  3.845104  4.125885  3.814690
TMIN KNOT~*~*~*~PH 2004-09-22 08:30:00 2004-09-22 08:31:00  2.052100  2.
↪183028  2.200011  2.206647  2.007717  2.048865  2.000910  2.248923  2.
↪207463  2.053240  2.234365  2.000021  2.025138  2.156905  2.0  2.155822  2.
↪054124  2.155325  2.053771  2.0  2.160025  2.207441  2.154995
TMAX KNOT~*~*~*~PH 2004-09-22 08:30:00 2004-09-22 08:31:00  2.302971  4.
↪085822  4.183360  4.221471  2.043288  2.283566  2.004937  4.411284  4.
↪225996  2.309655  4.391567  2.000133  2.141440  3.925947  2.0  3.919444  2.
↪314658  3.916576  2.312659  2.0  3.945080  4.225861  3.914667
>>> mx.dfVecAggs
                                                                                ␣
↪0           1             2           3           4           5           6␣
↪          7           8           9           10          11          12    ␣
↪      13          14          15          16          17          18        ␣
↪  19          20          21          22          23          24          ␣
↪25          26          27          28          29          30          31
TYPE Sir3sID                 TIMESTAMPL          TIMESTAMPR                   ␣
↪                                                                                ␣
↪                                                                                ␣
↪                                                                                ␣
↪                                                                                ␣
↪
TIME ROHR~*~*~*~QMI          2004-09-22 08:30:00 2004-09-22 08:30:00   -8.
↪509475   19.059780 -1.537890e+01    8.509476  -22.987946   22.987946   22.
↪987947   -3.928166   22.987947   15.378901    3.928167  -22.987946  -19.
↪059778   -3.928166    3.928167  -22.987946         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN     ␣
↪    NaN
    KNOT~*~*~*~P            2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283566    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819467    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814690         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    FWES~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN         ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN     ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN
    KNOT~*~*~*~HMAX_INST    2004-09-22 08:30:00 2004-09-22 08:30:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283566    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
```

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```
    ROHR~*~*~*~PVECMIN_INST 2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    3.309655  4.985845e+00    4.845103    3.283566    3.302971    4.
↪825970    4.819467    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314658    5.121495    5.083383    4.
↪845103    4.825970    4.819467    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
    VENT~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00   22.
↪987947   22.987946  2.199973e-06         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~H            2004-09-22 08:30:00 2004-09-22 08:30:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283566    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~SVEC         2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000   88.019997  0.000000e+00  405.959991    0.000000   83.550003    0.
↪000000   88.019997    0.000000   73.419998    0.000000  195.529999    0.
↪000000   68.599998    0.000000  109.769997    0.000000   76.400002    0.
↪000000   83.550003    0.000000  164.910004    0.000000  195.529999    0.
↪000000  405.959991    0.000000  164.910004    0.000000  109.769997    0.
↪000000   76.400002
    ROHR~*~*~*~PVEC         2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    3.309655  4.985845e+00    4.845103    3.283566    3.302971    4.
↪825970    4.819467    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314659    5.121495    5.083383    4.
↪845103    4.825970    4.819467    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
    ROHR~*~*~*~VI           2004-09-22 08:30:00 2004-09-22 08:30:00   -0.
↪266728    0.608561 -4.820491e-01    0.271700   -0.321646    0.733984    0.
↪327641   -0.123128    0.733984    0.491034    0.125423   -0.720553   -0.
↪597426   -0.123128    0.125423   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~T            2004-09-22 08:30:00 2004-09-22 08:30:00   60.
↪000000   90.000000  9.000000e+01   90.000000   60.000000   60.000000   60.
↪000000   60.000000   90.000000   60.000000   60.000000   60.000000   60.
↪000000   90.000000   60.000000   90.000000   60.000000   90.000000   60.
↪000000   60.000000   90.000000   90.000000   90.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KLAP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~RHOVEC       2004-09-22 08:30:00 2004-09-22 08:30:00  983.
↪700012  983.700012  9.657000e+02  965.700012  983.700012  983.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  965.700012  965.700012  983.700012  983.700012  983.
↪700012  983.700012  983.700012  983.700012  965.700012  965.700012  983.
↪700012  983.700012
```

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```
    ROHR~*~*~*~*~TVEC         2004-09-22 08:30:00 2004-09-22 08:30:00   60.
↪000000   60.000000  9.000000e+01   90.000000   60.000000   60.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   90.000000   90.000000   60.000000   60.000000   60.
↪000000   60.000000   60.000000   60.000000   90.000000   90.000000   60.
↪000000   60.000000
    FWES~*~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:30:00    1.
↪291413         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~*~MVEC         2004-09-22 08:30:00 2004-09-22 08:30:00   -2.
↪363743   -2.363743  5.294384e+00    5.294384   -4.271917   -4.271917    2.
↪363743    2.363743   -6.385540   -6.385540    6.385540    6.385540    6.
↪385541    6.385541   -1.091157   -1.091157    6.385540    6.385540    4.
↪271917    4.271917    1.091157    1.091157   -6.385540   -6.385540   -5.
↪294383   -5.294383   -1.091157   -1.091157    1.091157    1.091157   -6.
↪385540   -6.385540
    ROHR~*~*~*~*~VK           2004-09-22 08:30:00 2004-09-22 08:30:00   -0.
↪266728    0.608561 -4.820491e-01    0.271700   -0.321646    0.733984    0.
↪327641   -0.123128    0.733984    0.491034    0.125423   -0.720553   -0.
↪597426   -0.123128    0.125423   -0.720553         NaN         NaN
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↪    NaN
    PUMP~*~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~*~PDAMPF       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪199200    0.701100  7.011000e-01    0.701100    0.199200    0.199200    0.
↪199200    0.199200    0.701100    0.199200    0.199200    0.199200    0.
↪199200    0.701100    0.199200    0.701100    0.199200    0.701100    0.
↪199200    0.199200    0.701100    0.701100    0.701100         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~*~PVECMAX_INST 2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    3.309655  4.985845e+00    4.845104    3.283566    3.302971    4.
↪825970    4.819468    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314659    5.121495    5.083383    4.
↪845104    4.825970    4.819468    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
    VENT~*~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪374182    0.367335  3.163897e-07         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~*~HMIN_INST    2004-09-22 08:30:00 2004-09-22 08:30:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.285563    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
```

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    ROHR~*~*~*~QMK          2004-09-22 08:30:00 2004-09-22 08:30:00   -8.
↪509475   19.059780 -1.537890e+01    8.509476  -22.987946   22.987946   22.
↪987947   -3.928166   22.987947   15.378901    3.928167  -22.987946  -19.
↪059778   -3.928166    3.928167  -22.987946         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~PMIN_INST    2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283565    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819467    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KNOT~*~*~*~RHO          2004-09-22 08:30:00 2004-09-22 08:30:00  983.
↪700012  965.700012  9.657000e+02  965.700012  983.700012  983.700012  983.
↪700012  983.700012  965.700012  983.700012  983.700012  983.700012  983.
↪700012  965.700012  983.700012  965.700012  983.700012  965.700012  983.
↪700012  983.700012  965.700012  965.700012  965.700012         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~ZVEC         2004-09-22 08:30:00 2004-09-22 08:30:00   20.
↪000000   20.000000  2.000000e+01   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000
    KNOT~*~*~*~PMAX_INST    2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283566    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819468    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814691         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KLAP~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:30:00    1.
↪291413         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
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↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~WALTER       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪209004    0.161072  8.707356e-02    0.058160    0.403910    0.207487    0.
↪433363    0.496769    0.000000    0.286059    0.496769    0.496769    0.
↪328532    0.393637    0.000000    0.483626    0.000000    0.848857    0.
↪247643    0.496769    0.346373    0.000000    1.091969         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KNOT~*~*~*~PH           2004-09-22 08:30:00 2004-09-22 08:30:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283566    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    FWVB~*~*~*~W            2004-09-22 08:30:00 2004-09-22 08:30:00  160.
↪000000  200.000000  1.600000e+02  160.000000  120.000000         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
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↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
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    FWVB~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:30:00    3.
↪928166    6.869426  4.581308e+00    3.928166    3.680879         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~QMAV         2004-09-22 08:30:00 2004-09-22 08:30:00   -8.
↪509475   19.059780 -1.537890e+01    8.509476  -22.987946   22.987946   22.
↪987947   -3.928166   22.987947   15.378901    3.928167  -22.987946  -19.
↪059778   -3.928166    3.928167  -22.987946         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    ROHR~*~*~*~VAV          2004-09-22 08:30:00 2004-09-22 08:30:00   -0.
↪266728    0.608561 -4.820491e-01    0.271700   -0.321646    0.733984    0.
↪327641   -0.123128    0.733984    0.491034    0.125423   -0.720553   -0.
↪597426   -0.123128    0.125423   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    VENT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000    0.000000  0.000000e+00         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
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↪NaN
    KLAP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
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    PUMP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
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↪NaN
    FWVB~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
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↪NaN
```

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    FWES~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:30:00    0.
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↪NaN
TMIN ROHR~*~*~*~QMI          2004-09-22 08:30:00 2004-09-22 08:31:00   -8.
↪509475    7.394749 -1.537890e+01    3.256006  -22.987946    9.266180    9.
↪266181   -3.928166    9.266181    5.923044    1.496261  -22.987946  -19.
↪059778   -3.928166    1.496260  -22.987946         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
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↪    NaN
    KNOT~*~*~*~P            2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪052100    3.183028  3.200011e+00    3.206647    3.007717    3.048865    3.
↪000910    3.248923    3.207463    3.053240    3.234365    3.000021    3.
↪025138    3.156905    3.000000    3.155822    3.054124    3.155325    3.
↪053771    3.000000    3.160025    3.207441    3.154995         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    FWES~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    9.
↪266181         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
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    KNOT~*~*~*~HMAX_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283565    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~PVECMIN_INST 2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪052100    3.053240  3.183028e+00    3.160025    3.048864    3.052100    3.
↪156905    3.155822    3.000022    3.000910    3.200011    3.183028    3.
↪207441    3.206647    3.053771    3.054124    3.206647    3.200011    3.
↪160025    3.156905    3.155822    3.155326    3.007717    3.025139    3.
↪025139    3.048864    3.053240    3.053771    3.155326    3.154995    3.
↪000910    3.007717
    VENT~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    9.
↪266181    9.266179  2.199715e-06         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
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↪NaN
    KNOT~*~*~*~H            2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪052100    2.183028  2.200011e+00    2.206647    2.007717    2.048865    2.
↪000910    2.248923    2.207463    2.053240    2.234365    2.000021    2.
↪025138    2.156905    2.000000    2.155822    2.054124    2.155325    2.
↪053771    2.000000    2.160025    2.207441    2.154995         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~SVEC         2004-09-22 08:30:00 2004-09-22 08:31:00    0.
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↪000000   88.019997    0.000000   73.419998    0.000000  195.529999    0.
↪000000   68.599998    0.000000  109.769997    0.000000   76.400002    0.
↪000000   83.550003    0.000000  164.910004    0.000000  195.529999    0.
↪000000  405.959991    0.000000  164.910004    0.000000  109.769997    0.
↪000000   76.400002
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    ROHR~*~*~*~PVEC         2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪052100    3.053240  3.183028e+00    3.160025    3.048864    3.052100    3.
↪156905    3.155822    3.000022    3.000910    3.200011    3.183028    3.
↪207441    3.206647    3.053771    3.054124    3.206647    3.200011    3.
↪160025    3.156905    3.155822    3.155326    3.007717    3.025139    3.
↪025139    3.048864    3.053240    3.053771    3.155326    3.154995    3.
↪000910    3.007717
    ROHR~*~*~*~VI           2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪266728    0.236108 -4.820491e-01    0.103961   -0.321646    0.295861    0.
↪132068   -0.123128    0.295861    0.189117    0.047774   -0.720553   -0.
↪597426   -0.123128    0.047774   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
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↪    NaN
    KNOT~*~*~*~T            2004-09-22 08:30:00 2004-09-22 08:31:00   60.
↪000000   90.000000  9.000000e+01   90.000000   60.000000   60.000000   60.
↪000000   60.000000   90.000000   60.000000   60.000000   60.000000   60.
↪000000   90.000000   60.000000   90.000000   60.000000   90.000000   60.
↪000000   60.000000   90.000000   90.000000   90.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KLAP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    9.
↪266181         NaN           NaN         NaN         NaN         NaN
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↪NaN
    ROHR~*~*~*~RHOVEC       2004-09-22 08:30:00 2004-09-22 08:31:00  983.
↪700012  983.700012  9.657000e+02  965.700012  983.700012  983.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
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↪700012  983.700012
    ROHR~*~*~*~TVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   60.
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↪000000   60.000000
    FWES~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪520554         NaN           NaN         NaN         NaN         NaN
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↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~MVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   -2.
↪363743   -2.363743  2.054097e+00    2.054097   -4.271917   -4.271917    0.
↪904446    0.904446   -6.385540   -6.385540    2.573939    2.573939    2.
↪573939    2.573939   -1.091157   -1.091157    2.573939    2.573939    1.
↪645290    1.645290    0.415628    0.415628   -6.385540   -6.385540   -5.
↪294383   -5.294383   -1.091157   -1.091157    0.415628    0.415628   -6.
↪385540   -6.385540
    ROHR~*~*~*~VK           2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪266728    0.236108 -4.820491e-01    0.103961   -0.321646    0.295861    0.
↪132068   -0.123128    0.295861    0.189117    0.047774   -0.720553   -0.
↪597426   -0.123128    0.047774   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
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    PUMP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    9.
↪266181         NaN           NaN         NaN         NaN         NaN        ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN    ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN        ␣
↪NaN
    KNOT~*~*~*~PDAMPF       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪199200    0.701100  7.011000e-01    0.701100    0.199200    0.199200    0.
↪199200    0.199200    0.701100    0.199200    0.199200    0.199200    0.
↪199200    0.701100    0.199200    0.701100    0.199200    0.701100    0.
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↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~PVECMAX_INST 2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    3.309655  4.985845e+00    4.845104    3.283566    3.302971    4.
↪825970    4.819468    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314658    5.121495    5.083383    4.
↪845104    4.825970    4.819468    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
    VENT~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪150829    0.148069  3.163526e-07         NaN         NaN         NaN        ␣
↪ NaN         NaN         NaN         NaN         NaN         NaN         ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN    ␣
↪    NaN         NaN         NaN         NaN         NaN         NaN        ␣
↪NaN
    KNOT~*~*~*~HMIN_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪052100    2.183028  2.200011e+00    2.206647    2.007717    2.048865    2.
↪000910    2.248923    2.207463    2.053240    2.234365    2.000021    2.
↪025138    2.156905    2.000000    2.155822    2.054124    2.155325    2.
↪053771    2.000000    2.160025    2.207441    2.154995         NaN        ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~QMK          2004-09-22 08:30:00 2004-09-22 08:31:00   -8.
↪509475    7.394749 -1.537890e+01    3.256006  -22.987946    9.266180    9.
↪266181   -3.928166    9.266181    5.923044    1.496261  -22.987946  -19.
↪059778   -3.928166    1.496260  -22.987946         NaN         NaN        ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN    ␣
↪    NaN
    KNOT~*~*~*~PMIN_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪052100    3.183028  3.200011e+00    3.206647    3.007717    3.048865    3.
↪000910    3.248923    3.207463    3.053240    3.234365    3.000021    3.
↪025138    3.156905    3.000000    3.155822    3.054124    3.155325    3.
↪053771    3.000000    3.160025    3.207441    3.154995         NaN        ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KNOT~*~*~*~RHO          2004-09-22 08:30:00 2004-09-22 08:31:00  983.
↪700012  965.700012  9.657000e+02  965.700012  983.700012  983.700012  983.
↪700012  983.700012  965.700012  983.700012  983.700012  983.700012  983.
↪700012  965.700012  983.700012  965.700012  983.700012  965.700012  983.
↪700012  983.700012  965.700012  965.700012  965.700012         NaN        ␣
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~ZVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   20.
↪000000   20.000000  2.000000e+01   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000
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```
    KNOT~*~*~*~PMAX_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283565    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819468    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814691         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KLAP~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪520554         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~WALTER       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪209004    0.161072  8.707356e-02    0.058160    0.403910    0.207487    0.
↪433363    0.496769    0.000000    0.286059    0.496769    0.496769    0.
↪328532    0.393637    0.000000    0.483626    0.000000    0.848857    0.
↪247643    0.496769    0.346373    0.000000    1.091969         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    KNOT~*~*~*~PH           2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪052100    2.183028  2.200011e+00    2.206647    2.007717    2.048865    2.
↪000910    2.248923    2.207463    2.053240    2.234365    2.000021    2.
↪025138    2.156905    2.000000    2.155822    2.054124    2.155325    2.
↪053771    2.000000    2.160025    2.207441    2.154995         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    FWVB~*~*~*~W            2004-09-22 08:30:00 2004-09-22 08:31:00   76.
↪226158   77.649529  6.145824e+01   60.944885   47.978928         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWVB~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    1.
↪871431    2.667038  1.759745e+00    1.496260    1.471705         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~QMAV         2004-09-22 08:30:00 2004-09-22 08:31:00   -8.
↪509475    7.394749 -1.537890e+01    3.256006  -22.987946    9.266180    9.
↪266181   -3.928166    9.266181    5.923044    1.496261  -22.987946  -19.
↪059778   -3.928166    1.496260  -22.987946         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    ROHR~*~*~*~VAV          2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪266728    0.236108 -4.820491e-01    0.103961   -0.321646    0.295861    0.
↪132068   -0.123128    0.295861    0.189117    0.047774   -0.720553   -0.
↪597426   -0.123128    0.047774   -0.720553         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
```

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```
    ROHR~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    VENT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KLAP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    PUMP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWVB~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWES~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
TMAX ROHR~*~*~*~QMI         2004-09-22 08:30:00 2004-09-22 08:31:00   -3.
↪256005   19.059780 -5.923043e+00    8.509476   -9.266179   22.987946   22.
↪987947   -1.496260   22.987947   15.378901    3.928167   -9.266179   -7.
↪394748   -1.496260    3.928167   -9.266179         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~P            2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    5.085822  5.183360e+00    5.221471    3.043288    3.283566    3.
↪004937    5.411284    5.225996    3.309655    5.391567    3.000133    3.
↪141440    4.925947    3.000000    4.919444    3.314658    4.916576    3.
↪312659    3.000000    4.945080    5.225861    4.914667         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    FWES~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00   22.
↪987947         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
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```
    KNOT~*~*~*~HMAX_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    4.085822  4.183360e+00    4.221471    2.043288    2.283566    2.
↪004937    4.411284    4.225996    2.309655    4.391567    2.000133    2.
↪141440    3.925947    2.000000    3.919444    2.314658    3.916576    2.
↪312659    2.000000    3.945080    4.225861    3.914667         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~PVECMIN_INST 2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    3.309655  4.985845e+00    4.845104    3.283566    3.302971    4.
↪825970    4.819468    3.000133    3.004937    5.083383    4.985845    5.
↪125884    5.121495    3.312659    3.314658    5.121495    5.083383    4.
↪845104    4.825970    4.819468    4.816599    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.816599    4.814690    3.
↪004937    3.043288
    VENT~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00   22.
↪987947   22.987946  2.199973e-06         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~H            2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    4.085822  4.183360e+00    4.221471    2.043288    2.283566    2.
↪004937    4.411284    4.225996    2.309655    4.391567    2.000133    2.
↪141440    3.925947    2.000000    3.919444    2.314658    3.916576    2.
↪312659    2.000000    3.945080    4.225861    3.914667         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~SVEC         2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000   88.019997  0.000000e+00  405.959991    0.000000   83.550003    0.
↪000000   88.019997    0.000000   73.419998    0.000000  195.529999    0.
↪000000   68.599998    0.000000  109.769997    0.000000   76.400002    0.
↪000000   83.550003    0.000000  164.910004    0.000000  195.529999    0.
↪000000  405.959991    0.000000  164.910004    0.000000  109.769997    0.
↪000000   76.400002
    ROHR~*~*~*~PVEC         2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    3.309655  5.085822e+00    4.945080    3.283566    3.302971    4.
↪925946    4.919444    3.000133    3.004937    5.183360    5.085822    5.
↪225861    5.221471    3.312659    3.314659    5.221471    5.183360    4.
↪945080    4.925946    4.919444    4.916576    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.916576    4.914667    3.
↪004937    3.043288
    ROHR~*~*~*~VI           2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪102059    0.608561 -1.856568e-01    0.271700   -0.129652    0.733984    0.
↪327641   -0.046900    0.733984    0.491034    0.125423   -0.290447   -0.
↪231787   -0.046900    0.125423   -0.290447         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~T            2004-09-22 08:30:00 2004-09-22 08:31:00   60.
↪000000   90.000000  9.000000e+01   90.000000   60.000000   60.000000   60.
↪000000   60.000000   90.000000   60.000000   60.000000   60.000000   60.
↪000000   90.000000   60.000000   90.000000   60.000000   90.000000   60.
↪000000   60.000000   90.000000   90.000000   90.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KLAP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00   22.
↪987947         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
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    ROHR~*~*~*~RHOVEC       2004-09-22 08:30:00 2004-09-22 08:31:00  983.
↪700012  983.700012  9.657000e+02  965.700012  983.700012  983.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  983.700012  983.700012  965.700012  965.700012  965.
↪700012  965.700012  965.700012  965.700012  983.700012  983.700012  983.
↪700012  983.700012  983.700012  983.700012  965.700012  965.700012  983.
↪700012  983.700012
    ROHR~*~*~*~TVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   60.
↪000000   60.000000  9.000000e+01   90.000000   60.000000   60.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   60.000000   60.000000   90.000000   90.000000   90.
↪000000   90.000000   90.000000   90.000000   60.000000   60.000000   60.
↪000000   60.000000   60.000000   60.000000   90.000000   90.000000   60.
↪000000   60.000000
    FWES~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    1.
↪291413         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~MVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪904446   -0.904446  5.294384e+00    5.294384   -1.645290   -1.645290    2.
↪363743    2.363743   -2.573939   -2.573939    6.385540    6.385540    6.
↪385541    6.385541   -0.415628   -0.415628    6.385540    6.385540    4.
↪271917    4.271917    1.091157    1.091157   -2.573939   -2.573939   -2.
↪054097   -2.054097   -0.415628   -0.415628    1.091157    1.091157   -2.
↪573939   -2.573939
    ROHR~*~*~*~VK           2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪102059    0.608561 -1.856568e-01    0.271700   -0.129652    0.733984    0.
↪327641   -0.046900    0.733984    0.491034    0.125423   -0.290447   -0.
↪231787   -0.046900    0.125423   -0.290447         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    PUMP~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00   22.
↪987947         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~PDAMPF       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪199200    0.701100  7.011000e-01    0.701100    0.199200    0.199200    0.
↪199200    0.199200    0.701100    0.199200    0.199200    0.199200    0.
↪199200    0.701100    0.199200    0.701100    0.199200    0.701100    0.
↪199200    0.199200    0.701100    0.701100    0.701100         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~PVECMAX_INST 2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    3.309655  5.085822e+00    4.945080    3.283566    3.302971    4.
↪925946    4.919444    3.000133    3.004937    5.183360    5.085822    5.
↪225861    5.221471    3.312659    3.314659    5.221471    5.183360    4.
↪945080    4.925946    4.919444    4.916576    3.043288    3.141441    3.
↪141441    3.283566    3.309655    3.312659    4.916576    4.914667    3.
↪004937    3.043288
    VENT~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪374182    0.367335  3.163897e-07         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
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```
    KNOT~*~*~*~HMIN_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    3.985846  4.083384e+00    4.121495    2.043288    2.283565    2.
↪004937    4.311307    4.126019    2.309655    4.291591    2.000133    2.
↪141440    3.825970    2.000000    3.819467    2.314658    3.816599    2.
↪312659    2.000000    3.845104    4.125885    3.814690         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~QMK          2004-09-22 08:30:00 2004-09-22 08:31:00   -3.
↪256005   19.059780 -5.923043e+00    8.509476   -9.266179   22.987946   22.
↪987947   -1.496260   22.987947   15.378901    3.928167   -9.266179   -7.
↪394748   -1.496260    3.928167   -9.266179         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~PMIN_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    4.985846  5.083384e+00    5.121495    3.043288    3.283565    3.
↪004937    5.311307    5.126019    3.309655    5.291591    3.000133    3.
↪141440    4.825970    3.000000    4.819468    3.314658    4.816599    3.
↪312659    3.000000    4.845104    5.125885    4.814691         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KNOT~*~*~*~RHO          2004-09-22 08:30:00 2004-09-22 08:31:00  983.
↪700012  965.700012  9.657000e+02  965.700012  983.700012  983.700012  983.
↪700012  983.700012  965.700012  983.700012  983.700012  983.700012  983.
↪700012  965.700012  983.700012  965.700012  983.700012  965.700012  983.
↪700012  983.700012  965.700012  965.700012  965.700012         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    ROHR~*~*~*~ZVEC         2004-09-22 08:30:00 2004-09-22 08:31:00   20.
↪000000   20.000000  2.000000e+01   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000   20.000000   20.000000   20.000000   20.000000   20.
↪000000   20.000000
    KNOT~*~*~*~PMAX_INST    2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪302971    5.085822  5.183360e+00    5.221471    3.043288    3.283566    3.
↪004937    5.411284    5.225996    3.309655    5.391567    3.000133    3.
↪141440    4.925947    3.000000    4.919444    3.314658    4.916576    3.
↪312659    3.000000    4.945080    5.225861    4.914667         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KLAP~*~*~*~V            2004-09-22 08:30:00 2004-09-22 08:31:00    1.
↪291413         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KNOT~*~*~*~WALTER       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪542668    0.399595  2.160159e-01    0.144285    1.002038    0.534794    1.
↪075106    1.232407    0.000000    0.747607    1.232407    1.232407    0.
↪815037    0.999922    0.000000    1.235105    0.000000    2.193956    0.
↪650142    1.232407    0.877202    0.000000    2.832201         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
    KNOT~*~*~*~PH           2004-09-22 08:30:00 2004-09-22 08:31:00    2.
↪302971    4.085822  4.183360e+00    4.221471    2.043288    2.283566    2.
↪004937    4.411284    4.225996    2.309655    4.391567    2.000133    2.
↪141440    3.925947    2.000000    3.919444    2.314658    3.916576    2.
↪312659    2.000000    3.945080    4.225861    3.914667         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN␣
↪        NaN
```

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```
    FWVB~*~*~*~W            2004-09-22 08:30:00 2004-09-22 08:31:00  160.
↪000000  200.000000  1.600000e+02  160.000000  120.000000         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWVB~*~*~*~QM           2004-09-22 08:30:00 2004-09-22 08:31:00    3.
↪928166    6.869426  4.581308e+00    3.928166    3.680879         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    ROHR~*~*~*~QMAV         2004-09-22 08:30:00 2004-09-22 08:31:00   -3.
↪256005   19.059780 -5.923043e+00    8.509476   -9.266179   22.987946   22.
↪987947   -1.496260   22.987947   15.378901    3.928167   -9.266179   -7.
↪394748   -1.496260    3.928167   -9.266179         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    ROHR~*~*~*~VAV          2004-09-22 08:30:00 2004-09-22 08:31:00   -0.
↪102059    0.608561 -1.856568e-01    0.271700   -0.129652    0.733984    0.
↪327641   -0.046900    0.733984    0.491034    0.125423   -0.290447   -0.
↪231787   -0.046900    0.125423   -0.290447         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    KNOT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN
    ROHR~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000    0.000000    0.000000    0.
↪000000    0.000000    0.000000    0.000000         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN
    VENT~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    KLAP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    PUMP~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN         NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
```

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```
    FWVB~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000    0.000000  0.000000e+00    0.000000    0.000000         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
    FWES~*~*~*~IAKTIV       2004-09-22 08:30:00 2004-09-22 08:31:00    0.
↪000000         NaN           NaN         NaN         NaN         NaN
↪ NaN         NaN         NaN         NaN         NaN         NaN
↪NaN         NaN         NaN         NaN         NaN         NaN         NaN
↪        NaN         NaN         NaN         NaN         NaN         NaN
↪    NaN         NaN         NaN         NaN         NaN         NaN
↪NaN
```

**_checkMxsVecsFile** (*fullCheck=False*)

Returns (firstTime,lastTime,NOfTimes).

**Args:**

- **fullCheck (bool)**
  - False (default): only 1st and last h5Keys are read
  - True: all h5Keys are read

**Raises:** MxError

**_handleMxsVecsFileDeletion** (*mxsFile*, *newMxsVecsFile=False*)

Handles the deletion of mxsVecsFile.

**Args:**

- mxsFile
- newMxsVecsFile (default: False)

**mxsVecsFile is DELETED! if:**

- existing and older than mxsFile
- or newMxsVecsFile is True

**Raises:** MxError

**setResultsToMxsFile** (*mxsFile=None*, *add=False*, *NewH5Vec=False*, *maxRecords=None*)

Sets (default) or adds mxsFile-Content to .df.

**Args:**

- **mxsFile (str)**
  - None (default): .mxsFile is used
- add (bool): default: False: sets df to mxsFile-Content
- NewH5Vec: False (default); if True, an existing mxsVecsFile will be deleted even if it is newer than mxsFile
- maxRecords: Anzahl der max. zu lesenden Records

**Returns:**

- timesWrittenToMxsVecs

**.df**

- index: TIMESTAMP
- self.df.index.is_unique will be True
- because in SIR 3S'
- 1st Time twice (SNAPSHOTTYPE: STAT+TIME) and Last Time triple (SNAPSHOTTYPE: TIME+TMIN/TMAX)
  - **+TIME** is dropped (STAT is used in df)
  - +TMIN/TMAX are dropped (not used in df)
- and because resulting overlapping TIMESTAMPs due to intersections (add=True) are also dropped

.h5FileVecs

- is updated with mxsFile-Content
- **is DELETED! before if existing and**
  - older than mxsFile
  - or newMxsVecsFile

**Raises:** MxError

```
>>> mxDummy=mxs['GPipes']
>>> try:
...     mx=Mx(mx1File=mxDummy.mx1File,NoH5Read=True,NoMxsRead=True) # avoid␣
↪doing anything than just plain Init
... except:
...     from Mx import Mx
...     mx=Mx(mx1File=mxDummy.mx1File,NoH5Read=True,NoMxsRead=True) # avoid␣
↪doing anything than just plain Init
>>> mx.df
>>> mx.setResultsToMxsFile(NewH5Vec=True)
1
```

**setResultsToMxsZipFile**(*mxsZipFile=None*, *add=False*, *NewH5Vec=False*, *maxRecords=None*)

Sets (default) or adds mxsZipFile-Content to .df.

**Args:**

- **mxsZipFile (str)**
  - None (default): .mxsFile is used
- add (bool): default: False: sets df to mxsZipFile-Content
- NewH5Vec
- maxRecords: Anzahl der max. zu lesenden Records

**Returns:**

- timesWrittenToMxsVecsFromZip

**Raises:** MxError

**ToH5**(*h5File=None*)

Stores .mx1Df, .mx2Df, .df, .dfVecAggs to h5File.

**Args:** h5File(str): default: None: self.h5File is used

**.h5File**

- is !DELETED! before if existing

**Keys used**

- /MX1
- /MX2
- /MXS

**Raises:** MxError

**FromH5** (*h5File=None*)

Sets .mx1Df, .mx2Df, .df, .dfVecAggs to h5File-Content.

**Args:**

- **h5File(str)**
  - None (default): .h5File is used

**Keys expected**

- /MX1
- /MX2
- /MXS
- /VecAggs

**Raises:** MxError

**getMxsVecsFileData** (*timesReq=None*, *fastMode=False*)

Returns List of dfs with mxsVecsFileData. One TIMESTAMP (index) per df.

**Args:**

- **timesReq: List of TIMESTAMPs**
  - if None: a List with a single time only, the 1st Time, is constructed as timesReq
- fastMode (default: False): H5-Access with no Checks

**Returns:**

- List of dfs with mxsVecsFileData
- empty List if no TIMESTAMP could be found
- one df per TIMESTAMP
- index: TIMESTAMP

**Raises:** MxError

```
>>> mx=mxs['LocalHeatingNetwork']
>>> mx.delFiles()
>>> mx.setResultsToMxsFile() # reads TIMESTAMPS and constructs .vec.h5 while␣
↪reading; returns TIMESTAMPs read
5
>>> mxVecsFileDataLst=mx.getMxsVecsFileData()
>>> len(mxVecsFileDataLst)
1
```

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```
>>> mxVecsFileData=mxVecsFileDataLst[0]
>>> type(mxVecsFileData)
<class 'pandas.core.frame.DataFrame'>
>>> mxVecsFileData.index[0]
Timestamp('2004-09-22 08:30:00+0000', tz='UTC')
>>> vecsFileDataOneCol=mxVecsFileData['ROHR~*~*~*~SVEC']
>>> vecsFileDataOneColResult=vecsFileDataOneCol[0]
>>> vecsFileDataOneColResult[-1]
76.4000015258789
>>> mxVecsFileDataLst=mx.getMxsVecsFileData(fastMode=True)
>>> mxVecsFileData=mxVecsFileDataLst[0]
>>> vecsFileDataOneCol=mxVecsFileData['ROHR~*~*~*~SVEC']
>>> vecsFileDataOneColResult=vecsFileDataOneCol[0]
>>> vecsFileDataOneColResult[-1]
76.4000015258789
>>> import pandas as pd
>>> timeNotAva=mx.df.index[-1]+pd.Timedelta('666 milliseconds')
>>> timeNotAva
Timestamp('2004-09-22 08:31:00.666000+0000', tz='UTC')
>>> mxVecsFileDataLst=mx.getMxsVecsFileData(timesReq=[timeNotAva])
>>> len(mxVecsFileDataLst)
0
```

**unPackMxsVecsFileDataDf** (*mxVecsFileData*, *mIndex*, *returnMultiIndex=True*)

Unpacks mxVecsFileData-Content into a returned df.

**Args:**

- mxVecsFileData: the stuff returned by getMxsVecsFileData is: List of dfs with mxsVecsFileData. One TIMESTAMP (index) per df. One of these dfs.
- **mIndex: MultiIndex to be used**
  - 1st Level: Timestamps
  - 2nd Level: cols (Sir3sIds to be unpacked)
- **returnMultiIndex (default: True):**
  - if True, the data is unpacked row-wise (stored in cols) and the index is mIndex and the col-Labels are mx2Idx
  - if False, the data is unpacked col-wise (stored in rows) and the index is mx2Idx and th col-Labels are the cols (the Sir3sIds)

**Returns:**

- df

**Raises:** MxError

```
>>> mx=mxs['LocalHeatingNetwork']
>>> mx.delFiles()
>>> mx.setResultsToMxsFile() # reads 5 TIMESTAMPS and constructs .vec.h5␣
↪while reading
5
>>> timesReq=list(mx.df.index[:]) # all times
>>> mxVecsFileDataLst=mx.getMxsVecsFileData(timesReq=timesReq)
>>> len(mxVecsFileDataLst)
5
```

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```
>>> mxVecsFileData=mxVecsFileDataLst[0]
>>> # construct MultiIndex Start ... ---
>>> colsToBeUnpacked=['ROHR~*~*~*~SVEC','ROHR~*~*~*~QMAV','KNOT~*~*~*~PH'] #␣
↪mxVecsFileData.columns.tolist() for all columns
>>> arrays=[[mxVecsFileData.index[0]]*len(colsToBeUnpacked),colsToBeUnpacked]
>>> tuples = list(zip(*arrays))
>>> import pandas as pd
>>> mIndex = pd.MultiIndex.from_tuples(tuples, names=['Timestamp', 'Sir3sID'])
>>> mIndex
MultiIndex(levels=[[2004-09-22 08:30:00+00:00], ['KNOT~*~*~*~PH', 'ROHR~*~*~*~
↪QMAV', 'ROHR~*~*~*~SVEC']],
           labels=[[0, 0, 0], [2, 1, 0]],
           names=['Timestamp', 'Sir3sID'])
>>> colIdx=mIndex.get_level_values('Sir3sID')
>>> colIdx
Index(['ROHR~*~*~*~SVEC', 'ROHR~*~*~*~QMAV', 'KNOT~*~*~*~PH'], dtype='object',
↪ name='Sir3sID')
>>> colIdx.values.tolist()
['ROHR~*~*~*~SVEC', 'ROHR~*~*~*~QMAV', 'KNOT~*~*~*~PH']
>>> # construct MultiIndex End ... ---
>>> mx.unPackMxsVecsFileDataDf(mxVecsFileData,mIndex,returnMultiIndex=False)
    ROHR~*~*~*~SVEC  ROHR~*~*~*~QMAV  KNOT~*~*~*~PH
0          0.000000        -8.509475       2.302971
1         88.019997        19.059780       3.985846
2          0.000000       -15.378901       4.083384
3        405.959991         8.509476       4.121495
4          0.000000       -22.987946       2.043288
5         83.550003        22.987946       2.283565
6          0.000000        22.987947       2.004937
7         88.019997        -3.928166       4.311307
8          0.000000        22.987947       4.126019
9         73.419998        15.378901       2.309655
10         0.000000         3.928167       4.291591
11       195.529999       -22.987946       2.000133
12         0.000000       -19.059778       2.141440
13        68.599998        -3.928166       3.825970
14         0.000000         3.928167       2.000000
15       109.769997       -22.987946       3.819467
16         0.000000              NaN       2.314658
17        76.400002              NaN       3.816599
18         0.000000              NaN       2.312659
19        83.550003              NaN       2.000000
20         0.000000              NaN       3.845104
21       164.910004              NaN       4.125885
22         0.000000              NaN       3.814690
23       195.529999              NaN            NaN
24         0.000000              NaN            NaN
25       405.959991              NaN            NaN
26         0.000000              NaN            NaN
27       164.910004              NaN            NaN
28         0.000000              NaN            NaN
29       109.769997              NaN            NaN
30         0.000000              NaN            NaN
31        76.400002              NaN            NaN
>>> mx.unPackMxsVecsFileDataDf(mxVecsFileData,mIndex)
                                                 0           1           2
↪     3           4           5           6           7           8           9
↪     10          11          12          13          14          15          16
↪      17          18          19          20          21          22          23
↪24          25   26          27   28          29   30          31
```

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```
Timestamp                 Sir3sID                                                                 ␣
↪                                                                                                 ␣
↪                                                                                                 ␣
↪                                                                                                 ␣
↪
2004-09-22 08:30:00+00:00 ROHR~*~*~*~SVEC  0.000000  88.019997   0.000000 ␣
↪405.959991   0.000000  83.550003   0.000000  88.019997   0.000000  73.
↪419998  0.000000  195.529999   0.000000  68.599998  0.000000  109.769997  0.
↪000000  76.400002  0.000000  83.550003  0.000000  164.910004  0.00000  195.
↪529999  0.0  405.959991  0.0  164.910004  0.0  109.769997  0.0  76.400002
                          ROHR~*~*~*~QMAV -8.509475  19.059780 -15.378901   ␣
↪8.509476 -22.987946  22.987946  22.987947  -3.928166  22.987947  15.378901 ␣
↪3.928167  -22.987946 -19.059778  -3.928166  3.928167  -22.987946       NaN ␣
↪      NaN       NaN        NaN       NaN         NaN      NaN         NaN ␣
↪NaN         NaN  NaN         NaN  NaN         NaN  NaN        NaN
                          KNOT~*~*~*~PH    2.302971   3.985846   4.083384   ␣
↪4.121495   2.043288   2.283565   2.004937   4.311307   4.126019   2.309655 ␣
↪4.291591    2.000133   2.141440   3.825970  2.000000    3.819467  2.314658 ␣
↪ 3.816599  2.312659   2.000000  3.845104    4.125885  3.81469         NaN ␣
↪NaN         NaN  NaN         NaN  NaN         NaN  NaN        NaN
>>> dfs=[]
>>> for idx,mxVecsFileData in enumerate(mxVecsFileDataLst):
...     arrays=[[mxVecsFileData.index[0]]*len(colsToBeUnpacked),
↪colsToBeUnpacked]
...     tuples = list(zip(*arrays))
...     mIndex = pd.MultiIndex.from_tuples(tuples, names=['Timestamp',
↪'Sir3sID'])
...     dfs.append(mx.unPackMxsVecsFileDataDf(mxVecsFileData,mIndex))
>>> df=pd.concat(dfs)
>>> idx=pd.IndexSlice
>>> dfOneVecChannel=df.loc[(idx[:],'KNOT~*~*~*~PH'),0:22] # df.loc[(idx[:],
↪idx[:]),idx[:]]: everything
>>> dfOneVecChannel
                                                 0         1         2         ␣
↪3         4         5         6         7         8         9         10   ␣
↪    11        12        13   14        15        16        17        18   ␣
↪19        20        21        22
Timestamp                 Sir3sID                                                                 ␣
↪                                                                                                 ␣
↪                                                                                                 ␣
↪
2004-09-22 08:30:00+00:00 KNOT~*~*~*~PH  2.302971  3.985846  4.083384  4.
↪121495  2.043288  2.283565  2.004937  4.311307  4.126019  2.309655  4.
↪291591  2.000133  2.141440  3.825970  2.0  3.819467  2.314658  3.816599  2.
↪312659  2.0  3.845104  4.125885  3.814690
2004-09-22 08:30:15+00:00 KNOT~*~*~*~PH  2.272133  3.034820  3.123339  3.
↪157926  2.039330  2.255054  2.004493  3.331398  3.162040  2.277968  3.
↪311887  2.000120  2.128487  2.892818  2.0  2.887151  2.282320  2.884661  2.
↪280581  2.0  2.909634  3.161918  2.883003
2004-09-22 08:30:30+00:00 KNOT~*~*~*~PH  2.144123  2.528542  2.576208  2.
↪594833  2.021343  2.135238  2.002467  2.694164  2.597075  2.147196  2.
↪676144  2.000063  2.069655  2.455464  2.0  2.452505  2.149524  2.451183  2.
↪148594  2.0  2.464144  2.597010  2.450303
2004-09-22 08:30:45+00:00 KNOT~*~*~*~PH  2.052100  2.183028  2.200011  2.
↪206647  2.007717  2.048865  2.000910  2.248923  2.207463  2.053240  2.
↪234365  2.000021  2.025138  2.156905  2.0  2.155822  2.054124  2.155325  2.
↪053771  2.0  2.160025  2.207441  2.154995
```

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```
2004-09-22 08:31:00+00:00 KNOT~*~*~*~PH  2.302971  4.085822  4.183360  4.
↪221471  2.043288  2.283566  2.004937  4.411284  4.225996  2.309655  4.
↪391567  2.000133  2.141440  3.925947  2.0  3.919444  2.314658  3.916576  2.
↪312659  2.0  3.945080  4.225861  3.914667
>>> dfOneVecChannel.min()
0     2.052100
1     2.183028
2     2.200011
3     2.206647
4     2.007717
5     2.048865
6     2.000910
7     2.248923
8     2.207463
9     2.053240
10    2.234365
11    2.000021
12    2.025138
13    2.156905
14    2.000000
15    2.155822
16    2.054124
17    2.155325
18    2.053771
19    2.000000
20    2.160025
21    2.207441
22    2.154995
dtype: float64
>>> df.min(level=1)
                        0          1          2           3          4          ␣
↪ 5         6          7          8          9          10         11          ␣
↪12         13        14         15         16          17        18          ␣
↪19        20          21        22          23   24          25   26         ␣
↪  27   28          29   30         31
Sir3sID                                                                         ␣
↪                                                                               ␣
↪                                                                               ␣
↪                                                                               ␣
↪
ROHR~*~*~*~SVEC  0.000000  88.019997   0.000000  405.959991   0.000000  83.
↪550003  0.000000  88.019997  0.000000  73.419998  0.000000  195.529999   0.
↪000000  68.599998  0.00000  109.769997  0.000000  76.400002  0.000000  83.
↪550003  0.000000  164.910004  0.000000  195.529999  0.0  405.959991  0.0 ␣
↪164.910004  0.0  109.769997  0.0  76.400002
ROHR~*~*~*~QMAV -8.509475   7.394749 -15.378901    3.256006 -22.987946   9.
↪266180  9.266181  -3.928166  9.266181   5.923044  1.496261  -22.987946 -19.
↪059778  -3.928166  1.49626  -22.987946       NaN        NaN       NaN       ␣
↪ NaN       NaN         NaN       NaN         NaN  NaN         NaN  NaN       ␣
↪   NaN  NaN         NaN  NaN        NaN
KNOT~*~*~*~PH    2.052100   2.183028   2.200011    2.206647   2.007717   2.
↪048865  2.000910   2.248923  2.207463   2.053240  2.234365    2.000021   2.
↪025138   2.156905  2.00000    2.155822  2.054124   2.155325  2.053771   2.
↪000000  2.160025    2.207441  2.154995         NaN  NaN         NaN  NaN    ␣
↪     NaN  NaN         NaN  NaN        NaN
```

**getVecAggs**(*time1st=None*, *time1stIncluded=True*, *time2nd=None*, *time2ndIncluded=True*)

Gets (or calcs) Aggregates (MIN, MAX, . . . ) of mxsVecsFileData between the 2 Times.

- Newly calced Aggregates are stored in dfVecAggs.

**Args:**

- time1st: TIMESTAMP (first if None)
- time2nd: TIMESTAMP (last if None)

**Returns:**

- **dfs with MultiIndex:**
  - Level 0: 'MIN', 'MAX', . . .
  - Level 1: col (Sir3sID)
  - cols: mx2Idx
- TIMESTAMPL: left ScenTimeStamp included in calculating the Aggregate
- TIMESTAMPR: right ScenTimeStamp included in calculating the Aggregate

**Raises:** MxError

```
>>> mx=mxs['LocalHeatingNetwork']
>>> mx.delFiles()
>>> mx.setResultsToMxsFile() # reads 5 TIMESTAMPS and constructs .vec.h5␣
↪while reading
5
>>> # check dfVecAggs to demonstrate how getVecAggs stores to / reads from␣
↪getVecAggs
>>> Sir3sIDs=mx.dfVecAggs.index.unique(level=1).values
>>> len(Sir3sIDs)
41
>>> mx.dfVecAggs.index.unique(level=0).values
array(['TIME', 'TMIN', 'TMAX'], dtype=object)
>>> len(mx.dfVecAggs.columns.tolist())
32
>>> mx.dfVecAggs.shape # (3*41,32)
(123, 32)
>>> df,tL,tR=mx.getVecAggs()
>>> mx.dfVecAggs.index.unique(level=0).values
array(['TIME', 'TMIN', 'TMAX', 'MIN', 'MAX', 'DIF'], dtype=object)
>>> mx.dfVecAggs.shape
(246, 32)
>>> import pandas as pd
>>> #idx=pd.IndexSlice
>>> df.loc[(['MIN','MAX','DIF'],'KNOT~*~*~*~PH'),0:22].round(6) ## df.
↪loc[(slice(None),'KNOT~*~*~*~PH'),slice(None)] # df.loc[(idx[:],'KNOT~*~*~*~
↪PH'),idx[:]]
                          0         1         2         3         4         5␣
↪        6         7         8         9        10        11        12      ␣
↪  13   14        15        16        17        18   19        20        21 ␣
↪      22
TYPE Sir3sID                                                                 ␣
↪                                                                            ␣
↪                                                                            ␣
↪
MIN  KNOT~*~*~*~PH  2.052100  2.183028  2.200011  2.206647  2.007717  2.
↪048865  2.000910  2.248923  2.207463  2.053240  2.234365  2.000021  2.
↪025138  2.156905  2.0  2.155822  2.054124  2.155325  2.053771  2.0  2.
↪160025  2.207441  2.154995
```

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```
MAX  KNOT~*~*~*~PH  2.302971  4.085822  4.183360  4.221471  2.043288  2.
↪283566  2.004937  4.411284  4.225996  2.309655  4.391567  2.000133  2.
↪141440  3.925947  2.0  3.919444  2.314658  3.916576  2.312659  2.0  3.
↪945080  4.225861  3.914667
DIF  KNOT~*~*~*~PH  0.000000  0.099977  0.099977  0.099977  0.000000  0.
↪000000  0.000000  0.099977  0.099977  0.000000  0.099977  0.000000  0.
↪000000  0.099977  0.0  0.099977  0.000000  0.099977  0.000000  0.0  0.
↪099977  0.099977  0.099977
>>> # demonstrate how to transform an getVecAggs()-df-Result for Xm...
>>> dfT=df.loc[('MIN',df.index.get_level_values(1).tolist()),:].
↪transpose(copy=True) ## dfT=df.loc[('MIN',slice(None)),:].
↪transpose(copy=True)   # dfT=df.loc[('MIN',idx[:]),idx[:]].
↪transpose(copy=True)
>>> colIndex=dfT.columns.droplevel(level=0)
>>> colIndex.name=None
>>> pd.DataFrame(dfT.values,columns=colIndex)[['ROHR~*~*~*~SVEC', 'ROHR~*~*~*~
↪QMAV', 'KNOT~*~*~*~PH']]
    ROHR~*~*~*~SVEC  ROHR~*~*~*~QMAV  KNOT~*~*~*~PH
0          0.000000        -8.509475       2.052100
1         88.019997         7.394749       2.183028
2          0.000000       -15.378901       2.200011
3        405.959991         3.256006       2.206647
4          0.000000       -22.987946       2.007717
5         83.550003         9.266180       2.048865
6          0.000000         9.266181       2.000910
7         88.019997        -3.928166       2.248923
8          0.000000         9.266181       2.207463
9         73.419998         5.923044       2.053240
10         0.000000         1.496261       2.234365
11       195.529999       -22.987946       2.000021
12         0.000000       -19.059778       2.025138
13        68.599998        -3.928166       2.156905
14         0.000000         1.496260       2.000000
15       109.769997       -22.987946       2.155822
16         0.000000              NaN       2.054124
17        76.400002              NaN       2.155325
18         0.000000              NaN       2.053771
19        83.550003              NaN       2.000000
20         0.000000              NaN       2.160025
21       164.910004              NaN       2.207441
22         0.000000              NaN       2.154995
23       195.529999              NaN            NaN
24         0.000000              NaN            NaN
25       405.959991              NaN            NaN
26         0.000000              NaN            NaN
27       164.910004              NaN            NaN
28         0.000000              NaN            NaN
29       109.769997              NaN            NaN
30         0.000000              NaN            NaN
31        76.400002              NaN            NaN
>>> df,tL,tR=mx.getVecAggs()
>>> mx.dfVecAggs.shape
(246, 32)
>>> # demonstrate how to transform an AggEntry for Xm...
>>> df=mx.dfVecAggs.loc[('TMIN',slice(None),mx.df.index[0],mx.df.index[-1]),:]
>>> dfT=df.transpose(copy=True)
>>> colIndex=dfT.columns.droplevel(level=0)
```

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```
>>> colIndex=colIndex.droplevel(level=1)
>>> colIndex=colIndex.droplevel(level=1)
>>> colIndex.name=None
>>> pd.DataFrame(dfT.values,columns=colIndex)[['ROHR~*~*~*~SVEC', 'ROHR~*~*~*~
↪QMAV', 'KNOT~*~*~*~PH']]
    ROHR~*~*~*~SVEC  ROHR~*~*~*~QMAV  KNOT~*~*~*~PH
0          0.000000        -8.509475       2.052100
1         88.019997         7.394749       2.183028
2          0.000000       -15.378901       2.200011
3        405.959991         3.256006       2.206647
4          0.000000       -22.987946       2.007717
5         83.550003         9.266180       2.048865
6          0.000000         9.266181       2.000910
7         88.019997        -3.928166       2.248923
8          0.000000         9.266181       2.207463
9         73.419998         5.923044       2.053240
10         0.000000         1.496261       2.234365
11       195.529999       -22.987946       2.000021
12         0.000000       -19.059778       2.025138
13        68.599998        -3.928166       2.156905
14         0.000000         1.496260       2.000000
15       109.769997       -22.987946       2.155822
16         0.000000              NaN       2.054124
17        76.400002              NaN       2.155325
18         0.000000              NaN       2.053771
19        83.550003              NaN       2.000000
20         0.000000              NaN       2.160025
21       164.910004              NaN       2.207441
22         0.000000              NaN       2.154995
23       195.529999              NaN            NaN
24         0.000000              NaN            NaN
25       405.959991              NaN            NaN
26         0.000000              NaN            NaN
27       164.910004              NaN            NaN
28         0.000000              NaN            NaN
29       109.769997              NaN            NaN
30         0.000000              NaN            NaN
31        76.400002              NaN            NaN
```

**dumpInMxsFormat**(*mxsDumpFile=None*)

Dumps in MXS-Format to mxsDumpFile (for testing purposes).

**Returns:**

- **(TimeStampsDumped, TimeStampsFoundInH5)**
  - normally: TimeStampsDumped=TimeStampsFoundInH5
  - if TimesStamps in self.df are manipulated . . .
  - . . . the H5-Content remains unchanged
  - in effect the H5-Content can be different from self.df-Content . . .
  - während in self.df die Zeiten (Index) immer geordnet und voneinander verschieden sind
  - sind beim H5-Content nur die Keys voneinander verschieden
  - um pruefen zu koennen, ob alle Zeiten in self.df im H5-Content auch gefunden wurden, wird TimeStampsFoundInH5 mit ausgegeben

- gedumped werden immer alle Zeiten aus self.df
- * fuer jede im H5-Content nicht gefundene Zeit wird das Ergebnis der zuletzt zuvor gefundenen Zeit ausgegeben

**Raises:** MxError

**_getMx1DfAsOneString**(*regex='KNOT~\\S*~\\S*~\\d+~[P|Q]{1}[H|M]{1}$'*)

Returns .mx1Df-Content as one String (for Doctest-Purposes).

**Raises:** MxError

**_getDfAsOneString**(*regex='KNOT~\\S*~\\S*~\\d+~[P|Q]{1}[H|M]{1}$'*)

Returns .df-Content as one String (for Doctest-Purposes).

**Raises:** MxError

**_getDfVecAsOneString**(*df=None*, *regex='KNOT~\\S*~\\S*~\\S*~[P|Q]{1}[H|M]{1}$'*)

Returns dfVec-Content as one String (for Doctest-Purposes).

**Raises:** MxError

Mx.**getMicrosecondsFromRefTime**(*refTime*, *time*)

Returns time in microseconds since refTime.

**Args:**

- refTime
- time

**Raises:** MxError

```
>>> import pandas as pd
>>> timeReadFromMx=b'2019-01-01 00:00:12.500000      '
>>> timeRefMx=b'2019-01-01 00:00:00.000000      '
>>> timeStampTimeReadFromMx=pd.to_datetime(timeReadFromMx.decode(),utc=True)
>>> timeStampTimeRefFromMx=pd.to_datetime(timeRefMx.decode(),utc=True)
>>> timeDelta=timeStampTimeReadFromMx-timeStampTimeRefFromMx
>>> timeDelta.total_seconds()
12.5
>>> import Mx
>>> Mx.getMicrosecondsFromRefTime(timeStampTimeRefFromMx,timeStampTimeReadFromMx)
12500
```

**exception** Mx.**MxError**(*value*)

MxError.

CHAPTER

TWO

# XM

```
>>> # ---
>>> # Imports
>>> # ---
>>> import os
>>> import logging
>>> logger = logging.getLogger('PT3S.Xm')
>>> # ---
>>> # path
>>> # ---
>>> if __name__ == "__main__":
...   try:
...      dummy=__file__
...      logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ','path =␣
↪os.path.dirname(__file__)'," ."))
...      path = os.path.dirname(__file__)
...   except NameError:
...      logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ","path =
↪'.' because __file__ not defined: ","from Xm import Xm follows ..."))
...      path = '.'
...      from Xm import Xm
... else:
...    logger.debug("{0:s}{1:s}{2:s}{3:s}".format('DOCTEST: Not __main__ Context: ','_
↪_name__: ',__name__,"path = '."))
...    path = '.'
>>> try:
...    from PT3S import Mx
... except ImportError:
...    logger.debug("{0:s}{1:s}".format("DOCTEST: ImportError: from PT3S import Mx: ",
↪"- trying import Mx instead ... maybe pip install -e . is active ..."))
...    import Mx
>>> # ---
>>> # testDir
>>> # ---
>>> # globs={'testDir':'testdata'}
>>> try:
...    dummy= testDir
... except NameError:
...    testDir='testdata'
>>> import pandas as pd
>>> # ---
>>> # Clean Up
>>> # ---
>>> h5File=os.path.join(os.path.join(path,testDir),'OneLPipe.h5')
>>> if os.path.exists(h5File):
```

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```
...     os.remove(h5File)
>>> # ---
>>> # Init
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'OneLPipe.XML')
>>> xm=Xm(xmlFile=xmlFile)
>>> # ---
>>> # a View
>>> # ---
>>> v='vKNOT'
>>> v in xm.dataFrames
True
>>> isinstance(xm.dataFrames[v],pd.core.frame.DataFrame)
True
>>> # ---
>>> # ToH5
>>> # ---
>>> xm.ToH5()
>>> os.path.exists(xm.h5File)
True
>>> # ---
>>> # force Read H5 instead of Xml
>>> # ---
>>> os.rename(xm.xmlFile,xm.xmlFile+'.blind')
>>> xm=Xm(xmlFile=xmlFile)
>>> os.rename(xm.xmlFile+'.blind',xm.xmlFile)
>>> # ---
>>> vKNOT=xm.dataFrames['vKNOT']
>>> vKNOT[(vKNOT.KTYP.isin(['QKON','PKON'])) & (vKNOT.BESCHREIBUNG.fillna('').str.
↪startswith('Template Element')==False)].shape
(2, 41)
>>> vROHR=xm.dataFrames['vROHR']
>>> vROHR.shape
(1, 74)
>>> isinstance(vROHR['pXCors'],pd.core.series.Series)
True
>>> vROHR['pXCors'][0]
[0.0, 500.0]
>>> vROHR.pYCors[0]
[0.0, 0.0]
>>> # ---
>>> # getWDirModelDirModelName()
>>> # ---
>>> (wDir,modelDir,modelName,mx1File)=xm.getWDirModelDirModelName()
>>> modelName
'M-1-0-1'
>>> # ---
>>> # H5-Deletion if NoH5Read=True
>>> # ---
>>> if os.path.exists(xm.h5File):
...     os.remove(xm.h5File)
>>> xm=Xm(xmlFile=xmlFile)
>>> xm.ToH5()
>>> os.path.exists(xm.h5File)
True
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> os.path.exists(xm.h5File)
```

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```
False
>>> # ---
>>> # print-Options
>>> # ---
>>> pd.set_option('display.max_columns',None)
>>> pd.set_option('display.max_rows',None)
>>> pd.set_option('display.max_colwidth',666666)
>>> pd.set_option('display.width',666666666)
>>> # ---
>>> # vKNOT
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT',dropColList=['LFKT_NAME','LF','LF_min
↪','LF_max','PVAR_NAME','PH','PH_min','PH_max','PZON_NAME','FSTF_NAME','STOF_NAME',
↪'GMIX_NAME','UTMP_NAME','2L_NAME','2L_KVR','fkHYDR','fkFQPS']))
  NAME BESCHREIBUNG              IDREFERENZ      CONT CONT_ID  CONT_LFDNR  CONT_VKNO ␣
↪KTYP LFAKT    QM_EIN QVAR_NAME  QM  QM_min  QM_max KVR  TE  TM XKOR YKOR ZKOR      ␣
↪            pk                    tk  pXCor  pYCor
0    I          NaN  3S5642914844465475844  OneLPipe    1001         NaN        NaN ␣
↪QKON     1  176.7146       NaN NaN     NaN     NaN   0 NaN  10  300  600   10 ␣
↪5642914844465475844  5642914844465475844    0.0    0.0
1    K          NaN  3S5289899964753656852  OneLPipe    1001         NaN        NaN ␣
↪PKON     1         0       NaN NaN     NaN     NaN   0 NaN  10  800  600   10 ␣
↪5289899964753656852  5289899964753656852  500.0    0.0
>>> # ---
>>> # vROHR
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',dropColList=['NAME_i_2L','NAME_k_2L
↪']))
  BESCHREIBUNG              IDREFERENZ BAUJAHR HAL IPLANUNG KENNUNG      L LZU   RAU␣
↪ZAUS ZEIN ZUML JLAMBS LAMBDA0 ASOLL INDSCHALL KVR AUSFALLZEIT DA   DI   DN KT PN␣
↪REHABILITATION REPARATUR  S WSTEIG WTIEFE LTGR_NAME  LTGR_BESCHREIBUNG SICHTBARKEIT␣
↪VERLEGEART DTRO_NAME                        DTRO_BESCHREIBUNG        E fkSTRASSE␣
↪fkSRAT                   pk                   tk IRTRENN LECKSTART LECKEND␣
↪LECKMENGE LECKORT LECKSTATUS QSVB ZVLIMPTNZ KANTENZV      CONT CONT_ID  CONT_LFDNR␣
↪NAME_i KVR_i TM_i XKOR_i YKOR_i ZKOR_i NAME_k KVR_k TM_k XKOR_k YKOR_k ZKOR_k ␣
↪pXCor_i  pYCor_i  pXCor_k  pYCor_k        pXCors      pYCors    pWAYPXCors ␣
↪pWAYPYCors                                WAYP
0          NaN  3S4737064599036143765    2017   0        1       0  10000   0  0.25  ␣
↪ 0    0    0      1   0.025  1000         0   0            0  0  250  250  0  0    ␣
↪        0         0  0      0      0   STDROHR                NaN            1    ␣
↪999999   STDROHR  Standard-Druckrohre mit di = DN (DIN 2402)  2.1E+11        -1    ␣
↪-1  4737064599036143765  4737064599036143765       0         0       0         0   ␣
↪   0          0    0         0        0  OneLPipe    1001         NaN      I     0 ␣
↪ 10    300    600     10      K     0   10    800    600     10      0.0      0.0  ␣
↪ 500.0      0.0  [0.0, 500.0]  [0.0, 0.0]  [0.0, 500.0]  [0.0, 0.0]  [(300.0, 600.
↪0), (800.0, 600.0)]
>>> # ---
>>> # Clean Up
>>> # ---
>>> xm.delFiles()
>>> # ---
>>> # LocalHeatingNetwork
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'LocalHeatingNetwork.XML')
>>> xm=Xm(xmlFile=xmlFile)
>>> # ---
>>> # vKNOT
```

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```
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT',dropColList=['LFKT_NAME','LF','LF_min
↪','LF_max','PVAR_NAME','PH','PH_min','PH_max','PZON_NAME','FSTF_NAME','STOF_NAME',
↪'GMIX_NAME','UTMP_NAME','2L_NAME','2L_KVR','fkHYDR','fkFQPS']))
           NAME                        BESCHREIBUNG IDREFERENZ                                 ␣
↪           CONT CONT_ID CONT_LFDNR CONT_VKNO  KTYP LFAKT QM_EIN QVAR_NAME  QM  QM_
↪min  QM_max KVR  TE  TM     XKOR     YKOR ZKOR                   pk                          ␣
↪  tk   pXCor  pYCor
0        R-K004                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541539  5706361   20  4638663808856251977 ␣
↪4638663808856251977   799.0  152.0
1        V-K002                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541059  5706265   20  4731792362611615619 ␣
↪4731792362611615619   319.0   56.0
2        V-K001                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2540867  5706228   20  4756962427318766791 ␣
↪4756962427318766791   127.0   19.0
3        V-K000                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2540793  5706209   20  4766681917240867943 ␣
↪4766681917240867943    53.0    0.0
4        R-K001                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2540867  5706228   20  4807712987325933680 ␣
↪4807712987325933680   127.0   19.0
5        R-K003                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541457  5706345   20  4891048046264179170 ␣
↪4891048046264179170   717.0  136.0
6        R-K000                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2540793  5706209   20  4979785838440534851 ␣
↪4979785838440534851    53.0    0.0
7        R-K005                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541627  5706363   20  5183147862966701025 ␣
↪5183147862966701025   887.0  154.0
8           R-L                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1      BHKW  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2540740  5706225   20  5356267303828212700 ␣
↪5356267303828212700     0.0   16.0
9        R-K002                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541059  5706265   20  5364712333175450942 ␣
↪5364712333175450942   319.0   56.0
10       V-K004                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541539  5706361   20  5370423799772591808 ␣
↪5370423799772591808   799.0  152.0
11       V-K005                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541627  5706363   20  5444644492819213978 ␣
↪5444644492819213978   887.0  154.0
12       R-K007                             None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541899  5706325   20  5508992300317633799 ␣
↪5508992300317633799  1159.0  116.0
```

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13       V-K006                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541790  5706338   20  5515313800585145571 ␣
↪5515313800585145571  1050.0  129.0
14       R-K006                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60  2541790  5706338   20  5543326527366090679 ␣
↪5543326527366090679  1050.0  129.0
15       V-K003                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541457  5706345   20  5646671866542823796 ␣
↪5646671866542823796   717.0  136.0
16          V-L                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1      BHKW  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2540740  5706240   20  5736262931552588702 ␣
↪5736262931552588702     0.0   31.0
17       V-K007                            None         -1  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90  2541899  5706325   20  5741235692335544560 ␣
↪5741235692335544560  1159.0  116.0
18           R2                            None         -1
↪          BHKW    1002         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60      170       20   20  5002109894154139899 ␣
↪5002109894154139899   170.0   20.0
19          V-1                            None         -1
↪          BHKW    1002         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   1  10  90      140      160   20  5049461676240771430 ␣
↪5049461676240771430   140.0  160.0
20           R3                            None         -1
↪          BHKW    1002         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60      140       20   20  5219230031772497417 ␣
↪5219230031772497417   140.0   20.0
21  PKON-Knoten  Druckhaltung - 2 bar Ruhedruck         -1
↪          BHKW    1002         -1       NaN  PKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  60  60      200       40   20  5397990465339071638 ␣
↪5397990465339071638   200.0   40.0
22          R-1          Anbindung Druckhaltung         -1
↪          BHKW    1002         -1       NaN  QKON     1      0       NaN NaN    ␣
↪NaN     NaN   2  10  60      195       20   20  5557222628687032084 ␣
↪5557222628687032084   195.0   20.0
>>> # ---
>>> # vROHR
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',dropColList=['NAME_i_2L','NAME_k_2L
↪']))
   BESCHREIBUNG IDREFERENZ BAUJAHR HAL IPLANUNG KENNUNG       L LZU  RAU ZAUS ZEIN␣
↪ZUML JLAMBS LAMBDA0 ASOLL INDSCHALL KVR  AUSFALLZEIT     DA     DI   DN     KT  PN ␣
↪REHABILITATION  REPARATUR    S WSTEIG WTIEFE LTGR_NAME            LTGR_BESCHREIBUNG␣
↪SICHTBARKEIT VERLEGEART DTRO_NAME                        DTRO_BESCHREIBUNG        E␣
↪fkSTRASSE fkSRAT                    pk                    tk IRTRENN LECKSTART␣
↪LECKEND LECKMENGE LECKORT LECKSTATUS QSVB ZVLIMPTNZ KANTENZV                      ␣
↪              CONT CONT_ID CONT_LFDNR  NAME_i KVR_i TM_i   XKOR_i   YKOR_i ZKOR_i ␣
↪NAME_k KVR_k TM_k   XKOR_k   YKOR_k ZKOR_k  pXCor_i  pYCor_i  pXCor_k  pYCor_k    ␣
↪      pXCors          pYCors                                pWAYPXCors           ␣
↪                 pWAYPYCors                                                       ␣
↪                                     WAYP
0          None         -1    None   0        1       0   88.02   0  0.1    0    0 ␣
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.925  NaN ␣
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe ␣
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11 ␣
↪    -1     -1  4613782368750024999  4613782368750024999       0         0       0 ␣
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K004     2   60  2541539  5706361     20  R-
↪K005     2   60  2541627  5706363     20    799.0    152.0    887.0    154.0    ␣
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1          None         -1    None   0        1       0  405.96   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4614949065966596185  4614949065966596185       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K002     1   90  2541059  5706265     20  V-
↪K003     1   90  2541457  5706345     20    319.0     56.0    717.0    136.0
↪[319.0, 717.0]   [56.0, 136.0]                [319.0, 716.9500000001863]
↪   [56.049999999813735, 136.0]
↪[(2541059.0, 5706265.05), (2541456.95, 5706345.0)]
2          None         -1    None   0        1       0   83.55   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4637102239750163477  4637102239750163477       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K003     2   60  2541457  5706345     20  R-
↪K004     2   60  2541539  5706361     20    717.0    136.0    799.0    152.0
↪[717.0, 799.0]  [136.0, 152.0]    [725.8500000000931, 807.8999999999069]  [124.
↪04999999981374, 140.09999999962747]
↪[(2541465.85, 5706333.05), (2541547.9, 5706349.1)]
3          None         -1    None   0        1       0   88.02   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4713733238627697042  4713733238627697042       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K004     1   90  2541539  5706361     20  V-
↪K005     1   90  2541627  5706363     20    799.0    152.0    887.0    154.0
↪[799.0, 887.0]  [152.0, 154.0]                [799.0, 887.0499999998137]
↪                [152.0, 154.0]
↪[(2541539.0, 5706361.0), (2541627.05, 5706363.0)]
4          None         -1    None   0        1       0  195.53   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4789218195240364437  4789218195240364437       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K001     1   90  2540867  5706228     20  V-
↪K002     1   90  2541059  5706265     20    127.0     19.0    319.0     56.0
↪[127.0, 319.0]    [19.0, 56.0]                            [127.0, 319.0]
↪    [19.0, 56.049999999813735]
↪[(2540867.0, 5706228.0), (2541059.0, 5706265.05)]
5          None         -1    None   0        1       0  109.77   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4945727430885351042  4945727430885351042       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K006     2   60  2541790  5706338     20  R-
↪K007     2   60  2541899  5706325     20   1050.0    129.0   1159.0    116.0  [1050.
↪0, 1159.0]  [129.0, 116.0]  [1058.8500000000931, 1167.8999999999069]
↪[117.0, 104.09999999962747]
↪[(2541798.85, 5706326.0), (2541907.9, 5706313.1)]
6          None         -1    None   0        1       0    76.4   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4984202422877610920  4984202422877610920       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K000     1   90  2540793  5706209     20  V-
↪K001     1   90  2540867  5706228     20     53.0      0.0    127.0     19.0
↪[53.0, 127.0]     [0.0, 19.0]                [53.049999999813735, 127.0]
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7          None         -1    None   0        1       0   83.55   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5037777106796980248  5037777106796980248       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K003     1   90  2541457  5706345     20  V-
↪K004     1   90  2541539  5706361     20    717.0    136.0    799.0    152.0
↪[717.0, 799.0]  [136.0, 152.0]                [716.9500000001863, 799.0]
↪                [136.0, 152.0]
↪[(2541456.95, 5706345.0), (2541539.0, 5706361.0)]
8          None         -1    None   0        1       0  164.91   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5123819811204259837  5123819811204259837       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K005     1   90  2541627  5706363     20  V-
↪K006     1   90  2541790  5706338     20    887.0    154.0   1050.0    129.0   [887.
↪0, 1050.0]  [154.0, 129.0]   [887.0499999998137, 1049.9500000001863]
↪[154.0, 128.95000000018626]
↪[(2541627.05, 5706363.0), (2541789.95, 5706337.95)]
9          None         -1    None   0        1       0  195.53   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5266224553324203132  5266224553324203132       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K001     2   60  2540867  5706228     20  R-
↪K002     2   60  2541059  5706265     20    127.0     19.0    319.0     56.0
↪[127.0, 319.0]    [19.0, 56.0]  [135.89999999990687, 327.89999999990687]   [7.
↪0499999998137355, 44.09999999962747]
↪  [(2540875.9, 5706216.05), (2541067.9, 5706253.1)]
10         None         -1    None   0        1       0  405.96   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5379365049009065623  5379365049009065623       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K002     2   60  2541059  5706265     20  R-
↪K003     2   60  2541457  5706345     20    319.0     56.0    717.0    136.0
↪[319.0, 717.0]   [56.0, 136.0]   [327.89999999990687, 725.8500000000931]   [44.
↪09999999962747, 124.04999999981374]
↪[(2541067.9, 5706253.1), (2541465.85, 5706333.05)]
11         None         -1    None   0        1       0  164.91   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5611703699850694889  5611703699850694889       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K005     2   60  2541627  5706363     20  R-
↪K006     2   60  2541790  5706338     20    887.0    154.0   1050.0    129.0   [887.
↪0, 1050.0]  [154.0, 129.0]   [895.9500000001863, 1058.8500000000931]
↪[142.04999999981374, 117.0]
↪[(2541635.95, 5706351.05), (2541798.85, 5706326.0)]
12         None         -1    None   0        1       0  109.77   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5620197984230756681  5620197984230756681       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K006     1   90  2541790  5706338     20  V-
↪K007     1   90  2541899  5706325     20   1050.0    129.0   1159.0    116.0  [1050.
↪0, 1159.0]  [129.0, 116.0]              [1049.9500000001863, 1159.0]  [128.
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```
13         None         -1    None   0        1       0    76.4   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5647213228462830353  5647213228462830353       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K000     2   60  2540793  5706209     20  R-
↪K001     2   60  2540867  5706228     20     53.0      0.0    127.0     19.0
↪[53.0, 127.0]     [0.0, 19.0]  [61.950000000186265, 135.89999999990687]
↪  [-12.0, 7.0499999998137355]
↪[(2540801.95, 5706197.0), (2540875.9, 5706216.05)]
14         None         -1    None   0        1       0   73.42   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  168.3  160.3  150   0.45 NaN
↪        NaN        NaN    4    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4769996343148550485  4769996343148550485       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1     R-L     2   60  2540740  5706225     20  R-
↪K000     2   60  2540793  5706209     20      0.0     16.0     53.0      0.0
↪[0.0, 53.0]     [16.0, 0.0]     [0.0, 24.0, 45.0, 61.950000000186265]
↪ [16.0, 16.0, -12.0, -12.0]  [(2540740.0, 5706225.0), (2540764.0, 5706225.0),␣
↪(2540785.0, 5706197.0), (2540801.95, 5706197.0)]
15         None         -1    None   0        1       0    68.6   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  168.3  160.3  150   0.45 NaN
↪        NaN        NaN    4    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4939422678063487923  4939422678063487923       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1     V-L     1   90  2540740  5706240     20  V-
↪K000     1   90  2540793  5706209     20      0.0     31.0     53.0      0.0
↪[0.0, 53.0]     [31.0, 0.0]          [0.0, 30.0, 53.049999999813735]       [31.0,
↪31.0, -0.049999999813735485]                        [(2540740.0, 5706240.0),␣
↪(2540770.0, 5706240.0), (2540793.05, 5706208.95)]
>>> # ---
>>> # vWBLZ
>>> # ---
>>> print("'''{:s}'''".format(repr(xm.dataFrames['vWBLZ']).replace('\n','\n    ')))
'''   AKTIV BESCHREIBUNG IDIM       NAME OBJTYPE                OBJID
↪  pk
   0      1  Wärmebilanz    0      BLNZ1    KNOT  4731792362611615619
↪5579937562601803472
   1      1  Wärmebilanz    0      BLNZ1    KNOT  5364712333175450942
↪5579937562601803472
   2      1  Wärmebilanz    0    BLNZ1u5    KNOT  5183147862966701025
↪5187647097142898375
   3      1  Wärmebilanz    0    BLNZ1u5    KNOT  5444644492819213978
↪5187647097142898375
   4      1  Wärmebilanz    0    BLNZ1u5    KNOT  4731792362611615619
↪5187647097142898375
   5      1  Wärmebilanz    0    BLNZ1u5    KNOT  5364712333175450942
↪5187647097142898375
   6      1  Wärmebilanz    0  BLNZ1u5u7    KNOT  5183147862966701025
↪4694700216019268978
   7      1  Wärmebilanz    0  BLNZ1u5u7    KNOT  5444644492819213978
↪4694700216019268978
   8      1  Wärmebilanz    0  BLNZ1u5u7    KNOT  4731792362611615619
↪4694700216019268978
```

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   9      1  Wärmebilanz     0  BLNZ1u5u7    KNOT  5364712333175450942 ␣
↪4694700216019268978
   10     1  Wärmebilanz     0  BLNZ1u5u7    KNOT  5508992300317633799 ␣
↪4694700216019268978
   11     1  Wärmebilanz     0  BLNZ1u5u7    KNOT  5741235692335544560 ␣
↪4694700216019268978
   12     1  Wärmebilanz     0      BLNZ5    KNOT  5183147862966701025 ␣
↪5581152085151655438
   13     1  Wärmebilanz     0      BLNZ5    KNOT  5444644492819213978 ␣
↪5581152085151655438'''
>>> # ---
>>> # vAGSN
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vAGSN',end=7,dropColList=['nrObjIdTypeInAgsn
↪']))
  LFDNR                                     NAME AKTIV OBJTYPE                OBJID ␣
↪                 pk                    tk  nrObjIdInAgsn  Layer nextNODE  compNr
0     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4939422678063487923 ␣
↪5252525269080005909  5252525269080005909              1      1   V-K000       1
1     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4984202422877610920 ␣
↪5252525269080005909  5252525269080005909              2      1   V-K001       1
2     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4789218195240364437 ␣
↪5252525269080005909  5252525269080005909              3      1   V-K002       1
3     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4614949065966596185 ␣
↪5252525269080005909  5252525269080005909              4      1   V-K003       1
4     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  5037777106796980248 ␣
↪5252525269080005909  5252525269080005909              5      1   V-K004       1
5     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  4713733238627697042 ␣
↪5252525269080005909  5252525269080005909              6      1   V-K005       1
6     1  Netzdruckdiagramm VL/RL: BHKW - Netzende   101    ROHR  5123819811204259837 ␣
↪5252525269080005909  5252525269080005909              7      1   V-K006       1
>>> # ---
>>> # vFWVB
>>> # ---
>>> print("'''{:s}'''".format(repr(xm.dataFrames['vFWVB']).replace('\n','\n    ')))
'''  BESCHREIBUNG IDREFERENZ   W0  LFK  W0LFK  TVL0  TRS0  LFKT      W  W_min  W_max ␣
↪INDTR  TRSK  VTYP  DPHAUS  IMBG  IRFV                    pk                    tk ␣
↪NAME_i KVR_i TM_i   XKOR_i   YKOR_i ZKOR_i  pXCor_i  pYCor_i  NAME_k KVR_k TM_k  ␣
↪XKOR_k   YKOR_k ZKOR_k  pXCor_k  pYCor_k                                      CONT␣
↪CONT_ID CONT_LFDNR                         WBLZ
   0           1         -1  200  0.8  160.0    90    50  LFKT  160.0  160.0  160.0 ␣
↪    1    55    14     0.7     0   0.0  4643800032883366034  4643800032883366034  V-
↪K002     1   90  2541059  5706265     20    319.0     56.0  R-K002     2   60 ␣
↪2541059  5706265     20    319.0     56.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1  [BLNZ1, BLNZ1u5, BLNZ1u5u7]
   1           3         -1  200  1.0  200.0    90    65  LFKT  200.0  200.0  200.0 ␣
↪    1    65    14     0.7     0   0.0  4704603947372595298  4704603947372595298  V-
↪K004     1   90  2541539  5706361     20    799.0    152.0  R-K004     2   60 ␣
↪2541539  5706361     20    799.0    152.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1                            []
   2           4         -1  200  0.8  160.0    90    60  LFKT  160.0  160.0  160.0 ␣
↪    1    60    14     0.7     0   0.0  5121101823283893406  5121101823283893406  V-
↪K005     1   90  2541627  5706363     20    887.0    154.0  R-K005     2   60 ␣
↪2541627  5706363     20    887.0    154.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1  [BLNZ1u5, BLNZ1u5u7, BLNZ5]
   3           5         -1  200  0.8  160.0    90    55  LFKT  160.0  160.0  160.0 ␣
↪    1    55    14     0.7     0   0.0  5400405917816384862  5400405917816384862  V-
↪K007     1   90  2541899  5706325     20   1159.0    116.0  R-K007     2   60 ␣
↪2541899  5706325     20   1159.0    116.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1                   [BLNZ1u5u7]
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   4            2          -1  200  0.6  120.0    90    60  LFKT  120.0  120.0  120.0 ␣
↪    1    62    14     0.7     0   0.0  5695730293103267172  5695730293103267172  V-
↪K003     1   90  2541457  5706345     20    717.0    136.0  R-K003     2   60 ␣
↪2541457  5706345     20    717.0    136.0  Nahwärmenetz mit 1000 kW Anschlussleistu␣
↪   1001         -1                          []'''
>>> # ---
>>> # vLAYR
>>> # ---
>>> # print("'''{:s}'''".format(repr(xm.dataFrames['vLAYR'].sort_values(['LFDNR','NAME
↪','OBJTYPE','OBJID'],ascending=True)).replace('\n','\n    ')))
>>> print(xm._getvXXXXAsOneString(vXXXX='vLAYR'))
   LFDNR           NAME OBJTYPE                OBJID                    pk            ␣
↪      tk  nrObjInGroup  nrObjtypeInGroup
0      1        Vorlauf    FWES  5638756766880678918  5206516471428693478 ␣
↪5206516471428693478             1                  1
1      1        Vorlauf    KNOT  4731792362611615619  5206516471428693478 ␣
↪5206516471428693478             1                  1
2      1        Vorlauf    KNOT  4756962427318766791  5206516471428693478 ␣
↪5206516471428693478             1                  2
3      1        Vorlauf    KNOT  4766681917240867943  5206516471428693478 ␣
↪5206516471428693478             1                  3
4      1        Vorlauf    KNOT  5049461676240771430  5206516471428693478 ␣
↪5206516471428693478             1                  4
5      1        Vorlauf    KNOT  5370423799772591808  5206516471428693478 ␣
↪5206516471428693478             1                  5
6      1        Vorlauf    KNOT  5444644492819213978  5206516471428693478 ␣
↪5206516471428693478             1                  6
7      1        Vorlauf    KNOT  5515313800585145571  5206516471428693478 ␣
↪5206516471428693478             1                  7
8      1        Vorlauf    KNOT  5646671866542823796  5206516471428693478 ␣
↪5206516471428693478             1                  8
9      1        Vorlauf    KNOT  5736262931552588702  5206516471428693478 ␣
↪5206516471428693478             1                  9
10     1        Vorlauf    KNOT  5741235692335544560  5206516471428693478 ␣
↪5206516471428693478             1                 10
11     1        Vorlauf    ROHR  4614949065966596185  5206516471428693478 ␣
↪5206516471428693478             1                  1
12     1        Vorlauf    ROHR  4713733238627697042  5206516471428693478 ␣
↪5206516471428693478             1                  2
13     1        Vorlauf    ROHR  4789218195240364437  5206516471428693478 ␣
↪5206516471428693478             1                  3
14     1        Vorlauf    ROHR  4939422678063487923  5206516471428693478 ␣
↪5206516471428693478             1                  4
15     1        Vorlauf    ROHR  4984202422877610920  5206516471428693478 ␣
↪5206516471428693478             1                  5
16     1        Vorlauf    ROHR  5037777106796980248  5206516471428693478 ␣
↪5206516471428693478             1                  6
17     1        Vorlauf    ROHR  5123819811204259837  5206516471428693478 ␣
↪5206516471428693478             1                  7
18     1        Vorlauf    ROHR  5620197984230756681  5206516471428693478 ␣
↪5206516471428693478             1                  8
19     1        Vorlauf    VENT  4678923650983295610  5206516471428693478 ␣
↪5206516471428693478             1                  1
20     2       Rücklauf    KLAP  4801110583764519435  4693347477612662930 ␣
↪4693347477612662930             1                  1
21     2       Rücklauf    KNOT  4638663808856251977  4693347477612662930 ␣
↪4693347477612662930             1                  1
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22     2       Rücklauf    KNOT  4807712987325933680  4693347477612662930 ␣
↪4693347477612662930             1                  2
23     2       Rücklauf    KNOT  4891048046264179170  4693347477612662930 ␣
↪4693347477612662930             1                  3
24     2       Rücklauf    KNOT  4979785838440534851  4693347477612662930 ␣
↪4693347477612662930             1                  4
25     2       Rücklauf    KNOT  5002109894154139899  4693347477612662930 ␣
↪4693347477612662930             1                  5
26     2       Rücklauf    KNOT  5183147862966701025  4693347477612662930 ␣
↪4693347477612662930             1                  6
27     2       Rücklauf    KNOT  5219230031772497417  4693347477612662930 ␣
↪4693347477612662930             1                  7
28     2       Rücklauf    KNOT  5356267303828212700  4693347477612662930 ␣
↪4693347477612662930             1                  8
29     2       Rücklauf    KNOT  5364712333175450942  4693347477612662930 ␣
↪4693347477612662930             1                  9
30     2       Rücklauf    KNOT  5397990465339071638  4693347477612662930 ␣
↪4693347477612662930             1                 10
31     2       Rücklauf    KNOT  5508992300317633799  4693347477612662930 ␣
↪4693347477612662930             1                 11
32     2       Rücklauf    KNOT  5543326527366090679  4693347477612662930 ␣
↪4693347477612662930             1                 12
33     2       Rücklauf    KNOT  5557222628687032084  4693347477612662930 ␣
↪4693347477612662930             1                 13
34     2       Rücklauf    PUMP  5481331875203087055  4693347477612662930 ␣
↪4693347477612662930             1                  1
35     2       Rücklauf    ROHR  4613782368750024999  4693347477612662930 ␣
↪4693347477612662930             1                  1
36     2       Rücklauf    ROHR  4637102239750163477  4693347477612662930 ␣
↪4693347477612662930             1                  2
37     2       Rücklauf    ROHR  4769996343148550485  4693347477612662930 ␣
↪4693347477612662930             1                  3
38     2       Rücklauf    ROHR  4945727430885351042  4693347477612662930 ␣
↪4693347477612662930             1                  4
39     2       Rücklauf    ROHR  5266224553324203132  4693347477612662930 ␣
↪4693347477612662930             1                  5
40     2       Rücklauf    ROHR  5379365049009065623  4693347477612662930 ␣
↪4693347477612662930             1                  6
41     2       Rücklauf    ROHR  5611703699850694889  4693347477612662930 ␣
↪4693347477612662930             1                  7
42     2       Rücklauf    ROHR  5647213228462830353  4693347477612662930 ␣
↪4693347477612662930             1                  8
43     2       Rücklauf    VENT  4897018421024717974  4693347477612662930 ␣
↪4693347477612662930             1                  1
44     2       Rücklauf    VENT  5525310316015533093  4693347477612662930 ␣
↪4693347477612662930             1                  2
45     3  Kundenanlagen    FWVB  4643800032883366034  5003333277973347346 ␣
↪5003333277973347346             1                  1
46     3  Kundenanlagen    FWVB  4704603947372595298  5003333277973347346 ␣
↪5003333277973347346             1                  2
47     3  Kundenanlagen    FWVB  5121101823283893406  5003333277973347346 ␣
↪5003333277973347346             1                  3
48     3  Kundenanlagen    FWVB  5400405917816384862  5003333277973347346 ␣
↪5003333277973347346             1                  4
49     3  Kundenanlagen    FWVB  5695730293103267172  5003333277973347346 ␣
↪5003333277973347346             1                  5
50     4           BHKW    BSYM  5043395081363401573  5555393404073362943 ␣
↪5555393404073362943             1                  1
```

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```
51      4            BHKW     TEXT  5056836766824229789  5555393404073362943 ␣
↪5555393404073362943              1                   1
52      4            BHKW     TEXT  5329748935118523443  5555393404073362943 ␣
↪5555393404073362943              1                   2
53      5           Texte     ARRW  4664845735864571219  5394410243594912680 ␣
↪5394410243594912680              1                   1
54      5           Texte     ARRW  4902474974831811106  5394410243594912680 ␣
↪5394410243594912680              1                   2
55      5           Texte     ARRW  5026846801782366678  5394410243594912680 ␣
↪5394410243594912680              1                   3
56      5           Texte     ARRW  5688313372729413840  5394410243594912680 ␣
↪5394410243594912680              1                   4
57      5           Texte     NRCV  4681213816714574464  5394410243594912680 ␣
↪5394410243594912680              1                   1
58      5           Texte     NRCV  4857294696992797631  5394410243594912680 ␣
↪5394410243594912680              1                   2
59      5           Texte     NRCV  4914949875368816179  5394410243594912680 ␣
↪5394410243594912680              1                   3
60      5           Texte     NRCV  4946584950744559030  5394410243594912680 ␣
↪5394410243594912680              1                   4
61      5           Texte     NRCV  4968703141722117357  5394410243594912680 ␣
↪5394410243594912680              1                   5
62      5           Texte     NRCV  5091374651838464239  5394410243594912680 ␣
↪5394410243594912680              1                   6
63      5           Texte     NRCV  5097127385155151127  5394410243594912680 ␣
↪5394410243594912680              1                   7
64      5           Texte     NRCV  5179988968597313889  5394410243594912680 ␣
↪5394410243594912680              1                   8
65      5           Texte     NRCV  5281885868749421521  5394410243594912680 ␣
↪5394410243594912680              1                   9
66      5           Texte     NRCV  5410904806390050339  5394410243594912680 ␣
↪5394410243594912680              1                  10
67      5           Texte     NRCV  5476262878682325254  5394410243594912680 ␣
↪5394410243594912680              1                  11
68      5           Texte     NRCV  5557806245003742769  5394410243594912680 ␣
↪5394410243594912680              1                  12
69      5           Texte     RECT  4994817837124479818  5394410243594912680 ␣
↪5394410243594912680              1                   1
70      5           Texte     RPFL  5158870568935841216  5394410243594912680 ␣
↪5394410243594912680              1                   1
71      5           Texte     TEXT  4628671704393700430  5394410243594912680 ␣
↪5394410243594912680              1                   1
72      5           Texte     TEXT  4654104397990769217  5394410243594912680 ␣
↪5394410243594912680              1                   2
73      5           Texte     TEXT  4666644549022031339  5394410243594912680 ␣
↪5394410243594912680              1                   3
74      5           Texte     TEXT  4693143208412077585  5394410243594912680 ␣
↪5394410243594912680              1                   4
75      5           Texte     TEXT  4768731522550494423  5394410243594912680 ␣
↪5394410243594912680              1                   5
76      5           Texte     TEXT  4770844990228490264  5394410243594912680 ␣
↪5394410243594912680              1                   6
77      5           Texte     TEXT  4782197969172967134  5394410243594912680 ␣
↪5394410243594912680              1                   7
78      5           Texte     TEXT  4855692488683645764  5394410243594912680 ␣
↪5394410243594912680              1                   8
79      5           Texte     TEXT  4965628942555351751  5394410243594912680 ␣
↪5394410243594912680              1                   9
```

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80     5          Texte    TEXT  4995961504641886710  5394410243594912680 ␣
↪5394410243594912680             1                 10
81     5          Texte    TEXT  5017907661719368413  5394410243594912680 ␣
↪5394410243594912680             1                 11
82     5          Texte    TEXT  5028052147238787802  5394410243594912680 ␣
↪5394410243594912680             1                 12
83     5          Texte    TEXT  5036153631350515544  5394410243594912680 ␣
↪5394410243594912680             1                 13
84     5          Texte    TEXT  5054433315422452796  5394410243594912680 ␣
↪5394410243594912680             1                 14
85     5          Texte    TEXT  5108336975548011049  5394410243594912680 ␣
↪5394410243594912680             1                 15
86     5          Texte    TEXT  5262441422409836340  5394410243594912680 ␣
↪5394410243594912680             1                 16
87     5          Texte    TEXT  5297832234834839298  5394410243594912680 ␣
↪5394410243594912680             1                 17
88     5          Texte    TEXT  5370727463979416592  5394410243594912680 ␣
↪5394410243594912680             1                 18
89     5          Texte    TEXT  5421223289472778073  5394410243594912680 ␣
↪5394410243594912680             1                 19
90     5          Texte    TEXT  5501963349880613918  5394410243594912680 ␣
↪5394410243594912680             1                 20
91     5          Texte    TEXT  5502619581048467908  5394410243594912680 ␣
↪5394410243594912680             1                 21
92     5          Texte    TEXT  5540395812045688781  5394410243594912680 ␣
↪5394410243594912680             1                 22
93     5          Texte    TEXT  5550982489075668484  5394410243594912680 ␣
↪5394410243594912680             1                 23
94     5          Texte    TEXT  5610916400841895317  5394410243594912680 ␣
↪5394410243594912680             1                 24
95     5          Texte    TEXT  5646820849868629537  5394410243594912680 ␣
↪5394410243594912680             1                 25
96     5          Texte    TEXT  5696590398594231893  5394410243594912680 ␣
↪5394410243594912680             1                 26
97     5          Texte    TEXT  5697088036451277538  5394410243594912680 ␣
↪5394410243594912680             1                 27
>>> # ---
>>> # vGTXT
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vGTXT',sortList=['CONT_ID','pk'],index=False,
↪header=False))
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪numerische Anzeige:  4614148870174765680  4614148870174765680                    ␣
↪    (219.0, -278.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪Georeferenzpunkt 2  4628671704393700430  4628671704393700430              (1115.
↪9500000001863, -323.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪    Block  4666644549022031339  4666644549022031339                        (-58.
↪0, -77.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪numerische Anzeige  4693143208412077585  4693143208412077585                   ␣
↪    (1211.0, -9.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                           Knoten␣
↪und Rohre  4995961504641886710  4995961504641886710                          (570.
↪0, -49.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                              ␣
↪Vorlaufstrang  5017907661719368413  5017907661719368413  (358.20699999999960, 120.2
↪39499999955297)
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Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪LocalHeatingNetwork  5028052147238787802  5028052147238787802                ␣
↪    (1163.0, 536.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1   Tel. 05131 - 4980-0 ; Fax. 05131␣
↪- 4980-15  5054433315422452796  5054433315422452796                       (-230.0,
↪ -1143.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                   Kontrolle: DH-
↪Massenstrom  5100960407865990868  5100960407865990868                        (-
↪60.0, -160.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪Wärmebilanz: 3 Kunden  5150752151066924202  5150752151066924202              ␣
↪      (219.0, -318.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1  eMail. info@3SConsult.de ; www.
↪3SConsult.de  5370727463979416592  5370727463979416592                       (-
↪230.0, -1204.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1               Differenzdruck VL-/␣
↪RL-Knoten  5502619581048467908  5502619581048467908                         (1211.
↪0, -49.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪Kundenanlage  5540395812045688781  5540395812045688781  (1131.9500000001863, 283.
↪95000000018626)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪Fernwärmeverbraucher  5550982489075668484  5550982489075668484              ␣
↪     (1050.0, 239.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                          ␣
↪Rücklaufstrang  5610916400841895317  5610916400841895317                    ␣
↪ (570.0, -9.0)
Nahwärmenetz mit 1000 kW Anschlussleistu  1001  -1                         Knoten␣
↪und Rohre  5646820849868629537  5646820849868629537  (358.20699999993667, 174.
↪39499999955297)
                                  BHKW  1002  -1                            ␣
↪Fernwärmeeinspeiser  4654104397990769217  4654104397990769217              ␣
↪      (115.0, 80.0)
                                  BHKW  1002  -1                            ␣
↪    Pumpe  4768731522550494423  4768731522550494423                        ␣
↪(175.0, 25.0)
                                  BHKW  1002  -1                            ␣
↪Wärmebilanz Netz:  4770844990228490264  4770844990228490264                 ␣
↪    (90.0, 160.0)
                                  BHKW  1002  -1                            ␣
↪Speicherung  4782197969172967134  4782197969172967134                      ␣
↪(110.0, 140.0)
                                  BHKW  1002  -1                            ␣
↪Richtungspfeil  4855692488683645764  4855692488683645764                    ␣
↪(220.0, 105.0)
                                  BHKW  1002  -1                            ␣
↪ Verluste  4965628942555351751  4965628942555351751                         (110.
↪0, 145.0)
                                  BHKW  1002  -1                      (Element␣
↪verbinden)  5036153631350515544  5036153631350515544                        ␣
↪(150.0, 90.0)
                                  BHKW  1002  -1                BHKW Modul 1000␣
↪kW therm.  5056836766824229789  5056836766824229789                          ␣
↪(35.0, 55.0)
                                  BHKW  1002  -1                            ␣
↪   Ventil  5108336975548011049  5108336975548011049                         ␣
↪(205.0, 25.0)
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                                    BHKW  1002  -1                                                  ␣
↪Verbrauch  5262441422409836340  5262441422409836340                                (110.
↪0, 150.0)
                                    BHKW  1002  -1                                                    ␣
↪Einspeisung  5297832234834839298  5297832234834839298                                ␣
↪(110.0, 155.0)
                                    BHKW  1002  -1                                          ␣
↪Druckhaltung 2 bar  5329748935118523443  5329748935118523443                                  ␣
↪     (180.0, 65.0)
                                    BHKW  1002  -1                                          ␣
↪Numerische Anzeige  5421223289472778073  5421223289472778073                                  ␣
↪    (190.0, 115.0)
                                    BHKW  1002  -1                                             ␣
↪Verbindungslinie  5501963349880613918  5501963349880613918                                  ␣
↪   (150.0, 95.0)
                                    BHKW  1002  -1                                                ␣
↪   (Text)  5696590398594231893  5696590398594231893                                ␣
↪(35.0, 50.0)
                                    BHKW  1002  -1                                                ␣
↪   Klappe  5697088036451277538  5697088036451277538                                ␣
↪(145.0, 25.0)
>>> # ---
>>> # vNRCV
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vNRCV',end=14,dropColList=['DPGR','CONT_LFDNR
↪','pk_ROWS'],sortList=['OBJTYPE','fkOBJTYPE','ATTRTYPE','cRefLfdNr']))
   cRefLfdNr                                    CONT CONT_ID OBJTYPE                ␣
↪fkOBJTYPE ATTRTYPE              tk_ROWS                   pk                   tk   ␣
↪                              pXYLB
0          1                                    BHKW    1002    FWES ␣
↪5638756766880678918        W  5762106696740202356  4857294696992797631 ␣
↪4857294696992797631                          (90.0, 65.0)
1          1                                    BHKW    1002    KNOT ␣
↪5049461676240771430        T  4723443975311885965  5097127385155151127 ␣
↪5097127385155151127                          (90.0, 95.0)
2          1                                    BHKW    1002    KNOT ␣
↪5219230031772497417        T  5602301870151014230  5557806245003742769 ␣
↪5557806245003742769                          (90.0, 35.0)
3          1                                    BHKW    1002    KNOT ␣
↪5356267303828212700       PH  5000989080893535213  4968703141722117357 ␣
↪4968703141722117357                         (220.0, 25.0)
4          1  Nahwärmenetz mit 1000 kW Anschlussleistu    1001    KNOT ␣
↪5397990465339071638       QM  5134531789044068877  5410059595276504750 ␣
↪5410059595276504750                         (91.0, -94.0)
5          2                                    BHKW    1002    KNOT ␣
↪5397990465339071638       QM  5134531789044068877  5357021981944933535 ␣
↪5357021981944933535  (184.999999464624, 57.99999953107601)
6          1                                    BHKW    1002    KNOT ␣
↪5736262931552588702       PH  4754881272083464445  4681213816714574464 ␣
↪4681213816714574464                         (220.0, 85.0)
7          1  Nahwärmenetz mit 1000 kW Anschlussleistu    1001    KNOT ␣
↪5741235692335544560       DP  4949183695502554728  4914949875368816179 ␣
↪4914949875368816179                        (1234.0, 83.0)
8          1                                    BHKW    1002    PUMP ␣
↪5481331875203087055        N  5563842594211689762  5091374651838464239 ␣
↪5091374651838464239                         (170.0, 45.0)
9          1                                    BHKW    1002    VENT ␣
↪4678923650983295610       QM  5126307362398248950  5410904806390050339 ␣
↪5410904806390050339                        (200.0, 110.0)
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10          1  Nahwärmenetz mit 1000 kW Anschlussleistu    1001    WBLZ ␣
↪4694700216019268978      WVB  4778244458749966216  4991097791264453745 ␣
↪4991097791264453745                        (354.0, -225.0)
11          1                                   BHKW    1002    WBLZ ␣
↪5262603207038486299      WES  5690691957596882133  5179988968597313889 ␣
↪5179988968597313889                          (90.0, 155.0)
12          1                                   BHKW    1002    WBLZ ␣
↪5262603207038486299    WSPEI  5153847813311339683  4946584950744559030 ␣
↪4946584950744559030                          (90.0, 140.0)
13          1                                   BHKW    1002    WBLZ ␣
↪5262603207038486299      WVB  5214984699859365639  5281885868749421521 ␣
↪5281885868749421521                          (90.0, 150.0)
>>> # ---
>>> # MxSync() - without Mx-Object
>>> # ---
>>> vROHR=xm.dataFrames['vROHR']
>>> vROHR.shape
(16, 74)
>>> 'vNRCV_Mx1' in xm.dataFrames
False
>>> mx=xm.MxSync()
>>> 'vNRCV_Mx1' in xm.dataFrames
True
>>> vROHR.shape
(16, 76)
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',dropColList=['NAME_i_2L','NAME_k_2L
↪']))
   BESCHREIBUNG IDREFERENZ BAUJAHR HAL IPLANUNG KENNUNG       L LZU  RAU ZAUS ZEIN␣
↪ZUML JLAMBS LAMBDA0 ASOLL INDSCHALL KVR  AUSFALLZEIT     DA     DI   DN     KT  PN ␣
↪REHABILITATION  REPARATUR    S WSTEIG WTIEFE LTGR_NAME           LTGR_BESCHREIBUNG␣
↪SICHTBARKEIT VERLEGEART DTRO_NAME                      DTRO_BESCHREIBUNG        E␣
↪fkSTRASSE fkSRAT                   pk                    tk IRTRENN LECKSTART␣
↪LECKEND LECKMENGE LECKORT LECKSTATUS QSVB ZVLIMPTNZ KANTENZV                    ␣
↪              CONT CONT_ID CONT_LFDNR  NAME_i KVR_i TM_i   XKOR_i   YKOR_i ZKOR_i ␣
↪NAME_k KVR_k TM_k   XKOR_k   YKOR_k ZKOR_k  pXCor_i  pYCor_i  pXCor_k  pYCor_k    ␣
↪      pXCors          pYCors                                pWAYPXCors           ␣
↪                 pWAYPYCors                                                       ␣
↪                                      WAYP  mx2NofPts  mx2Idx
0          None         -1    None   0        1       0   88.02   0  0.1    0    0 ␣
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN  ␣
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe  ␣
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11  ␣
↪    -1     -1  4613782368750024999  4613782368750024999       0         0       0 ␣
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K004     2   60  2541539  5706361     20  R-
↪K005     2   60  2541627  5706363     20    799.0    152.0    887.0    154.0   ␣
↪[799.0, 887.0]  [152.0, 154.0]    [807.8999999999069, 895.9500000001863]  [140.
↪09999999962747, 142.04999999981374]                                             ␣
↪[(2541547.9, 5706349.1), (2541635.95, 5706351.05)]          2       0
1          None         -1    None   0        1       0  405.96   0  0.1    0    0 ␣
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN  ␣
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe  ␣
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11  ␣
↪    -1     -1  4614949065966596185  4614949065966596185       0         0       0 ␣
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K002     1   90  2541059  5706265     20  V-
↪K003     1   90  2541457  5706345     20    319.0     56.0    717.0    136.0   ␣
↪[319.0, 717.0]   [56.0, 136.0]                [319.0, 716.9500000001863]       ␣
↪   [56.049999999813735, 136.0]                                                  ␣
↪[(2541059.0, 5706265.05), (2541456.95, 5706345.0)]          2       1
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2          None         -1    None   0        1       0   83.55   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4637102239750163477  4637102239750163477       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K003     2   60  2541457  5706345     20  R-
↪K004     2   60  2541539  5706361     20    717.0    136.0    799.0    152.0
↪[717.0, 799.0]  [136.0, 152.0]    [725.8500000000931, 807.8999999999069]  [124.
↪04999999981374, 140.09999999962747]
↪[(2541465.85, 5706333.05), (2541547.9, 5706349.1)]          2       2
3          None         -1    None   0        1       0   88.02   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4713733238627697042  4713733238627697042       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K004     1   90  2541539  5706361     20  V-
↪K005     1   90  2541627  5706363     20    799.0    152.0    887.0    154.0
↪[799.0, 887.0]  [152.0, 154.0]                [799.0, 887.0499999998137]
↪                [152.0, 154.0]
↪[(2541539.0, 5706361.0), (2541627.05, 5706363.0)]          2       3
4          None         -1    None   0        1       0  195.53   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4789218195240364437  4789218195240364437       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K001     1   90  2540867  5706228     20  V-
↪K002     1   90  2541059  5706265     20    127.0     19.0    319.0     56.0
↪[127.0, 319.0]    [19.0, 56.0]                            [127.0, 319.0]
↪    [19.0, 56.049999999813735]
↪[(2540867.0, 5706228.0), (2541059.0, 5706265.05)]          2       5
5          None         -1    None   0        1       0  109.77   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4945727430885351042  4945727430885351042       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  R-K006     2   60  2541790  5706338     20  R-
↪K007     2   60  2541899  5706325     20   1050.0    129.0   1159.0    116.0  [1050.
↪0, 1159.0]  [129.0, 116.0]  [1058.8500000000931, 1167.8999999999069]
↪[117.0, 104.09999999962747]
↪[(2541798.85, 5706326.0), (2541907.9, 5706313.1)]          2       7
6          None         -1    None   0        1       0    76.4   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  4984202422877610920  4984202422877610920       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K000     1   90  2540793  5706209     20  V-
↪K001     1   90  2540867  5706228     20     53.0      0.0    127.0     19.0
↪[53.0, 127.0]     [0.0, 19.0]                [53.049999999813735, 127.0]
↪[-0.049999999813735485, 19.0]
↪[(2540793.05, 5706208.95), (2540867.0, 5706228.0)]          2       8
7          None         -1    None   0        1       0   83.55   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5037777106796980248  5037777106796980248       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu    1001         -1  V-K003     1   90  2541457  5706345     20  V-
↪K004     1   90  2541539  5706361     20    717.0    136.0    799.0    152.0
↪[717.0, 799.0]  [136.0, 152.0]                [716.9500000001863, 799.0]
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8          None         -1    None   0        1       0  164.91   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5123819811204259837  5123819811204259837       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K005     1   90  2541627  5706363     20  V-
↪K006     1   90  2541790  5706338     20    887.0    154.0   1050.0    129.0   [887.
↪0, 1050.0]  [154.0, 129.0]   [887.0499999998137, 1049.9500000001863]
↪[154.0, 128.95000000018626]
↪[(2541627.05, 5706363.0), (2541789.95, 5706337.95)]          2      10
9          None         -1    None   0        1       0  195.53   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5266224553324203132  5266224553324203132       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K001     2   60  2540867  5706228     20  R-
↪K002     2   60  2541059  5706265     20    127.0     19.0    319.0     56.0
↪[127.0, 319.0]    [19.0, 56.0]  [135.89999999990687, 327.89999999990687]   [7.
↪0499999998137355, 44.09999999962747]
↪  [(2540875.9, 5706216.05), (2541067.9, 5706253.1)]          2      11
10         None         -1    None   0        1       0  405.96   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5379365049009065623  5379365049009065623       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K002     2   60  2541059  5706265     20  R-
↪K003     2   60  2541457  5706345     20    319.0     56.0    717.0    136.0
↪[319.0, 717.0]   [56.0, 136.0]   [327.89999999990687, 725.8500000000931]   [44.
↪09999999962747, 124.04999999981374]
↪[(2541067.9, 5706253.1), (2541465.85, 5706333.05)]          2      12
11         None         -1    None   0        1       0  164.91   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5611703699850694889  5611703699850694889       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K005     2   60  2541627  5706363     20  R-
↪K006     2   60  2541790  5706338     20    887.0    154.0   1050.0    129.0   [887.
↪0, 1050.0]  [154.0, 129.0]   [895.9500000001863, 1058.8500000000931]
↪[142.04999999981374, 117.0]
↪[(2541635.95, 5706351.05), (2541798.85, 5706326.0)]          2      13
12         None         -1    None   0        1       0  109.77   0  0.1    0    0
↪0      1   0.025  1000         0   1          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5620197984230756681  5620197984230756681       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  V-K006     1   90  2541790  5706338     20  V-
↪K007     1   90  2541899  5706325     20   1050.0    129.0   1159.0    116.0  [1050.
↪0, 1159.0]  [129.0, 116.0]             [1049.9500000001863, 1159.0]  [128.
↪95000000018626, 116.04999999981374]
↪[(2541789.95, 5706337.95), (2541899.0, 5706325.05)]          2      14
13         None         -1    None   0        1       0    76.4   0  0.1    0    0
↪0      1   0.025  1000         0   2          NaN  114.3  107.1  100  0.325 NaN
↪        NaN        NaN  3.6    NaN    NaN   KUMANRO  Beschreibung Leitungsgruppe
↪        1     999999   KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪    -1     -1  5647213228462830353  5647213228462830353       0         0       0
↪      0       0          0    0       NaN      NaN  Nahwärmenetz mit 1000 kW
↪Anschlussleistu    1001         -1  R-K000     2   60  2540793  5706209     20  R-
↪K001     2   60  2540867  5706228     20     53.0      0.0    127.0     19.0
↪[53.0, 127.0]     [0.0, 19.0]  [61.950000000186265, 135.89999999990687]
```

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14          None          -1     None    0         1        0    73.42    0  0.1     0     0
↪0       1    0.025  1000          0    2           NaN  168.3  160.3  150    0.45 NaN
↪         NaN         NaN     4     NaN     NaN    KUMANRO  Beschreibung Leitungsgruppe
↪         1      999999    KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪     -1      -1  4769996343148550485  4769996343148550485        0          0        0
↪       0        0           0     0        NaN       NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu     1001          -1      R-L      2    60  2540740  5706225      20  R-
↪K000      2    60  2540793  5706209      20       0.0      16.0      53.0       0.0
↪[0.0, 53.0]      [16.0, 0.0]      [0.0, 24.0, 45.0, 61.950000000186265]
↪ [16.0, 16.0, -12.0, -12.0]  [(2540740.0, 5706225.0), (2540764.0, 5706225.0),␣
↪(2540785.0, 5706197.0), (2540801.95, 5706197.0)]           2        4
15          None          -1     None    0         1        0    68.6    0  0.1     0     0
↪0       1    0.025  1000          0    1           NaN  168.3  160.3  150    0.45 NaN
↪         NaN         NaN     4     NaN     NaN    KUMANRO  Beschreibung Leitungsgruppe
↪         1      999999    KUMANRO  Kunststoffmantelrohr DN20-800 PANISOVIT  2.1E+11
↪     -1      -1  4939422678063487923  4939422678063487923        0          0        0
↪       0        0           0     0        NaN       NaN  Nahwärmenetz mit 1000 kW␣
↪Anschlussleistu     1001          -1      V-L      1    90  2540740  5706240      20  V-
↪K000      1    90  2540793  5706209      20       0.0      31.0      53.0       0.0
↪[0.0, 53.0]      [31.0, 0.0]            [0.0, 30.0, 53.049999999813735]        [31.0,␣
↪31.0, -0.049999999813735485]                          [(2540740.0, 5706240.0),␣
↪(2540770.0, 5706240.0), (2540793.05, 5706208.95)]           2        6
>>> # ---------
>>> # MxSync()
>>> # ---------
>>> xm=Xm(xmlFile=xmlFile)
>>> vROHR=xm.dataFrames['vROHR']
>>> vROHR.shape
(16, 74)
>>> 'vNRCV_Mx1' in xm.dataFrames
False
>>> (wDir,modelDir,modelName,mx1File)=xm.getWDirModelDirModelName()
>>> mx=Mx.Mx(mx1File=mx1File)
>>> xm.MxSync(mx=mx)
>>> vROHR.shape
(16, 76)
>>> 'vNRCV_Mx1' in xm.dataFrames
True
>>> # ---
>>> # vNRCV_Mx1
>>> # ---
>>> import re
>>> f=lambda s: re.match('(\S+)~(\S*)~(\S*)~(\d+)~(\S+)',s).group(1)+'~~~'+re.match(
↪'(\S+)~(\S*)~(\S*)~(\d+)~(\S+)',s).group(4)+'~'+re.match('(\S+)~(\S*)~(\S*)~(\d+)~
↪(\S+)',s).group(5)
>>> print(xm._getvXXXXAsOneString(vXXXX='vNRCV_Mx1',end=14,dropColList=['DPGR','CONT_
↪LFDNR','pk_ROWS'],mapFunc={'Sir3sID':f},sortList=['Sir3sID'],fmtFunc={'Sir3sID':f},
↪index=False,header=False))
FWES~~~5638756766880678918~W  1                                        BHKW  1002  FWES␣
↪ 5638756766880678918      W  5762106696740202356  4857294696992797631 ␣
↪4857294696992797631     (90.0, 65.0)
    KNOT~~~5049461676240771430~T  1                                        BHKW  1002 ␣
↪KNOT  5049461676240771430      T  4723443975311885965  5097127385155151127 ␣
↪5097127385155151127     (90.0, 95.0)
    KNOT~~~5219230031772497417~T  1                                        BHKW  1002 ␣
↪KNOT  5219230031772497417      T  5602301870151014230  5557806245003742769 ␣
↪5557806245003742769     (90.0, 35.0)
```

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   KNOT~~~5356267303828212700~PH  1                                      BHKW  1002 ␣
↪KNOT  5356267303828212700     PH  5000989080893535213  4968703141722117357 ␣
↪4968703141722117357    (220.0, 25.0)
   KNOT~~~5397990465339071638~QM  1  Nahwärmenetz mit 1000 kW Anschlussleistu  1001 ␣
↪KNOT  5397990465339071638     QM  5134531789044068877  5410059595276504750 ␣
↪5410059595276504750    (91.0, -94.0)
   KNOT~~~5736262931552588702~PH  1                                      BHKW  1002 ␣
↪KNOT  5736262931552588702     PH  4754881272083464445  4681213816714574464 ␣
↪4681213816714574464    (220.0, 85.0)
   KNOT~~~5741235692335544560~DP  1  Nahwärmenetz mit 1000 kW Anschlussleistu  1001 ␣
↪KNOT  5741235692335544560     DP  4949183695502554728  4914949875368816179 ␣
↪4914949875368816179   (1234.0, 83.0)
    PUMP~~~5481331875203087055~N  1                                      BHKW  1002 ␣
↪PUMP  5481331875203087055      N  5563842594211689762  5091374651838464239 ␣
↪5091374651838464239    (170.0, 45.0)
   VENT~~~4678923650983295610~QM  1                                      BHKW  1002 ␣
↪VENT  4678923650983295610     QM  5126307362398248950  5410904806390050339 ␣
↪5410904806390050339   (200.0, 110.0)
  WBLZ~~~4694700216019268978~WVB  1  Nahwärmenetz mit 1000 kW Anschlussleistu  1001 ␣
↪WBLZ  4694700216019268978    WVB  4778244458749966216  4991097791264453745 ␣
↪4991097791264453745  (354.0, -225.0)
  WBLZ~~~5262603207038486299~WES  1                                      BHKW  1002 ␣
↪WBLZ  5262603207038486299    WES  5690691957596882133  5179988968597313889 ␣
↪5179988968597313889    (90.0, 155.0)
WBLZ~~~5262603207038486299~WSPEI  1                                      BHKW  1002 ␣
↪WBLZ  5262603207038486299  WSPEI  5153847813311339683  4946584950744559030 ␣
↪4946584950744559030    (90.0, 140.0)
  WBLZ~~~5262603207038486299~WVB  1                                      BHKW  1002 ␣
↪WBLZ  5262603207038486299    WVB  5214984699859365639  5281885868749421521 ␣
↪5281885868749421521    (90.0, 150.0)
WBLZ~~~5262603207038486299~WVERL  1                                      BHKW  1002 ␣
↪WBLZ  5262603207038486299  WVERL  4722863010266870887  5476262878682325254 ␣
↪5476262878682325254    (90.0, 145.0)
>>> # ---
>>> # vKNOT
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT',end=1,dropColList=['LFKT_NAME','LF',
↪'LF_min','LF_max','PVAR_NAME','PH','PH_min','PH_max','PZON_NAME','FSTF_NAME','STOF_
↪NAME','GMIX_NAME','UTMP_NAME','2L_NAME','2L_KVR','fkHYDR','fkFQPS']))
     NAME BESCHREIBUNG IDREFERENZ                                    CONT CONT_ID␣
↪CONT_LFDNR CONT_VKNO  KTYP LFAKT QM_EIN QVAR_NAME  QM  QM_min  QM_max KVR  TE  TM  ␣
↪  XKOR     YKOR ZKOR                  pk                  tk  pXCor  pYCor  mx2Idx
0  R-K004         None         -1  Nahwärmenetz mit 1000 kW Anschlussleistu    1001  ␣
↪      -1       NaN  QKON     1      0       NaN NaN     NaN     NaN   2  10  60 ␣
↪2541539  5706361   20  4638663808856251977  4638663808856251977  799.0  152.0      ␣
↪0
>>> print(xm._getvXXXXAsOneString(vXXXX='vFWVB',dropColList=['BESCHREIBUNG',
↪'IDREFERENZ','W0','W','IRFV','LFK','TVL0','TRS0','LFKT','W_min','W_max','INDTR',
↪'TRSK','VTYP','DPHAUS','IMBG', 'pk','tk','KVR_i','TM_i', 'XKOR_i','YKOR_i','ZKOR_i',
↪'pXCor_i','pYCor_i','KVR_k','TM_k', 'XKOR_k','YKOR_k','ZKOR_k','pXCor_k', 'pYCor_k',
↪'CONT','CONT_ID', 'CONT_LFDNR']))
   W0LFK  NAME_i  NAME_k                        WBLZ  mx2Idx
0  160.0  V-K002  R-K002  [BLNZ1, BLNZ1u5, BLNZ1u5u7]       0
1  200.0  V-K004  R-K004                          []       1
2  160.0  V-K005  R-K005  [BLNZ1u5, BLNZ1u5u7, BLNZ5]       2
3  160.0  V-K007  R-K007                  [BLNZ1u5u7]       3
4  120.0  V-K003  R-K003                          []       4
```

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```
>>> # ---
>>> # vXXXX
>>> # ---
>>> xm.dataFrames['vVBEL_forTestOnly']=xm.dataFrames['vVBEL'].reset_
↪index(inplace=False) # Multiindex to Cols
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL_forTestOnly',index=True,dropColList=[
↪'Z_i','pk_i','CONT_i','CONT_VKNO_i','Z_k','pk_k','CONT_k','CONT_VKNO_k','IDREFERENZ
↪','tk']))
   OBJTYPE                OBJID                   BESCHREIBUNG       NAME_i  NAME_k  ␣
↪          LAYR       L      D  mx2Idx
0     FWES  5638756766880678918  BHKW - Modul - 1000 kW therm.           R3     V-1  ␣
↪     [Vorlauf]       0     80       0
1     FWVB  4643800032883366034                               1       V-K002  R-K002 ␣
↪[Kundenanlagen]       0    NaN       0
2     FWVB  4704603947372595298                               3       V-K004  R-K004 ␣
↪[Kundenanlagen]       0    NaN       1
3     FWVB  5121101823283893406                               4       V-K005  R-K005 ␣
↪[Kundenanlagen]       0    NaN       2
4     FWVB  5400405917816384862                               5       V-K007  R-K007 ␣
↪[Kundenanlagen]       0    NaN       3
5     FWVB  5695730293103267172                               2       V-K003  R-K003 ␣
↪[Kundenanlagen]       0    NaN       4
6     KLAP  4801110583764519435                            None           R2      R3  ␣
↪    [Rücklauf]       0     80       0
7     PGRP  4986517622672493603                     Pumpengruppe          R-1      R3  ␣
↪            []       0    NaN       0
8     PUMP  5481331875203087055                      Umwälzpumpe          R-1      R2  ␣
↪    [Rücklauf]       0    NaN       0
9     ROHR  4613782368750024999                            None       R-K004  R-K005  ␣
↪    [Rücklauf]   88.02  107.1       0
10    ROHR  4614949065966596185                            None       V-K002  V-K003  ␣
↪     [Vorlauf]  405.96  107.1       1
11    ROHR  4637102239750163477                            None       R-K003  R-K004  ␣
↪    [Rücklauf]   83.55  107.1       2
12    ROHR  4713733238627697042                            None       V-K004  V-K005  ␣
↪     [Vorlauf]   88.02  107.1       3
13    ROHR  4769996343148550485                            None          R-L  R-K000  ␣
↪    [Rücklauf]   73.42  160.3       4
14    ROHR  4789218195240364437                            None       V-K001  V-K002  ␣
↪     [Vorlauf]  195.53  107.1       5
15    ROHR  4939422678063487923                            None          V-L  V-K000  ␣
↪     [Vorlauf]    68.6  160.3       6
16    ROHR  4945727430885351042                            None       R-K006  R-K007  ␣
↪    [Rücklauf]  109.77  107.1       7
17    ROHR  4984202422877610920                            None       V-K000  V-K001  ␣
↪     [Vorlauf]    76.4  107.1       8
18    ROHR  5037777106796980248                            None       V-K003  V-K004  ␣
↪     [Vorlauf]   83.55  107.1       9
19    ROHR  5123819811204259837                            None       V-K005  V-K006  ␣
↪     [Vorlauf]  164.91  107.1      10
20    ROHR  5266224553324203132                            None       R-K001  R-K002  ␣
↪    [Rücklauf]  195.53  107.1      11
21    ROHR  5379365049009065623                            None       R-K002  R-K003  ␣
↪    [Rücklauf]  405.96  107.1      12
22    ROHR  5611703699850694889                            None       R-K005  R-K006  ␣
↪    [Rücklauf]  164.91  107.1      13
23    ROHR  5620197984230756681                            None       V-K006  V-K007  ␣
↪     [Vorlauf]  109.77  107.1      14
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```
24    ROHR  5647213228462830353                          None       R-K000  R-K001  ␣
↪    [Rücklauf]    76.4  107.1      15
25    VENT  4678923650983295610                          None          V-1     V-L  ␣
↪     [Vorlauf]       0    150       0
26    VENT  4897018421024717974                          None          R-L     R-1  ␣
↪    [Rücklauf]       0    150       1
27    VENT  5525310316015533093                          None  PKON-Knoten     R-1  ␣
↪    [Rücklauf]       0     50       2
>>> # ---
>>> # vRART
>>> # ---
>>> print(xm._getvXXXXAsOneString(vXXXX='vRART',index=True,sortList=['INDSTD','NAME
↪']))
  NAME              BESCHREIBUNG                                  INDSTD_TXT  INDSTD␣
↪  DWDT WSOSTD                  pk NAME_KREF1 NAME_KREF2 NAME_SWVT
0   dp  Bezeichnung Regelungsart  Differenzdruck Druckseite, Sollwert Tabelle      55␣
↪ 1E+20      0  5552938346422332788     V-K007     R-K007      SWVT
>>> # ------
>>> # MxAdd
>>> # ------
>>> if 'vNRCV_Mx1' in xm.dataFrames:
...     del xm.dataFrames['vNRCV_Mx1'] # delete MxSync-Result to force MxSync-Call in␣
↪MxAdd
>>> oldShape=xm.dataFrames['vKNOT'].shape
>>> mx=xm.MxAdd()
>>> firstShape=xm.dataFrames['vKNOT'].shape
>>> oldShape[1]<firstShape[1]
True
>>> xm.MxAdd(mx=mx)
>>> secondShape=xm.dataFrames['vKNOT'].shape
>>> secondShape==firstShape
True
>>> xm.MxAdd(mx=mx)
>>> thirdShape=xm.dataFrames['vKNOT'].shape
>>> thirdShape==firstShape
True
>>> xm.dataFrames['vKNOT_forTestOnly']=xm.dataFrames['vKNOT'].rename(columns={'KNOT~*~
↪*~*~PH':'Druck'})
>>> if 'Druck' not in xm.dataFrames['vKNOT_forTestOnly']:
...     xm.dataFrames['vKNOT_forTestOnly'].rename(columns={'KNOT~*~*~*~H':'Druck'},
↪inplace=True)
>>> if 'Druck' not in xm.dataFrames['vKNOT_forTestOnly']:
...     xm.dataFrames['vKNOT_forTestOnly'].rename(columns={'KNOT~*~~*~PH':'Druck'},
↪inplace=True) #09
>>> f = lambda x: round(x,1) if x != None else None
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT_forTestOnly',filterColList=['mx2Idx',
↪'KVR','NAME','Druck'],mapFunc={'Druck':f},index=True))
    mx2Idx KVR         NAME  Druck
0        0   2       R-K004    2.3
1        1   1       V-K002    4.0
2        2   1       V-K001    4.1
3        3   1       V-K000    4.1
4        4   2       R-K001    2.0
5        5   2       R-K003    2.3
6        6   2       R-K000    2.0
7        9   2       R-K005    2.3
8       11   2          R-L    2.0
```

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```
9       12   2       R-K002    2.1
10      13   1       V-K004    3.8
11      15   1       V-K005    3.8
12      16   2       R-K007    2.3
13      17   1       V-K006    3.8
14      18   2       R-K006    2.3
15      20   1       V-K003    3.8
16      21   1          V-L    4.1
17      22   1       V-K007    3.8
18       7   2           R2    4.3
19       8   1          V-1    4.1
20      10   2           R3    4.3
21      14   2  PKON-Knoten    2.0
22      19   2          R-1    2.0
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',filterColList=['mx2Idx','L','KVR',
↪'NAME_i','NAME_k','ROHR~*~*~*~QMAV'],mapFunc={'ROHR~*~*~*~QMAV':f},sortList=['ROHR~
↪*~*~*~QMAV','NAME_i'],index=True))
    mx2Idx       L KVR  NAME_i  NAME_k  ROHR~*~*~*~QMAV
13      15    76.4   2  R-K000  R-K001            -23.0
9       11  195.53   2  R-K001  R-K002            -23.0
14       4   73.42   2     R-L  R-K000            -23.0
10      12  405.96   2  R-K002  R-K003            -19.1
2        2   83.55   2  R-K003  R-K004            -15.4
0        0   88.02   2  R-K004  R-K005             -8.5
11      13  164.91   2  R-K005  R-K006             -3.9
5        7  109.77   2  R-K006  R-K007             -3.9
8       10  164.91   1  V-K005  V-K006              3.9
12      14  109.77   1  V-K006  V-K007              3.9
3        3   88.02   1  V-K004  V-K005              8.5
7        9   83.55   1  V-K003  V-K004             15.4
1        1  405.96   1  V-K002  V-K003             19.1
6        8    76.4   1  V-K000  V-K001             23.0
4        5  195.53   1  V-K001  V-K002             23.0
15       6    68.6   1     V-L  V-K000             23.0
>>> print(xm._getvXXXXAsOneString(vXXXX='vFWVB',filterColList=['mx2Idx','NAME_i',
↪'NAME_k','FWVB~*~*~*~W'],mapFunc={'FWVB~*~*~*~W':f},sortList=['FWVB~*~*~*~W','NAME_i
↪'],index=True))
   mx2Idx  NAME_i  NAME_k  FWVB~*~*~*~W
4       4  V-K003  R-K003         120.0
0       0  V-K002  R-K002         160.0
2       2  V-K005  R-K005         160.0
3       3  V-K007  R-K007         160.0
1       1  V-K004  R-K004         200.0
>>> xm.dataFrames['vVBEL_forTestOnly2']=xm.dataFrames['vVBEL'].loc[['ROHR','FWVB'],:].
↪reset_index(inplace=False) # Multiindex to Cols
>>> xm.dataFrames['vVBEL_forTestOnly2'].rename(columns={'KNOT~*~*~*~PH_i':'Druck_i'},
↪inplace=True)
>>> if 'Druck_i' not in xm.dataFrames['vVBEL_forTestOnly2']:
...     xm.dataFrames['vVBEL_forTestOnly2'].rename(columns={'KNOT~*~*~*~H_i':'Druck_i
↪'},inplace=True)
>>> if 'Druck_i' not in xm.dataFrames['vVBEL_forTestOnly2']:
...     xm.dataFrames['vVBEL_forTestOnly2'].rename(columns={'KNOT~*~~*~PH_i':'Druck_i
↪'},inplace=True) #09
>>> f = lambda x: round(x,1) if x != None else None
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL_forTestOnly2',filterColList=['OBJTYPE',
↪'mx2Idx','L','D','NAME_i','NAME_k','Druck_i','Q'],mapFunc={'Druck_i':f,'Q':f},
↪sortList=['OBJTYPE','NAME_i','Q'],index=True))
```

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```
   OBJTYPE  mx2Idx       L      D  NAME_i  NAME_k  Druck_i     Q
0     FWVB       0       0    NaN  V-K002  R-K002      4.0   3.9
4     FWVB       4       0    NaN  V-K003  R-K003      3.8   3.7
1     FWVB       1       0    NaN  V-K004  R-K004      3.8   6.9
2     FWVB       2       0    NaN  V-K005  R-K005      3.8   4.6
3     FWVB       3       0    NaN  V-K007  R-K007      3.8   3.9
20    ROHR      15    76.4  107.1  R-K000  R-K001      2.0 -23.0
16    ROHR      11  195.53  107.1  R-K001  R-K002      2.0 -23.0
17    ROHR      12  405.96  107.1  R-K002  R-K003      2.1 -19.1
7     ROHR       2   83.55  107.1  R-K003  R-K004      2.3 -15.4
5     ROHR       0   88.02  107.1  R-K004  R-K005      2.3  -8.5
18    ROHR      13  164.91  107.1  R-K005  R-K006      2.3  -3.9
12    ROHR       7  109.77  107.1  R-K006  R-K007      2.3  -3.9
9     ROHR       4   73.42  160.3     R-L  R-K000      2.0 -23.0
13    ROHR       8    76.4  107.1  V-K000  V-K001      4.1  23.0
10    ROHR       5  195.53  107.1  V-K001  V-K002      4.1  23.0
6     ROHR       1  405.96  107.1  V-K002  V-K003      4.0  19.1
14    ROHR       9   83.55  107.1  V-K003  V-K004      3.8  15.4
8     ROHR       3   88.02  107.1  V-K004  V-K005      3.8   8.5
15    ROHR      10  164.91  107.1  V-K005  V-K006      3.8   3.9
19    ROHR      14  109.77  107.1  V-K006  V-K007      3.8   3.9
11    ROHR       6    68.6  160.3     V-L  V-K000      4.1  23.0
>>> # ---
>>> # Clean Up LocalHeatingNetwork Xm and Mx
>>> # ---
>>> xm.delFiles()
>>> mx.delFiles()
>>> # ---
>>> # TinyWDN
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'TinyWDN.XML')
>>> xm=Xm(xmlFile=xmlFile)
>>> # ---
>>> # GPipe
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'GPipe.XML')
>>> xm=Xm(xmlFile=xmlFile)
```

**class** Xm.**Xm**(*xmlFile*, *NoH5Read=False*)

SIR 3S modelFile to pandas DataFrames.

**Args:**

- xmlFile (str): SIR 3S modelFile
- **NoH5Read (bool):**

  **False (default):**

  - An existing _and newer h5File will be read _instead of xmlFile.
  - xmlFile will _not be read (it does even not have to exist)

  **True:**

  - An existing h5File will be deleted.
  - xmlFile will be read.

**Attributes:**

- **states**
  - h5Read: True, if read from H5
- **fileNames**
  - xmlFile
  - constructed from MX during Init and Usage:
  - h5File: corresponding h5File(name) derived from xmlFile(name)
- **dataFrames**
  - dict with pandas DataFrames
  - one pandas DataFrame per SIR 3S Objecttype (i.e. KNOT, ROHR, ...)
  - keys: KNOT, ROHR, ... and vKNOT, vROHR, ...
  - **some Views as pandas DataFrames**
    * i.e. vKNOT, vROHR, ...
    * The Views are designed to deal with tedious groundwork.
    * The Views are aggregated somewhat arbitrary.
    * However: Usage of SIR 3S Modeldata is more convenient and efficient with appropriate Views.
- pXCorZero, pYCorZero

**Raises:** XmError

**classmethod constructNewMultiindexFromCols**(*df=None, mColNames=['OBJTYPE', 'pk'], mIdxNames=['OBJTYPE', 'OBJID']*)

Constructs a new Multiindex from existing cols and returns the constructed df.

**Args:**

- df: dataFrame without Multiindex
- mColNames: list of columns which shall be used as Multiindex; the columns must exist; the columns will be droped
- mIdxNames: list of names for the indices for the Cols above

**Returns:**

- df with Multiindex
- empty DataFrame is returned if an Error occurs

```
>>> d = {'OBJTYPE': ['ROHR', 'VENT'], 'pk': [123, 345], 'data': ['abc', 'def
↪']}
>>> import pandas as pd
>>> df = pd.DataFrame(data=d)
>>> from Xm import Xm
>>> df=Xm.constructNewMultiindexFromCols(df=df,mColNames=['OBJTYPE','pk'],
↪mIdxNames=['OBJTYPE','OBJID'])
>>> df['data']
OBJTYPE  OBJID
ROHR     123      abc
VENT     345      def
Name: data, dtype: object
```

**classmethod constructShortestPathFromNodeList**(*df=None*, *sourceCol='NAME_i'*, *targetCol='NAME_k'*, *nl=None*, *weight=None*, *query=None*, *fmask=None*)

Returns a DataFrame with Edges (one per row) implementing the shortest Path over NodeList.

**Args:**

- df: DataFrame with (all) Edges (one per row)
- nl: NodeList
- weight: columnName of the weight attribute

  # Der kürzeste Weg zwischen zwei Knoten in einem # zusammenhängenden Graphen ist derjenige, bei dem die # Summe der Gewichte über die durchlaufenen Kanten den # kleinstmöglichen Wert annimmt.

  # also bei konstantem Kantengewicht die kleinste Kantenanzahl # kürzeste Weglaenge: L als Gewicht # Durchflussstärkster Weg: 1 / Abs(Q) (Flüsse mit 0 oder Kanten ohne Flusswert müssen vorher eliminiert werden ... # ... birgt die Gefahr, dass es dann keinen Weg mehr gibt) # Durchflussschwächster Weg: Q

  - **examples for weight:**
    - 'L': converted to float before usage here
    - **if 'Q' is already a column in df:**
      - non Null Q-rows filtered here - then:
      - 'QAbs': constructed here
      - 'QAbsInv': constructed here
  - query: query to filter vVBEL (to filter Edges) before constructing the Graph
  - fmask: function to filter vVBEL (to filter Edges) before constructing the Graph
  - query and fmask are used both if not None

**Returns:**

- df: DataFrame with Edges (one per row) implementing the shortest Path over NodeList
- empty DataFrame is returned if an error occurs
- columns
  - OBJTYPE
  - OBJID
  - nextNODE
  - compNr
    - starts with 1
    - the number of the connected component
    - 1 for all edges if all nodes in NodeList are connected

```
>>> xmlFile=ms['GPipes']
>>> from Xm import Xm
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['GL','GR'])
```

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```
  OBJTYPE                OBJID nextNODE  compNr
0    VENT  5309992331398639768       G1       1
1    ROHR  5244313507655010738      GKS       1
2    VENT  5116489323526156845      GKD       1
3    ROHR  5114681686941855110       G3       1
4    ROHR  4979507900871287244       G4       1
5    VENT  5745097345184516675       GR       1
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['GL','GR'],weight='QAbsInv')   #␣
↪durchflussstaerkster Weg
  OBJTYPE                OBJID nextNODE  compNr
0    VENT  5309992331398639768       G1       1
1    ROHR  5244313507655010738      GKS       1
2    VENT  5508684139418025293      GKD       1
3    ROHR  5114681686941855110       G3       1
4    ROHR  4979507900871287244       G4       1
5    VENT  5745097345184516675       GR       1
>>> xmlFile=ms['LocalHeatingNetwork']
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['V-L','V-K07'])
Empty DataFrame
Columns: []
Index: []
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['V-L','V-K007'])
  OBJTYPE                OBJID nextNODE  compNr
0    ROHR  4939422678063487923   V-K000       1
1    ROHR  4984202422877610920   V-K001       1
2    ROHR  4789218195240364437   V-K002       1
3    ROHR  4614949065966596185   V-K003       1
4    ROHR  5037777106796980248   V-K004       1
5    ROHR  4713733238627697042   V-K005       1
6    ROHR  5123819811204259837   V-K006       1
7    ROHR  5620197984230756681   V-K007       1
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['V-K007','R-K007'])
  OBJTYPE                OBJID nextNODE  compNr
0    FWVB  5400405917816384862   R-K007       1
>>> f=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2' else False
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['V-K007','R-K007'],fmask=f)
Empty DataFrame
Columns: []
Index: []
>>> xm.constructShortestPathFromNodeList(df=xm.
↪getvVBELwithNodeAttributeAdded(),nl=['R-K007','V-K007'],query="OBJTYPE not␣
↪in ['FWVB','PGRP']")
   OBJTYPE                OBJID nextNODE  compNr
0     ROHR  4945727430885351042   R-K006       1
1     ROHR  5611703699850694889   R-K005       1
2     ROHR  4613782368750024999   R-K004       1
3     ROHR  4637102239750163477   R-K003       1
4     ROHR  5379365049009065623   R-K002       1
5     ROHR  5266224553324203132   R-K001       1
```

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```
6     ROHR  5647213228462830353   R-K000       1
7     ROHR  4769996343148550485      R-L       1
8     VENT  4897018421024717974      R-1       1
9     PUMP  5481331875203087055       R2       1
10    KLAP  4801110583764519435       R3       1
11    FWES  5638756766880678918      V-1       1
12    VENT  4678923650983295610      V-L       1
13    ROHR  4939422678063487923   V-K000       1
14    ROHR  4984202422877610920   V-K001       1
15    ROHR  4789218195240364437   V-K002       1
16    ROHR  4614949065966596185   V-K003       1
17    ROHR  5037777106796980248   V-K004       1
18    ROHR  4713733238627697042   V-K005       1
19    ROHR  5123819811204259837   V-K006       1
20    ROHR  5620197984230756681   V-K007       1
```

**delFiles**()

Deletes Files constructed by XM during Init and Usage.

**_xmlRead**()

Reads the SIR 3S modelFile.

- Performs fixes and basic conversions inplace the dataFrames read from modelFile: _convertAndFix()
- Creates some Views: _vXXXX()

**Raises:** XmError

**FromH5**(*h5File=None*)

Reads all dataFrames stored in h5File into self.DataFrames.

**Args:**

**h5File:**

- (str): the h5File(name) to be read
- (None): self.h5File will be read

- Reads all keys.
- Existing keys in self.dataFrames are overwritten.

Note that after .FromH5() the content of self.dataFrames may differ from the content given by an existing self.xmlFile.

**Raises:** XmError

**ToH5**(*h5File=None*)

Stores self.dataFrames to h5File.

**Args:**

**h5File:**

- (str): the h5File(name) to be used
- (None): self.h5File will be used

- Stores all keys.

- Existing keys in h5File are overwritten.

**Raises:** XmError

**_convertAndFix**()

Performs fixes and basic conversions inplace the dataFrames read from self.xmlFile.

- Fixes and conversions here are integrity-oriented.
- Usage-oriented conversions (i.e. pd.to_numeric and base64.b64decode) - if any - are done in the ._vXXXX-methods.
- Vorgehen in den Sichten: Anwendungs-orientierte Konvertierung von pandas Object in ein spezifisches Format nur wenn sinnvoll bzw. erforderlich

**Conversions:**

- , > . (converted in: SWVT_ROWT, LFKT_ROWT, QVAR_ROWT, PVAR_ROWT)

**Fixes:**

- **No SWVT_ROWT, LFKT_ROWT, QVAR_ROWT, PVAR_ROWT?!**
  - * SWVT, LFKT, QVAR, PVAR are constructed to
- 1st Time without Value?! (fixed in: SWVT_ROWT, LFKT_ROWT, QVAR_ROWT, PVAR_ROWT)
- Template Node(s)?!
- **in new Models constructed from SIR 3S**
  - **not all Objectattributes are written?!**
    - * KMOT/TE
    - * FWVB/LFK
    - * LTGR/BESCHREIBUNG
    - * **DTRO_ROWD**
      - · AUSFALLZEIT
      - · PN
      - · REHABILITATION
      - · REPARATUR
      - · WSTEIG, WTIEFE
    - * **RSLW**
      - · WMIN
      - · WMAX
  - **not all Objecttypes are written?!**
    - * CONT
- Models with no PZONs . . .
- Models with no GMIXs . . .
- Models with no STOFs . . .

- empty WBLZ OBJS-BLOBs
- empty LAYR OBJS-BLOBs
- BESCHREIBUNG nicht in RLVG?...
- BESCHREIBUNG nicht in RADD?...

**Raises:** XmError

**_constructEmptyDf**(*cols=['DummyCol1', 'DummyCol2']*)

Constructs an empty df with cols.

**Args:**

- cols: list of colNames

**Returns:** df: constructed df

**Raises:** XmError

**getWDirModelDirModelName**()

Returns (wDir,modelDir,modelName,mx1FileName).

**Returns:** (wDir,modelDir,modelName,mx1FileName)

**wDir** If wDir as given literally in .xmlFile is not a valid Dir or such a wDir relative to .xmlFile-Path exists the wDir relative is returned.

**mx1FileName** mx1FileName is assumed to be: .:...WD...B...V...BZ...M... .MX1 If not existing: .*.MX1 (first match is returned) If a suitable mx1File is not existing an INFO-Message is generated.

**Raises:** XmError

**getVersion**(*type='BASIS'*)

Returns VERSION-String i.e. Sir3S-90-10.

**Args:**

- type: BASIS or VARIANTE or BZ; the DATENEBENEn-TYPe from which the VERSION-String is requested

**Returns:**

- VERSION-String i.e. Sir3S-90-10; Sir3S-90-09 is returned if Attribute VERSION is not available

**Raises:** XmError

```
>>> xm=xms['OneLPipe']
>>> vStr=xm.getVersion()
>>> import re
>>> m=re.search('Sir(?P<Db3s>[DBdb3Ss]{2})-(?P<Major>\d+)-(?P<Minor>\d+)$',
↪vStr) # i.e. Sir3S-90-10
>>> int(m.group('Major')[0])
9
```

**_getvXXXXAsOneString**(*vXXXX=None, start=0, end=-1, dropColList=None, filterColList=None, mapFunc={}, sortList=None, ascending=True, fmtFunc={}, index=True, header=True*)

Returns vXXXX-Content as one String (for Doctest-Purposes).

**Args:**

- vXXXX: df=self.dataFrames[vXXXX]
- start

- end
- dropColList
- filterColList
- mapFunc: col:func: df[col].map(func)
- sortList
- ascending
- fmtFunc: col:func: passed to df.to_string(formatters=fmtFunc, . . . )
- index
- header

**Returns:**

- df.to_string(formatters=fmtFunc,index=index,header=header)

**Raises:** XmError

**_vXXXX** ( )

Creates all Views.

**Views created:**

- **BLOB-Data**
  - vLAYR
  - vWBLZ
  - vAGSN u. vAGSN_raw
- **Timeseries**
  - vLFKT
  - vQVAR
  - vPVAR
  - vSWVT
- **Signalmodel**
  - vRUES: RUES-Nodes of R
  - vRXXX: Nodes of R but RUES-Nodes
  - vREdges: die Kanten des Knoten-Kanten-Signalmodells
  - vRSLW
  - vRART
  - vRSTN
- **Hydraulicmodel**
  - **Nodes**
    - vVKNO: CONT-Nodes (also called Block-Nodes)
    - vKNOT
    - pXCorZero, pYCorZero

- **Edges**
  - vROHR: Pipes
  - vFWVB: Housestations (district heating)
- **all Edges (all; implemented Edges see vVBEL_edges)**
  - vVBEL

- **Annotations**
  - vNRCV
  - vGTXT

**Raises:** XmError

**_vLAYR**()

One row per LAYR and OBJ.

**Returns:**

**columns**

**LAYR (also called 'Group')**

- LFDNR
- NAME

**from SIR 3S OBJ BLOB collection:**

- OBJTYPE: type (i.e.ROHR) of a LAYR OBJ
- OBJID: pk (or tk?!) of a LAYR OBJ

**LAYR IDs**

- pk, tk

**ANNOTATION**

- nrObjInGroup: Element Nr. in LAYR (LFDNR) - should be 1 otherwise the same OBJ occurs in the same LAYR multiple times
- nrObjtypeInGroup: Element Nr. of OBJTYPE in LAYR (LFDNR)

**SORTING** LFDNR,NAME,OBJTYPE,OBJID

**Raises:** XmError

**_vWBLZ**()

One row per WBLZ and OBJ.

**Returns:**

**columns**

**WBLZ**

- AKTIV
- BESCHREIBUNG
- IDIM
- NAME

**from SIR 3S OBJ BLOB collection:**

- OBJTYPE: type (always KNOT?!)
- OBJID: pk (or tk?!)

**WBLZ IDs**

- pk

**SORTING** NAME,pk

**Raises:** XmError

**_vAGSN**()

One row per AGSN and OBJ.

**Returns:**

**columns**

**AGSN**

- LFDNR
- NAME
- AKTIV
- from SIR 3S OBJ BLOB collection:
  - OBJTYPE: type (i.e.ROHR)
  - OBJID: pk (or tk?!)

**AGSN IDs**

- pk, tk

**Sequence**

- **Model**
  - therefore nrObjIdInAgsn (see ANNOTATION below) should be the realwolrd sequence

**ANNOTATION**

- nrObjIdInAgsn: lfd.Nr. (in Schnittreihenfolge) Obj. (der Kante) in AGSN (AGSN is defined by LFDNR not by NAME)
- **nrObjIdTypeInAgsn: should be 1 determined by raw data**
  - nrObjIdTypeInAgsn>1 - if any - are not part of the view
  - the 1st occurance is in the view
- **Layer** 0=undef bei Netztyp 21: 1=VL, 2=RL, 0=undef wenn keine BN-Trennzeile gefunden wird, wird VL angenommen und gesetzt die BN-Trennzeile wird dem VL (1) zugerechnet
- **nextNODE: node which is connected by the edge**
  - the cut-direction is defined (per cut and comp) by edge-sequence
  - the cut node-sequence ist the (longest shortest) path between the nodes of the 1st and last edge
  - in case of 1 edge cut-direction is edge-definition and cut node-sequence is edge-definition

  - the nextNODEs are the node-sequence omitting the start-node ...
  - ... nextNODE of an edge is the node connected by this edge in cut-direction; so nextNODE might be the i-node (the source-node) of the edge
  - if edge-direction is cut-direction nextNODE is the k-node (the sink-node) of the edge
- **compNr**
  - all 1 if all edges in the cut are connected
  - otherwise the compNr (starting with 1) the edge belongs to
  - the comp-Sequence is defined by the edge-sequence
  - the nodes of the 1st and last edge in cut-definition of the comp are defining the node-Sequence of the (longest shortest) path in the comp
- **parallel Edges**
  - are omitted in the cut-Result; the 1st edge in cut-definition is in the edge
- **Abzweige**
  - are omitted in the cut-Result
  - the nodes of the 1st and last edge in cut-definition are defining the node-Sequence of the (longest shortest) path (comp-wise)
  - only edges implementing this path are in the cut-Result

**Raises:** XmError

```
>>> xmlFile=ms['GPipes']
>>> from Xm import Xm
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> vAGSN=xm.dataFrames['vAGSN']
>>> schnitt=vAGSN[vAGSN['NAME']=='LR']
>>> xm.dataFrames['schnitt']=schnitt.reset_index()
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',index=True))
   index LFDNR NAME AKTIV OBJTYPE                OBJID                    pk
↪                tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE compNr
0      7    14   LR   101    VENT  5309992331398639768  5625063016896368599
↪5625063016896368599              1                  1      0       G1      1
1      8    14   LR   101    ROHR  5244313507655010738  5625063016896368599
↪5625063016896368599              2                  1      0      GKS      1
2      9    14   LR   101    VENT  5508684139418025293  5625063016896368599
↪5625063016896368599              3                  1      0      GKD      1
3     10    14   LR   101    ROHR  5114681686941855110  5625063016896368599
↪5625063016896368599              4                  1      0       G3      1
4     11    14   LR   101    ROHR  4979507900871287244  5625063016896368599
↪5625063016896368599              5                  1      0       G4      1
5     12    14   LR   101    VENT  5745097345184516675  5625063016896368599
↪5625063016896368599              6                  1      0       GR      1
>>> schnitt=vAGSN[vAGSN['NAME']=='LR-Lücke']
>>> xm.dataFrames['schnitt']=schnitt.reset_index()
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',index=True))
   index LFDNR      NAME AKTIV OBJTYPE                OBJID
↪pk                   tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE
↪compNr
0     13    16  LR-Lücke   101    VENT  5309992331398639768
↪5630543731618051887  5630543731618051887              1                  1
↪    0       G1      1
```

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```
1     14    16  LR-Lücke   101    ROHR  5244313507655010738 ␣
↪5630543731618051887  5630543731618051887             2                  1 ␣
↪    0      GKS      1
2     15    16  LR-Lücke   101    ROHR  5114681686941855110 ␣
↪5630543731618051887  5630543731618051887             3                  1 ␣
↪    0       G3      2
3     16    16  LR-Lücke   101    ROHR  4979507900871287244 ␣
↪5630543731618051887  5630543731618051887             4                  1 ␣
↪    0       G4      2
4     17    16  LR-Lücke   101    VENT  5745097345184516675 ␣
↪5630543731618051887  5630543731618051887             5                  1 ␣
↪    0       GR      2
>>> schnitt=vAGSN[vAGSN['NAME']=='LR-Flansch']
>>> xm.dataFrames['schnitt']=schnitt.reset_index()
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',index=True))
   index LFDNR        NAME AKTIV OBJTYPE                OBJID                ␣
↪  pk                   tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE␣
↪compNr
0     18    18  LR-Flansch   101    VENT  5309992331398639768 ␣
↪5134530907542044265  5134530907542044265             1                  1 ␣
↪    0       G1      1
1     19    18  LR-Flansch   101    ROHR  5244313507655010738 ␣
↪5134530907542044265  5134530907542044265             2                  1 ␣
↪    0      GKS      1
2     20    18  LR-Flansch   101    VENT  5508684139418025293 ␣
↪5134530907542044265  5134530907542044265             3                  1 ␣
↪    0      GKD      1
3     21    18  LR-Flansch   101    ROHR  5114681686941855110 ␣
↪5134530907542044265  5134530907542044265             4                  1 ␣
↪    0       G3      1
4     22    18  LR-Flansch   101    ROHR  4979507900871287244 ␣
↪5134530907542044265  5134530907542044265             5                  1 ␣
↪    0       G4      1
5     24    18  LR-Flansch   101    VENT  5745097345184516675 ␣
↪5134530907542044265  5134530907542044265             7                  1 ␣
↪    0       GR      1
>>> schnitt=vAGSN[vAGSN['NAME']=='LR-Parallel']
>>> xm.dataFrames['schnitt']=schnitt.reset_index()
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',index=True))
   index LFDNR         NAME AKTIV OBJTYPE                 OBJID                 ␣
↪   pk                   tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer␣
↪nextNODE compNr
0     25    20  LR-Parallel   101    VENT  5309992331398639768 ␣
↪4694969854935170169  4694969854935170169             1                  1 ␣
↪    0       G1      1
1     26    20  LR-Parallel   101    ROHR  5244313507655010738 ␣
↪4694969854935170169  4694969854935170169             2                  1 ␣
↪    0      GKS      1
2     27    20  LR-Parallel   101    VENT  5116489323526156845 ␣
↪4694969854935170169  4694969854935170169             3                  1 ␣
↪    0      GKD      1
3     29    20  LR-Parallel   101    ROHR  5114681686941855110 ␣
↪4694969854935170169  4694969854935170169             5                  1 ␣
↪    0       G3      1
4     30    20  LR-Parallel   101    ROHR  4979507900871287244 ␣
↪4694969854935170169  4694969854935170169             6                  1 ␣
↪    0       G4      1
```

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```
5     31    20  LR-Parallel   101    VENT  5745097345184516675 ␣
↪4694969854935170169  4694969854935170169              7                  1 ␣
↪    0       GR      1
```

**getvVBELwithNodeAttributeAdded**(*nodeAttribute='KVR'*, *preserveMultiindex=False*)

Adds a nodeAttribute (not already in vVBEL) to vVBEL and returns the df.

**Args:**

- nodeAttribute (default: 'KVR'): the Node Attribute which shall be added
- preserveMultiindex (default: False): if True an existing Multiindex will be preserved; False: existing Index(Indices) will be col(s)

**Returns:** df (which might be incomplete, corrupt or empty if an error occurs; vVBEL is unchanged)

**Raises:** XmError

```
>>> xmlFile=ms['LocalHeatingNetwork']
>>> from Xm import Xm
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> df=xm.getvVBELwithNodeAttributeAdded()
>>> xm.dataFrames['tmp']=df
>>> print(xm._getvXXXXAsOneString(vXXXX='tmp',index=True,filterColList=[
↪'OBJTYPE','OBJID','NAME_i','NAME_k','KVR_i','KVR_k']))
   OBJTYPE                OBJID       NAME_i  NAME_k KVR_i KVR_k
0     FWES  5638756766880678918           R3     V-1     2     1
1     FWVB  4643800032883366034       V-K002  R-K002     1     2
2     ROHR  5266224553324203132       R-K001  R-K002     2     2
3     ROHR  4614949065966596185       V-K002  V-K003     1     1
4     FWVB  4704603947372595298       V-K004  R-K004     1     2
5     ROHR  4637102239750163477       R-K003  R-K004     2     2
6     ROHR  4713733238627697042       V-K004  V-K005     1     1
7     FWVB  5121101823283893406       V-K005  R-K005     1     2
8     ROHR  4613782368750024999       R-K004  R-K005     2     2
9     ROHR  5123819811204259837       V-K005  V-K006     1     1
10    FWVB  5400405917816384862       V-K007  R-K007     1     2
11    ROHR  4945727430885351042       R-K006  R-K007     2     2
12    FWVB  5695730293103267172       V-K003  R-K003     1     2
13    ROHR  5379365049009065623       R-K002  R-K003     2     2
14    ROHR  5037777106796980248       V-K003  V-K004     1     1
15    KLAP  4801110583764519435           R2      R3     2     2
16    PGRP  4986517622672493603          R-1      R3     2     2
17    PUMP  5481331875203087055          R-1      R2     2     2
18    ROHR  4769996343148550485          R-L  R-K000     2     2
19    VENT  4897018421024717974          R-L     R-1     2     2
20    VENT  5525310316015533093  PKON-Knoten     R-1     2     2
21    ROHR  4789218195240364437       V-K001  V-K002     1     1
22    ROHR  4939422678063487923          V-L  V-K000     1     1
23    ROHR  4984202422877610920       V-K000  V-K001     1     1
24    ROHR  5611703699850694889       R-K005  R-K006     2     2
25    ROHR  5620197984230756681       V-K006  V-K007     1     1
26    ROHR  5647213228462830353       R-K000  R-K001     2     2
27    VENT  4678923650983295610          V-1     V-L     1     1
>>> len(df.columns.tolist())
20
>>> df.index.names
FrozenList([None])
```

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```
>>> df=xm.getvVBELwithNodeAttributeAdded(preserveMultiindex=True)
>>> len(df.columns.tolist())
18
>>> df.index.names
FrozenList(['OBJTYPE', 'OBJID'])
```

**vAGSN_Add**(*nl=None*, *weight=None*, *Layer=0*, *AKTIV=None*, *NAME='NEU'*, *query=None*, *fmask=None*)

Adds a new User-defined Cut to the Model-defined Cuts.

**Arguments:**

**see constructShortestPathFromNodeList:**

- nl: NodeList for the Cut
- weight: columnName of the weight attribute
- query: mask to filter vVBEL (to filter Edges) before constructing the Graph
- fmask: function to filter vVBEL (to filter Edges) before constructing the Graph
- query and fmask are used both if not None

- Layer (to use in constructed cut)
- AKTIV (to use in constructed cut)
- NAME (to use in constructed cut): the cut will NOT be constructed if such a NAME already exists

**Raises:** XmError

```
>>> xmlFile=ms['GPipes']
>>> from Xm import Xm
>>> xm=Xm(xmlFile=xmlFile,NoH5Read=True)
>>> xm.vAGSN_Add(nl=['GL','GR'])
>>> import pandas as pd
>>> pd.set_option('display.max_columns',None)
>>> pd.set_option('display.max_rows',None)
>>> pd.set_option('display.max_colwidth',666666)
>>> pd.set_option('display.width',666666666)
>>> xm.dataFrames['vAGSN_raw']
   LFDNR         NAME AKTIV OBJTYPE                OBJID                   pk␣
↪                  tk  nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE␣
↪compNr
0      2       1 Rohr   101    ROHR  5244313507655010738  5015814781412926392␣
↪ 5015814781412926392              1                  1      0      GKS      ␣
↪1
1      6       V-Rohr   101    VENT  5309992331398639768  5396484903084432138␣
↪ 5396484903084432138              1                  1      0       G1      ␣
↪1
2      6       V-Rohr   101    ROHR  5244313507655010738  5396484903084432138␣
↪ 5396484903084432138              2                  1      0      GKS      ␣
↪1
3      8       Rohr-V   101    ROHR  4979507900871287244  4989935433418681990␣
↪ 4989935433418681990              1                  1      0       G4      ␣
↪1
4      8       Rohr-V   101    VENT  5745097345184516675  4989935433418681990␣
↪ 4989935433418681990              2                  1      0       GR      ␣
↪1
```

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```
5     12      2 Rohre   101    ROHR  5114681686941855110  5748019382126004712␣
↪ 5748019382126004712              1                   1      0       G3      ␣
↪1
6     12      2 Rohre   101    ROHR  4979507900871287244  5748019382126004712␣
↪ 5748019382126004712              2                   1      0       G4      ␣
↪1
7     14           LR   101    VENT  5309992331398639768  5625063016896368599␣
↪ 5625063016896368599              1                   1      0       G1      ␣
↪1
8     14           LR   101    ROHR  5244313507655010738  5625063016896368599␣
↪ 5625063016896368599              2                   1      0      GKS      ␣
↪1
9     14           LR   101    VENT  5508684139418025293  5625063016896368599␣
↪ 5625063016896368599              3                   1      0      GKD      ␣
↪1
10    14           LR   101    ROHR  5114681686941855110  5625063016896368599␣
↪ 5625063016896368599              4                   1      0       G3      ␣
↪1
11    14           LR   101    ROHR  4979507900871287244  5625063016896368599␣
↪ 5625063016896368599              5                   1      0       G4      ␣
↪1
12    14           LR   101    VENT  5745097345184516675  5625063016896368599␣
↪ 5625063016896368599              6                   1      0       GR      ␣
↪1
13    16     LR-Lücke   101    VENT  5309992331398639768  5630543731618051887␣
↪ 5630543731618051887              1                   1      0       G1      ␣
↪1
14    16     LR-Lücke   101    ROHR  5244313507655010738  5630543731618051887␣
↪ 5630543731618051887              2                   1      0      GKS      ␣
↪1
15    16     LR-Lücke   101    ROHR  5114681686941855110  5630543731618051887␣
↪ 5630543731618051887              3                   1      0       G3      ␣
↪2
16    16     LR-Lücke   101    ROHR  4979507900871287244  5630543731618051887␣
↪ 5630543731618051887              4                   1      0       G4      ␣
↪2
17    16     LR-Lücke   101    VENT  5745097345184516675  5630543731618051887␣
↪ 5630543731618051887              5                   1      0       GR      ␣
↪2
18    18   LR-Flansch   101    VENT  5309992331398639768  5134530907542044265␣
↪ 5134530907542044265              1                   1      0       G1      ␣
↪1
19    18   LR-Flansch   101    ROHR  5244313507655010738  5134530907542044265␣
↪ 5134530907542044265              2                   1      0      GKS      ␣
↪1
20    18   LR-Flansch   101    VENT  5508684139418025293  5134530907542044265␣
↪ 5134530907542044265              3                   1      0      GKD      ␣
↪1
21    18   LR-Flansch   101    ROHR  5114681686941855110  5134530907542044265␣
↪ 5134530907542044265              4                   1      0       G3      ␣
↪1
22    18   LR-Flansch   101    ROHR  4979507900871287244  5134530907542044265␣
↪ 5134530907542044265              5                   1      0       G4      ␣
↪1
23    18   LR-Flansch   101    VENT  5745097345184516675  5134530907542044265␣
↪ 5134530907542044265              7                   1      0       GR      ␣
↪1
```

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```
24    20  LR-Parallel   101    VENT  5309992331398639768  4694969854935170169␣
↪ 4694969854935170169              1                  1      0       G1      ␣
↪1
25    20  LR-Parallel   101    ROHR  5244313507655010738  4694969854935170169␣
↪ 4694969854935170169              2                  1      0      GKS      ␣
↪1
26    20  LR-Parallel   101    VENT  5116489323526156845  4694969854935170169␣
↪ 4694969854935170169              3                  1      0      GKD      ␣
↪1
27    20  LR-Parallel   101    ROHR  5114681686941855110  4694969854935170169␣
↪ 4694969854935170169              5                  1      0       G3      ␣
↪1
28    20  LR-Parallel   101    ROHR  4979507900871287244  4694969854935170169␣
↪ 4694969854935170169              6                  1      0       G4      ␣
↪1
29    20  LR-Parallel   101    VENT  5745097345184516675  4694969854935170169␣
↪ 4694969854935170169              7                  1      0       GR      ␣
↪1
30    21          NEU  None    VENT  5309992331398639768                 PT3S␣
↪                PT3S              1                  1      0       G1      ␣
↪1
31    21          NEU  None    ROHR  5244313507655010738                 PT3S␣
↪                PT3S              2                  1      0      GKS      ␣
↪1
32    21          NEU  None    VENT  5116489323526156845                 PT3S␣
↪                PT3S              3                  1      0      GKD      ␣
↪1
33    21          NEU  None    ROHR  5114681686941855110                 PT3S␣
↪                PT3S              4                  1      0       G3      ␣
↪1
34    21          NEU  None    ROHR  4979507900871287244                 PT3S␣
↪                PT3S              5                  1      0       G4      ␣
↪1
35    21          NEU  None    VENT  5745097345184516675                 PT3S␣
↪                PT3S              6                  1      0       GR      ␣
↪1
>>> # Test if the same NAME is _not constructed twice ...
>>> xm.vAGSN_Add(nl=['GL','GR'])
>>> xm.dataFrames['vAGSN_raw'].shape
(36, 12)
>>> # test if re-use works without erors ...
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> # Test weight-Option ...
>>> xm.vAGSN_Add(nl=['GL','GR'],weight='QAbsInv',NAME='GL-GR w')   #␣
↪durchflussstaerksten Weg erzwingen
>>> df=xm.dataFrames['vAGSN_raw']
>>> df.query("LFDNR in [21,22] and nextNODE=='GKD'")
   LFDNR     NAME AKTIV OBJTYPE                OBJID    pk    tk ␣
↪nrObjIdInAgsn  nrObjIdTypeInAgsn  Layer nextNODE compNr
32    21      NEU  None    VENT  5116489323526156845  PT3S  PT3S             ␣
↪3                  1      0      GKD      1
38    22  GL-GR w  None    VENT  5508684139418025293  PT3S  PT3S             ␣
↪3                  1      0      GKD      1
```

**_vRART**()

One row per RART.

**Returns:**

**columns:**

**RART**

- NAME
- BESCHREIBUNG
- INDSTD_TXT
- INDSTD (numeric)
- DWDT

**RART_BZ**

- WSOSTD

**ID**

- pk

**References**

- NAME_KREF1
- NAME_KREF2
- NAME_SWVT
- [NAME_RCPL] - only if RCPLs exist

sequence: Model

**Raises:** XmError

**_OBJS** (*dfName*)

Decode a column OBJS (a BLOB containing a SIR 3S OBJ collection).

**Args:** dfName: Name of a dataFrame with column OBJS

**columns used (in self.dataFrames[dfName]):**

- OBJS (BLOB): i.e.: KNOT~4668229590574507160 . . .
- pk: ID (of the row)
- None is returned if these columns are missing
- in this case no changes in column OBJS in self.dataFrames[dfName]

**Returns:** column OBJS in self.dataFrames[dfName] is decoded to 'XXXX~' if OBJS was None

**dfOBJS: dataFrame with one row per OBJ in OBJS:**

**columns added (compared to self.dataFrames[dfName]):**

- OBJTYPE
- OBJID

**rows missing (compared to self.dataFrames[dfName]):**

- rows with OBJS None

**Raises:** XmError

**_vLFKT** ()

One row per Loadfactor Timeseries.

**Returns:**

**columns**

**LFKT**

- NAME
- BESCHREIBUNG
- INTPOL
- ZEITOPTION

**SERIES**

- LF: 1st Value
- LF_min
- LF_max

**LFKT ID**

- pk

**Raises:** XmError

**_vNRCV** ( )

One row per NRCV (NumeRiCal Value).

**Returns:**

**columns**

**ANNOTATIONS**

- cRefLfdNr

**CONT**

- CONT
- CONT_ID
- CONT_LFDNR

**DP**

**Datapointgroup**

- DPGR

**Datapoint**

- OBJTYPE
- fkOBJTYPE
- ATTRTYPE

**Datapoint IDs**

- pk_ROWS
- tk_ROWS

**NRCV IDs**

- pk

- tk

**PLot Coordinates**

- pXYLB: (X,Y): Left,Bottom

**Raises:** XmError

**_vGTXT** ()

One row per GTXT (Graphic TeXT).

**Returns:**

**columns**

**GTXT**

- GRAFTEXT (the text)

**CONT**

- CONT
- CONT_ID
- CONT_LFDNR

**GTXT IDs**

- pk
- tk

**PLot Coordinates**

- pXYLB: (X,Y): Left,Bottom

**Raises:** XmError

**_vSWVT** ()

One row per Timeseries.

**Returns:**

**columns**

**SWVT**

- NAME
- BESCHREIBUNG
- INTPOL
- ZEITOPTION

**SERIES**

- W: 1st Value
- W_min
- W_max

**SWVT ID**

- pk

**Raises:** XmError

**_vRUES** ()

One row per RUES (per Übergangssymbol).

1 Zeile für jede Definition(jeden Eingang). 1 Zeile für jede Referenz einer Definition (jeden Ausgang). Der Eingang muss nicht mit einem Signal versorgt sein. Der Ausgang muss nicht per Signal verwendet sein.

**Returns:**

**columns**

**RUES**

- IDUE: Name (eindeutige ID) der Übergangsstelle; bei IOTYP=3: IDUE kann offenbar undefiniert sein oder einen anderen nicht nachvollziehbaren Wert tragen; eingebbar in der GUI ist IDUE nur bei IOTYP=1
- IOTYP: 0=undefiniert|1=Eingang|3=Ausgang
- rkRUES: bei IOTYP=3: Verweis auf referenzierte Übergangsstelle; -1 sonst
- IDUE_rkRUES: bei IOTYP=3: Name (eindeutige ID) der referenzierten Übergangsstelle
- IOTYP_rkRUES: bei IOTYP=3: Typ der referenzierten Übergangsstelle
- Kn: Knotenname der RUES im Sinne eines Knoten-Kanten-Modells; IDUE bei IOTYP=1 und IDUE_rkRUES bei IOTYP=3

**CONT**

- CONT
- ID
- CONT_rkRUES: CONT der referenzierten Übergangsstelle
- ID_rkRUES: CONT ID der referenzierten Übergangsstelle

**RUES IDs**

- pk
- tk

**Raises:** XmError

```
>>> import pandas as pd
>>> # ---
>>> xm=xms['DHNetwork']
>>> # ---
>>> vRUES=xm.dataFrames['vRUES']
>>> vRUES.sort_values(by=['Kn','IOTYP'])[[ # pro Kn oten steht zuerst die␣
↪Definition, dann die Referenz(en)
...      'Kn'
...     ,'IOTYP'
...     ,'IDUE' # zur Kontrolle
...     ,'CONT'
...     ]]
               Kn IOTYP           IDUE                 CONT
28              0     1              0  Diverse Steuerungen
47              0     3     NTRx1xEin  Diverse Steuerungen
48              0     3     NTRx1xEin  Diverse Steuerungen
50              0     3     NTRx1xAus  Diverse Steuerungen
52              0     3     NTRx1xAus  Diverse Steuerungen
```

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```
54              0     3     NTRx3xAus  Diverse Steuerungen
56              0     3     NTRx3xAus  Diverse Steuerungen
58              0     3     NTRx3xEin  Diverse Steuerungen
60              0     3     NTRx3xEin  Diverse Steuerungen
62              0     3     NTRx2xAus  Diverse Steuerungen
64              0     3     NTRx2xAus  Diverse Steuerungen
66              0     3     NTRx2xEin  Diverse Steuerungen
68              0     3     NTRx2xEin  Diverse Steuerungen
105             0     3           NaN  Diverse Steuerungen
107             0     3           NaN  Diverse Steuerungen
109             0     3           NaN  Diverse Steuerungen
29              1     1             1  Diverse Steuerungen
40              1     3           NaN  Diverse Steuerungen
41              1     3           NaN  Diverse Steuerungen
42              1     3           NaN  Diverse Steuerungen
43              1     3           NaN  Diverse Steuerungen
44              1     3           NaN  Diverse Steuerungen
45              1     3           NaN  Diverse Steuerungen
80              1     3           NaN  Diverse Steuerungen
82              1     3           NaN  Diverse Steuerungen
83            100     1           100  Diverse Steuerungen
152           100     3           NaN        Sekundärwerte
155           100     3           NaN        Sekundärwerte
158           100     3           NaN        Sekundärwerte
30           ADum     1          ADum  Diverse Steuerungen
31           ADum     3           NaN  Diverse Steuerungen
32           ADum     3           NaN  Diverse Steuerungen
33           ADum     3           NaN  Diverse Steuerungen
46           ADum     3          ADum  Diverse Steuerungen
49           ADum     3          ADum  Diverse Steuerungen
51           ADum     3          ADum  Diverse Steuerungen
53           ADum     3          ADum  Diverse Steuerungen
55           ADum     3          ADum  Diverse Steuerungen
57           ADum     3          ADum  Diverse Steuerungen
59           ADum     3          ADum  Diverse Steuerungen
61           ADum     3          ADum  Diverse Steuerungen
63           ADum     3          ADum  Diverse Steuerungen
65           ADum     3          ADum  Diverse Steuerungen
67           ADum     3          ADum  Diverse Steuerungen
69           ADum     3          ADum  Diverse Steuerungen
86           ADum     3           NaN  Diverse Steuerungen
87           ADum     3           NaN  Diverse Steuerungen
90           ADum     3           NaN  Diverse Steuerungen
91           ADum     3           NaN  Diverse Steuerungen
96           ADum     3           NaN  Diverse Steuerungen
97           ADum     3           NaN  Diverse Steuerungen
98           ADum     3           NaN  Diverse Steuerungen
99           ADum     3           NaN  Diverse Steuerungen
104          ADum     3           NaN  Diverse Steuerungen
106          ADum     3           NaN  Diverse Steuerungen
108          ADum     3           NaN  Diverse Steuerungen
110          ADum     3           NaN  Diverse Steuerungen
114          ADum     3           NaN                    A
116          ADum     3           NaN                    A
122          ADum     3           NaN                    B
124          ADum     3           NaN                    B
130          ADum     3           NaN                    C
```

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```
131          ADum     3           NaN                    C
0      Leck_1_Ein     1    Leck_1_Ein    AGFW Symposium DH
84     Leck_1_Ein     3           NaN  Diverse Steuerungen
88     Leck_1_Ein     3           NaN  Diverse Steuerungen
1      Leck_2_Ein     1    Leck_2_Ein    AGFW Symposium DH
93     Leck_2_Ein     3           NaN  Diverse Steuerungen
94     Leck_2_Ein     3           NaN  Diverse Steuerungen
2      Leck_3_Ein     1    Leck_3_Ein    AGFW Symposium DH
101    Leck_3_Ein     3           NaN  Diverse Steuerungen
102    Leck_3_Ein     3           NaN  Diverse Steuerungen
25     Leck_Menge     1    Leck_Menge    AGFW Symposium DH
34     Leck_Menge     3           NaN  Diverse Steuerungen
35     Leck_Menge     3           NaN  Diverse Steuerungen
36     Leck_Menge     3           NaN  Diverse Steuerungen
37     Leck_Menge     3           NaN  Diverse Steuerungen
38     Leck_Menge     3           NaN  Diverse Steuerungen
39     Leck_Menge     3           NaN  Diverse Steuerungen
27        Leck_RL     1       Leck_RL    AGFW Symposium DH
89        Leck_RL     3           NaN  Diverse Steuerungen
92        Leck_RL     3           NaN  Diverse Steuerungen
100       Leck_RL     3           NaN  Diverse Steuerungen
26        Leck_VL     1       Leck_VL    AGFW Symposium DH
85        Leck_VL     3           NaN  Diverse Steuerungen
95        Leck_VL     3           NaN  Diverse Steuerungen
103       Leck_VL     3           NaN  Diverse Steuerungen
135        QDHGes     1        QDHGes        Sekundärwerte
146       dLastMW     1       dLastMW        Sekundärwerte
160       dUWMMax     1       dUWMMax        Sekundärwerte
159       dUWMMin     1       dUWMMin        Sekundärwerte
12            dpA     1           dpA    AGFW Symposium DH
117           dpA     3           NaN                    A
13             qB     1            qB    AGFW Symposium DH
125            qB     3           NaN                    B
14             qC     1            qC    AGFW Symposium DH
133            qC     3           NaN                    C
6        vorOrtNA     1      vorOrtNA    AGFW Symposium DH
15       vorOrtNA     3           NaN    AGFW Symposium DH
17       vorOrtNB     1      vorOrtNB    AGFW Symposium DH
18       vorOrtNB     3           NaN    AGFW Symposium DH
123      vorOrtNB     3           NaN                    B
126      vorOrtNB     3           NaN                    B
20       vorOrtNC     1      vorOrtNC    AGFW Symposium DH
21       vorOrtNC     3           NaN    AGFW Symposium DH
115      vorOrtNC     3           NaN                    A
118      vorOrtNC     3           NaN                    A
132      vorOrtNC     3           NaN                    C
134      vorOrtNC     3           NaN                    C
11    wDH_BA_A_MD     1   wDH_BA_A_MD    AGFW Symposium DH
78    wDH_BA_A_MD     3           NaN  Diverse Steuerungen
9     wDH_BA_A_RD     1   wDH_BA_A_RD    AGFW Symposium DH
10    wDH_BA_A_RD     3           NaN    AGFW Symposium DH
76    wDH_BA_A_RD     3           NaN  Diverse Steuerungen
8        wDH_MD_A     1      wDH_MD_A    AGFW Symposium DH
71       wDH_MD_A     3           NaN  Diverse Steuerungen
72   wDH_MD_A_ERO     1  wDH_MD_A_ERO  Diverse Steuerungen
77   wDH_MD_A_ERO     3           NaN  Diverse Steuerungen
7        wDH_RD_A     1      wDH_RD_A    AGFW Symposium DH
```

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```
73        wDH_RD_A     3           NaN  Diverse Steuerungen
74    wDH_RD_A_ERO     1  wDH_RD_A_ERO  Diverse Steuerungen
75    wDH_RD_A_ERO     3           NaN  Diverse Steuerungen
23           wLast     1         wLast    AGFW Symposium DH
79           wLast     3           NaN  Diverse Steuerungen
142        wLastMW     1       wLastMW        Sekundärwerte
145        wLastMW     3           NaN        Sekundärwerte
3              wNA     1           wNA    AGFW Symposium DH
112            wNA     3           NaN                    A
16          wNAEin     1        wNAEin    AGFW Symposium DH
111         wNAEin     3           NaN                    A
4              wNB     1           wNB    AGFW Symposium DH
120            wNB     3           NaN                    B
19          wNBEin     1        wNBEin    AGFW Symposium DH
113         wNBEin     3           NaN                    A
119         wNBEin     3           NaN                    B
121         wNBEin     3           NaN                    B
5              wNC     1           wNC    AGFW Symposium DH
128            wNC     3           NaN                    C
22          wNCEin     1        wNCEin    AGFW Symposium DH
127         wNCEin     3           NaN                    C
129         wNCEin     3           NaN                    C
24           wTRST     1         wTRST    AGFW Symposium DH
81           wTRST     3           NaN  Diverse Steuerungen
147           yAMW     1          yAMW        Sekundärwerte
151           yAMW     3           NaN        Sekundärwerte
148           yBMW     1          yBMW        Sekundärwerte
154           yBMW     3           NaN        Sekundärwerte
149           yCMW     1          yCMW        Sekundärwerte
157           yCMW     3           NaN        Sekundärwerte
136      yDH_dp2_A     1     yDH_dp2_A        Sekundärwerte
70       yDH_dp2_A     3           NaN  Diverse Steuerungen
137      yDH_dp2_A     3           NaN        Sekundärwerte
138      yDH_pMD_A     1     yDH_pMD_A        Sekundärwerte
139      yDH_pRL_A     1     yDH_pRL_A        Sekundärwerte
140      yDH_pRL_A     3           NaN        Sekundärwerte
143        yLastMW     1       yLastMW        Sekundärwerte
144        yLastMW     3           NaN        Sekundärwerte
150        yLastMW     3           NaN        Sekundärwerte
153        yLastMW     3           NaN        Sekundärwerte
156        yLastMW     3           NaN        Sekundärwerte
141           yUWM     1          yUWM        Sekundärwerte
162           yUWM     3           NaN        Sekundärwerte
161       yUWMLast     1      yUWMLast        Sekundärwerte
>>> # ---
>>> vRUESDefs=vRUES.loc[vRUES['IOTYP']=='1']
>>> # für Defs die Originaldefinition finden ...
>>> vRUESDefsCrgl=pd.merge(vRUESDefs,xm.dataFrames['CRGL'],left_on='pk',right_
↪on='fkKk',suffixes=('','_CRGL'),how='left') # für alle sollte eine Referenz␣
↪gefunden werden ...
>>> vRUESDefsCrgl.sort_values(by=['Kn'])[[
...      'Kn'
...     ,'CONT'
...     ,'fkKi'
...     ]]
              Kn                 CONT                  fkKi
24             0  Diverse Steuerungen  5486870913514090048
```

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```
25             1  Diverse Steuerungen  5377084992102722959
29           100  Diverse Steuerungen  5055797784689898209
26          ADum  Diverse Steuerungen  5408457159782566744
0     Leck_1_Ein    AGFW Symposium DH  5706111677806224290
1     Leck_2_Ein    AGFW Symposium DH  4704869532416514405
2     Leck_3_Ein    AGFW Symposium DH  4808434710442736644
21    Leck_Menge    AGFW Symposium DH  5390061625789905096
23       Leck_RL    AGFW Symposium DH  5644481773793849108
22       Leck_VL    AGFW Symposium DH  4880440884169110259
30        QDHGes        Sekundärwerte  5345716897595312355
37       dLastMW        Sekundärwerte  4611793887272861500
42       dUWMMax        Sekundärwerte  4672771372882677276
41       dUWMMin        Sekundärwerte  5463544828758888616
11           dpA    AGFW Symposium DH  4849866990207957614
12            qB    AGFW Symposium DH  4771725364091629759
13            qC    AGFW Symposium DH  4978409087288292434
6       vorOrtNA    AGFW Symposium DH  5194343043762135519
15      vorOrtNB    AGFW Symposium DH  4705080808435797677
17      vorOrtNC    AGFW Symposium DH  5620348872583735825
10   wDH_BA_A_MD    AGFW Symposium DH  4873987359791313088
9    wDH_BA_A_RD    AGFW Symposium DH  5322890886142492590
8       wDH_MD_A    AGFW Symposium DH  5093705160009582980
27  wDH_MD_A_ERO  Diverse Steuerungen  5729434727271745948
7       wDH_RD_A    AGFW Symposium DH  4622192786925004485
28  wDH_RD_A_ERO  Diverse Steuerungen  4980847179402621205
19         wLast    AGFW Symposium DH  5741660563170722352
35       wLastMW        Sekundärwerte  4833634373103605497
3            wNA    AGFW Symposium DH  4991855568438544033
14        wNAEin    AGFW Symposium DH  4742316320267545359
4            wNB    AGFW Symposium DH  4658075570394029953
16        wNBEin    AGFW Symposium DH  5013654033692161674
5            wNC    AGFW Symposium DH  5240575308071562858
18        wNCEin    AGFW Symposium DH  5670691593026035398
20         wTRST    AGFW Symposium DH  5547011912763631199
38          yAMW        Sekundärwerte  4726758453134789052
39          yBMW        Sekundärwerte  5528896084200811302
40          yCMW        Sekundärwerte  5274276049082272588
31     yDH_dp2_A        Sekundärwerte  5512879293670562022
32     yDH_pMD_A        Sekundärwerte  5255402486218254174
33     yDH_pRL_A        Sekundärwerte  4639451967914783278
36       yLastMW        Sekundärwerte  4817923247686815456
34          yUWM        Sekundärwerte  5008805081156446169
43      yUWMLast        Sekundärwerte  5574611204646558662
>>> vRUESDefsCrglRuesDef=pd.merge(vRUESDefsCrgl,vRUES,left_on='fkKi',right_on=
↪'pk',suffixes=('','_vRUES'),how='inner') # für die RUES-definierten RUES␣
↪sollte eine Referenz gefunden werden ...
>>> vRUESDefsCrglRuesDef.sort_values(by=['Kn'])[[
...      'Kn'
...     ,'CONT'
...     ,'fkKi'
...     ,'Kn_vRUES'
...     ]]
            Kn                 CONT                 fkKi  Kn_vRUES
0  wDH_RD_A_ERO  Diverse Steuerungen  4980847179402621205  wDH_RD_A
```

**_vRSLW**(*vSWVT=None*)

One row per RSLW.

**Args:**

- vSWVT

**Returns:**

**columns**

**RSLW**

- KA
- BESCHREIBUNG
- INDWBG
- WMIN
- WMAX
- INDWNO

**RSLW BZ**

- INDSLW
- SLWKON

**CONT**

- CONT
- ID
- CONT_PARENT

**SWVT**

- SWVT
- SWVT_Count (Anzahl der RSLW-Referenzierungen der SWVT; 0, wenn keine SWVT angegeben; INDSLW wird bei der Ermittlung nicht ausgewertet)
- BESCHREIBUNG_SWVT
- INTPOL
- ZEITOPTION

**SERIES**

- W: 1st Value
- W_min
- W_max

**RSLW IDs**

- pk
- tk

**Raises:** XmError

```
>>> import pandas as pd
>>> # ---
>>> xm=xms['DHNetwork']
>>> # ---
>>> vRSLW=xm.dataFrames['vRSLW']
>>> vRSLW[[
...  'KA'
... ,'BESCHREIBUNG'
... ,'INDSLW'
... ,'CONT'
... ,'CONT_PARENT'
... ,'SWVT'
... ]].sort_values(by=['KA'])
             KA          BESCHREIBUNG INDSLW                 CONT
↪CONT_PARENT        SWVT
20            0                  None      0  Diverse Steuerungen    AGFW␣
↪Symposium DH         NaN
21            1                  None      0  Diverse Steuerungen    AGFW␣
↪Symposium DH         NaN
23          100                  None      0  Diverse Steuerungen    AGFW␣
↪Symposium DH         NaN
22         ADum           Analog Dummy      0  Diverse Steuerungen    AGFW␣
↪Symposium DH         NaN
0    Leck_1_Ein            Leck_1_Ein      1    AGFW Symposium DH    AGFW␣
↪Symposium DH      zLeck1
1    Leck_2_Ein            Leck_2_Ein      1    AGFW Symposium DH    AGFW␣
↪Symposium DH      zLeck2
2    Leck_3_Ein            Leck_3_Ein      1    AGFW Symposium DH    AGFW␣
↪Symposium DH      zLeck3
11   Leck_Menge            Leck_Menge      1    AGFW Symposium DH    AGFW␣
↪Symposium DH  zLeckMenge
13      Leck_RL               Leck_RL      1    AGFW Symposium DH    AGFW␣
↪Symposium DH    zLeck_RL
12      Leck_VL               Leck_VL      1    AGFW Symposium DH    AGFW␣
↪Symposium DH    zLeck_VL
24           cp                   NaN      0        Sekundärwerte  Diverse␣
↪Steuerungen         NaN
17          dpA                   dpA      1    AGFW Symposium DH    AGFW␣
↪Symposium DH         dpA
18           qB                    qB      1    AGFW Symposium DH    AGFW␣
↪Symposium DH          qB
19           qC                    qC      1    AGFW Symposium DH    AGFW␣
↪Symposium DH          qC
6      vorOrtNA              vorOrtNA      1    AGFW Symposium DH    AGFW␣
↪Symposium DH    vorOrtNA
7      vorOrtNB              vorOrtNB      1    AGFW Symposium DH    AGFW␣
↪Symposium DH    vorOrtNB
8      vorOrtNC              vorOrtNC      1    AGFW Symposium DH    AGFW␣
↪Symposium DH    vorOrtNC
16  wDH_BA_A_RD  wDH_BA_A; 1=RD; 0=MD      1    AGFW Symposium DH    AGFW␣
↪Symposium DH    wDH_BA_A
15     wDH_MD_A              wDH_MD_A      1    AGFW Symposium DH    AGFW␣
↪Symposium DH    wDH_MD_A
14     wDH_RD_A              wDH_RD_A      1    AGFW Symposium DH    AGFW␣
↪Symposium DH    wDH_RD_A
9         wLast                 wLast      1    AGFW Symposium DH    AGFW␣
↪Symposium DH       wLast
```

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```
3            wNA                    wNA       1    AGFW Symposium DH    AGFW␣
↪Symposium DH         wNA
4            wNB                    wNB       1    AGFW Symposium DH    AGFW␣
↪Symposium DH         wNB
5            wNC                    wNC       1    AGFW Symposium DH    AGFW␣
↪Symposium DH         wNC
10         wTRST                  wTRST       1    AGFW Symposium DH    AGFW␣
↪Symposium DH       wTRSP
```

**_vRSTN** (*vSWVT=None*)

One row per RSTN.

**Returns:** columns: see code

**Raises:** XmError

```
>>> import pandas as pd
>>> # ---
>>> xm=xms['DHNetwork']
>>> # ---
>>> vRSTN=xm.dataFrames['vRSTN']
>>> vRSTN['RART_TYP']=vRSTN['RART_TYP'].str[:10]+'...' # zu lange␣
↪Ausgabezeile vermeiden
>>> vRSTN[[
...  'CONT'
... ,'CONT_PARENT'
... ,'KA'
... ,'BESCHREIBUNG'
... ,'ITYP_OBJTYPE'
... ,'ITYP_OBJATTR'
... ,'Chk'
... ,'ik_Chk'
... ,'OBJTYPE'
... ,'NAME_i'
... ,'NAME_k'
... ,'CONT_i'
... ,'TABL_Chk'
... ,'TABL'
... ,'KNOT'
... ,'RART'
... ,'RART_TYP'
... ,'RARTPG'
... ,'RCPL'
... ,'RCPL_KNOT1'
... ,'RCPL_KNOT2'
... ,'NAME_i_PUMP'
... ,'NAME_k_PUMP'
... ]].sort_values(by=['ITYP_OBJTYPE','ITYP_OBJATTR','CONT','KA'])
                   CONT        CONT_PARENT        KA    BESCHREIBUNG ITYP_
↪OBJTYPE ITYP_OBJATTR  Chk  ik_Chk OBJTYPE  NAME_i    NAME_k            ␣
↪CONT_i  TABL_Chk  TABL       KNOT     RART       RART_TYP  RARTPG      RCPL␣
↪RCPL_KNOT1 RCPL_KNOT2 NAME_i_PUMP NAME_k_PUMP
44  Diverse Steuerungen  AGFW Symposium DH   KA-0054             NaN       ␣
↪KNOT        PSOLL    1     NaN     NaN     NaN       NaN                NaN␣
↪      NaN   NaN  A_DH_pDef      NaN            NaN     NaN       NaN      ␣
↪NaN        NaN         NaN         NaN
45  Diverse Steuerungen  AGFW Symposium DH   KA-0055             NaN       ␣
↪KNOT        PSOLL    1     NaN     NaN     NaN       NaN                NaN␣
↪      NaN   NaN  A_DH_pDef      NaN            NaN     NaN       NaN      ␣
↪NaN        NaN         NaN         NaN
```

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```
12  Diverse Steuerungen  AGFW Symposium DH   KA-0004              NaN
↪LFKT          SOLL    1     NaN     NaN     NaN       NaN                NaN␣
↪      1.0  LFKT        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
2                     A  AGFW Symposium DH   KA-0045              NaN
↪PGRP        AKTIV    1     1.0    PGRP  R-A-SS  R-A-DS-2                A␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
6                     B  AGFW Symposium DH   KA-0057              NaN
↪PGRP        AKTIV    1     1.0    PGRP  R-B-SS  R-B-DS-2                B␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
10                    C  AGFW Symposium DH   KA-0060              NaN
↪PGRP        AKTIV    1     1.0    PGRP  R-C-SS  R-C-DS-2                C␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
1                     A  AGFW Symposium DH   KA-0044              NaN
↪PGRP        DEAKT    1     1.0    PGRP  R-A-SS  R-A-DS-2                A␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
5                     B  AGFW Symposium DH   KA-0053              NaN
↪PGRP        DEAKT    1     1.0    PGRP  R-B-SS  R-B-DS-2                B␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
9                     C  AGFW Symposium DH   KA-0059              NaN
↪PGRP        DEAKT    1     1.0    PGRP  R-C-SS  R-C-DS-2                C␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
48  Diverse Steuerungen  AGFW Symposium DH   KA-0050  Test ohne Fkt.
↪PGRP        PUDEA    1     1.0    PGRP  R-A-SS  R-A-DS-2                A␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN      R-A-SS      R-A-DS
47  Diverse Steuerungen  AGFW Symposium DH   KA-0049  Test ohne Fkt.
↪PGRP         RART    1     1.0    PGRP  R-A-SS  R-A-DS-2                A␣
↪      NaN   NaN        NaN      NaN            NaN  A_dpdS       NaN
↪NaN        NaN         NaN         NaN
0                     A  AGFW Symposium DH  wNA_RSTN              NaN
↪PUMP            N    1     1.0    PUMP  R-A-SS    R-A-DS                A␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
4                     B  AGFW Symposium DH  wNB_RSTN              NaN
↪PUMP            N    1     1.0    PUMP  R-B-SS    R-B-DS                B␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
8                     C  AGFW Symposium DH  wNC_RSTN              NaN
↪PUMP            N    1     1.0    PUMP  R-C-SS    R-C-DS                C␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
3                     A  AGFW Symposium DH   KA-0046              NaN
↪RART          SOLL    1     NaN     NaN     NaN       NaN                NaN␣
↪      NaN   NaN        NaN   A_dpdS  Differenzd...     NaN       NaN
↪NaN        NaN         NaN         NaN
7                     B  AGFW Symposium DH   KA-0058              NaN
↪RART          SOLL    1     NaN     NaN     NaN       NaN                NaN␣
↪      NaN   NaN        NaN  B_Menge  Durchfluss...     NaN       NaN
↪NaN        NaN         NaN         NaN
11                    C  AGFW Symposium DH   KA-0061              NaN
↪RART          SOLL    1     NaN     NaN     NaN       NaN                NaN␣
↪      NaN   NaN        NaN  C_Menge  Durchfluss...     NaN       NaN
↪NaN        NaN         NaN         NaN
```

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```
46  Diverse Steuerungen  AGFW Symposium DH   KA-0048  Test ohne Fkt.
↪RCPL         ROWT    1     NaN     NaN     NaN       NaN                NaN
↪      NaN   NaN        NaN      NaN            NaN     NaN  RCPL_ADS     V-
↪1131     R-1131         NaN         NaN
34  Diverse Steuerungen  AGFW Symposium DH   KA-0034    NTR_1_VL_Ein
↪ROHR          AUF    1     1.0    ROHR  V-1905    V-1906  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
35  Diverse Steuerungen  AGFW Symposium DH   KA-0035    NTR_1_RL_Ein
↪ROHR          AUF    1     1.0    ROHR  R-1905    R-1906  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
36  Diverse Steuerungen  AGFW Symposium DH   KA-0036    NTR_3_Aus_VL
↪ROHR          AUF    1     1.0    ROHR  V-3008    V-3007  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
37  Diverse Steuerungen  AGFW Symposium DH   KA-0037    NTR_3_Aus_RL
↪ROHR          AUF    1     1.0    ROHR  R-3008    R-3007  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
40  Diverse Steuerungen  AGFW Symposium DH   KA-0040    NTR_2_Aus_VL
↪ROHR          AUF    1     1.0    ROHR  V-1110    V-1111  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
41  Diverse Steuerungen  AGFW Symposium DH   KA-0041    NTR_2_Aus_RL
↪ROHR          AUF    1     1.0    ROHR  R-1110    R-1111  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
17  Diverse Steuerungen  AGFW Symposium DH   KA-0007             NaN
↪ROHR      LECKAUS    1     1.0    ROHR  R-1905    R-1906  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
15  Diverse Steuerungen  AGFW Symposium DH   KA-0008             NaN
↪ROHR      LECKAUS    1     1.0    ROHR  V-1905    V-1906  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
20  Diverse Steuerungen  AGFW Symposium DH   KA-0015             NaN
↪ROHR      LECKAUS    1     1.0    ROHR  V-1110    V-1111  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
21  Diverse Steuerungen  AGFW Symposium DH   KA-0016             NaN
↪ROHR      LECKAUS    1     1.0    ROHR  R-1110    R-1111  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
24  Diverse Steuerungen  AGFW Symposium DH   KA-0023             NaN
↪ROHR      LECKAUS    1     1.0    ROHR  V-3008    V-3007  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
25  Diverse Steuerungen  AGFW Symposium DH   KA-0024             NaN
↪ROHR      LECKAUS    1     1.0    ROHR  R-3008    R-3007  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
16  Diverse Steuerungen  AGFW Symposium DH   KA-0003             NaN
↪ROHR      LECKEIN    1     1.0    ROHR  R-1905    R-1906  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
14  Diverse Steuerungen  AGFW Symposium DH   KA-0006             NaN
↪ROHR      LECKEIN    1     1.0    ROHR  V-1905    V-1906  AGFW Symposium DH
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
```

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```
18  Diverse Steuerungen  AGFW Symposium DH   KA-0013             NaN
↪ROHR      LECKEIN    1     1.0    ROHR  V-1110    V-1111  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
19  Diverse Steuerungen  AGFW Symposium DH   KA-0014             NaN
↪ROHR      LECKEIN    1     1.0    ROHR  R-1110    R-1111  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
22  Diverse Steuerungen  AGFW Symposium DH   KA-0021             NaN
↪ROHR      LECKEIN    1     1.0    ROHR  V-3008    V-3007  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
23  Diverse Steuerungen  AGFW Symposium DH   KA-0022             NaN
↪ROHR      LECKEIN    1     1.0    ROHR  R-3008    R-3007  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
26  Diverse Steuerungen  AGFW Symposium DH   KA-0025             NaN
↪ROHR    LECKMENGE    1     1.0    ROHR  V-1905    V-1906  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
27  Diverse Steuerungen  AGFW Symposium DH   KA-0027             NaN
↪ROHR    LECKMENGE    1     1.0    ROHR  R-1905    R-1906  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
28  Diverse Steuerungen  AGFW Symposium DH   KA-0028             NaN
↪ROHR    LECKMENGE    1     1.0    ROHR  V-1110    V-1111  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
29  Diverse Steuerungen  AGFW Symposium DH   KA-0029             NaN
↪ROHR    LECKMENGE    1     1.0    ROHR  R-1110    R-1111  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
30  Diverse Steuerungen  AGFW Symposium DH   KA-0030             NaN
↪ROHR    LECKMENGE    1     1.0    ROHR  V-3008    V-3007  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
31  Diverse Steuerungen  AGFW Symposium DH   KA-0031             NaN
↪ROHR    LECKMENGE    1     1.0    ROHR  R-3008    R-3007  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
32  Diverse Steuerungen  AGFW Symposium DH   KA-0032    NTR_1_RL_Ein
↪ROHR           ZU    1     1.0    ROHR  R-1905    R-1906  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
33  Diverse Steuerungen  AGFW Symposium DH   KA-0033    NTR_1_VL_Ein
↪ROHR           ZU    1     1.0    ROHR  V-1905    V-1906  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
38  Diverse Steuerungen  AGFW Symposium DH   KA-0038    NTR_3_Ein_VL
↪ROHR           ZU    1     1.0    ROHR  V-3008    V-3007  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
39  Diverse Steuerungen  AGFW Symposium DH   KA-0039    NTR_3_Ein_RL
↪ROHR           ZU    1     1.0    ROHR  R-3008    R-3007  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
42  Diverse Steuerungen  AGFW Symposium DH   KA-0042    NTR_2_Ein_VL
↪ROHR           ZU    1     1.0    ROHR  V-1110    V-1111  AGFW Symposium DH␣
↪      NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
```

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```
43  Diverse Steuerungen  AGFW Symposium DH   KA-0043    NTR_2_Ein_RL
↪ROHR           ZU    1     1.0    ROHR  R-1110    R-1111  AGFW Symposium DH␣
↪       NaN   NaN        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
13  Diverse Steuerungen  AGFW Symposium DH   KA-0005             NaN
↪TEVT          SOLL    1     NaN     NaN     NaN       NaN                NaN␣
↪       2.0  TRST        NaN      NaN            NaN     NaN       NaN
↪NaN        NaN         NaN         NaN
```

**_vQVAR**()

One row per Timeseries.

**Returns:**

**columns**

**QVAR**

- NAME
- BESCHREIBUNG
- INTPOL
- ZEITOPTION

**SERIES**

- QM: 1st Value
- QM_min
- QM_max

**QVAR ID**

- pk

**Raises:** XmError

**_vPVAR**()

One row per Timeseries.

**Returns:**

**columns**

**PVAR**

- NAME
- BESCHREIBUNG
- INTPOL
- ZEITOPTION

**SERIES**

- PH: 1st Value
- PH_min
- PH_max

**QVAR ID**

- pk

**Raises:** XmError

**_vVKNO**()

One row per Blocknode.

**Returns:**

**columns**

- NAME
- CONT (Blockname)
- fkKNOT
- fkCONT

**Raises:** XmError

**_vKNOT**(*vVKNO=None*, *vQVAR=None*, *vPVAR=None*, *vLFKT=None*)

One row per Node (KNOT).

**Args:**

- vVKNO
- vQVAR
- vPVAR
- vLFKT

**Returns:**

**rows** sequence: Xml

**columns**

**KNOT**

- NAME
- BESCHREIBUNG
- IDREFERENZ

**CONT**

- CONT
- CONT_ID
- CONT_LFDNR
- CONT_VKNO (name of the Block/Container for NAME is a Blocknode)

**KNOT**

- KTYP
- LFAKT (Umrechnungsfaktor)
- QM_EIN

**QVAR**

- QVAR_NAME
- QM, QM_min, QM_max

**LFKT**

- LFKT_NAME
- LF, LF_min, LF_max

**PVAR**

- PVAR_NAME
- PH, PH_min, PH_max

**Zugehoerigkeit**

- PZON_NAME
- FSTF_NAME,STOF_NAME,GMIX_NAME
- UTMP_NAME

**2L**

- 2L_NAME
- 2L_KVR

**KNOT**

- KVR
- TE
- TM

**KNOT**

- XKOR, YKOR, ZKOR

**KNOT IDs**

- pk, tk

**KNOT: plot-Coordinates**

- pXCor: X-pXCorZero
- pYCor: Y-pYCorZero

**Raises:** XmError

**_vROHR**(*vKNOT=None*)

One row per Pipe (ROHR).

**Args:** vKNOT

**Returns:**

**rows** sequence: Xml

**columns**

**ROHR**

- BESCHREIBUNG
- IDREFERENZ

**ROHR**

- BAUJAHR, HAL

- IPLANUNG, KENNUNG

**ROHR**

- L, LZU, RAU, ZAUS, ZEIN, ZUML
- JLAMBS, LAMBDA0

**ROHR**

- ASOLL, INDSCHALL

**ROHR FW**

- NAME_i_2L
- NAME_k_2L
- KVR

**DTRO_ROWD**

- AUSFALLZEIT, DA , DI , DN , KT , PN , REHABILITATION , REPARATUR , S , WSTEIG , WTIEFE

**LTGR**

- LTGR_NAME, LTGR_BESCHREIBUNG , SICHTBARKEIT , VERLEGEART

**DTRO**

- DTRO_NAME, DTRO_BESCHREIBUNG, E

**REF**

- fkSTRASSE, fkSRAT

**ROHR IDs**

- pk, tk

**ROHR BZ**

- ITRENN
- LECKSTART, LECKEND, LECKMENGE, LECKORT, LECKSTATUS

**Rest**

- QSVB, ZVLIMPTNZ, KANTENZV

**CONT**

- CONT
- CONT_ID
- CONT_LFDNR

**vKNOT**

**KI**

- NAME_i
- KVR_i, TM_i
- XKOR_i, YKOR_i, ZKOR_i

**KK**

- NAME_k
- KVR_k, TM_k
- XKOR_k, YKOR_k, ZKOR_k

pXCor_i, pYCor_i pXCor_k, pYCor_k

**PLOT**

- pXCors, pYCors
- pWAYPXCors, pWAYPYCors

**Raises:** XmError

**_vFWVB**(*vKNOT=None*, *vLFKT=None*, *vWBLZ=None*)

One row per DistrictHeatingHousestation (FWVB).

**Args:**

- vKNOT
- vLFKT
- wWBLZ

**Returns:**

**columns**

**FWVB**

- BESCHREIBUNG
- IDREFERENZ
- W0
- LFK
- W0LFK
- TVL0, TRS0

**vLFKT**

- LFKT
- W, W_min, W_max

**FWVB contd.**

- INDTR, TRSK
- VTYP
- DPHAUS, IMBG, IRFV

**FWVB IDs**

- pk, tk

**vKNOT**

**Ki**

- NAME_i
- KVR_i, TM_i

- XKOR_i, YKOR_i, ZKOR_i
- pXCor_i, pYCor_i

**Kk**

- NAME_k
- KVR_k, TM_i
- XKOR_k, YKOR_k, ZKOR_i
- pXCor_k, pYCor_i

**vCONT**

- CONT
- CONT_ID
- CONT_LFDNR

**vWBLZ**

- **['BLZ1','BLZ2',...]** list of the WBLZ-Names of the FWVB in alphabetical Order; empty list, if FWVB is not a WBLZ-Member

**Raises:** XmError

**_vVBEL**(*vKNOT=None, edges=['ROHR', 'VENT', 'FWVB', 'FWES', 'PUMP', 'KLAP', 'REGV', 'PREG', 'MREG', 'DPRG', 'PGRP'], edgesD=['', 'DN', '', 'DN', '', 'DN', 'DN', 'DN', 'DN', 'DN', ''], mColNames=['OBJTYPE', 'pk'], mIdxNames=['OBJTYPE', 'OBJID']*)

One row per Edge.

**Args:**

- vKNOT: df
- edges: list of strs
- edgesD: list of strs
- mColNames: list of columns which shall be used as MIndex; the columns will be droped; the columns must be delivered by _vVBEL_XXXX
- mIdxNames: list of names for the indices for the columns above

**Returns:** Edge-df returned Edge-df is None if an exception occurs

**rows:**

- sequence edges: edges
- sequence within edges: Xml

**Mindices:**

- OBJTYPE: str: 'ROHR','VENT',... [default a MIndex not a column]
- OBJID [default a MIndex not a column]

**columns:**

- LAYR
- L in m (0 if edge <> ROHR)
- D in mm (NaN if no Diameter could be determined)

**columns:**

- see _vVBEL_XXXX

**_vVBEL_XXXX** (*vKNOT=None*, *OBJTYPE=None*)

One row per Edge.

**Args:**

- vKNOT: df
- **OBJTYPE: str ('ROHR','VENT',...)** self.dataFrames[OBJTYPE] is used to build with vKNOT the returned Edge-df

**Returns:** Edge-df None is returned if an exception occurs

**columns:**

- OBJTYPE: str: ROHR,VENT,...
- BESCHREIBUNG
- IDREFERENZ
- pk
- tk
- NAME_i
- CONT_i
- CONT_VKNO_i
- Z_i
- pk_i
- NAME_k
- CONT_k
- CONT_VKNO_k
- Z_k
- pk_k

**_vRXXX** (*nodes=['RSLW', 'RMES', 'RHYS', 'RLVG', 'RLSR', 'RMMA', 'RADD', 'RMUL', 'RDIV', 'RTOT', 'RPT1', 'RINT', 'RPID', 'RFKT', 'RSTN']*)

One row per R-Node.

**Args:**

- nodes: List of all R-Node Elements but RUES

**Returns:** R-Node df returned R-Node df is None if an exception occurs

**rows:**

- sequence nodes: nodes
- sequence within nodes: Xml

**columns:**

- see _vRXXX_XXXX

```
>>> import pandas as pd
>>> # ---
>>> xm=xms['DHNetwork']
>>> # ---
>>> vRXXX=xm.dataFrames['vRXXX']
>>> vRXXX
   OBJTYPE          BESCHREIBUNG            KA                  CONT
↪        pk                   tk
0     RSLW            Leck_1_Ein    Leck_1_Ein    AGFW Symposium DH
↪5706111677806224290  5706111677806224290
1     RSLW            Leck_2_Ein    Leck_2_Ein    AGFW Symposium DH
↪4704869532416514405  4704869532416514405
2     RSLW            Leck_3_Ein    Leck_3_Ein    AGFW Symposium DH
↪4808434710442736644  4808434710442736644
3     RSLW                   wNA           wNA    AGFW Symposium DH
↪4991855568438544033  4991855568438544033
4     RSLW                   wNB           wNB    AGFW Symposium DH
↪4658075570394029953  4658075570394029953
5     RSLW                   wNC           wNC    AGFW Symposium DH
↪5240575308071562858  5240575308071562858
6     RSLW              vorOrtNA      vorOrtNA    AGFW Symposium DH
↪5194343043762135519  5194343043762135519
7     RSLW              vorOrtNB      vorOrtNB    AGFW Symposium DH
↪4705080808435797677  4705080808435797677
8     RSLW              vorOrtNC      vorOrtNC    AGFW Symposium DH
↪5620348872583735825  5620348872583735825
9     RSLW                  wLast         wLast    AGFW Symposium DH
↪5741660563170722352  5741660563170722352
10    RSLW                  wTRST         wTRST    AGFW Symposium DH
↪5547011912763631199  5547011912763631199
11    RSLW            Leck_Menge    Leck_Menge    AGFW Symposium DH
↪5390061625789905096  5390061625789905096
12    RSLW               Leck_VL       Leck_VL    AGFW Symposium DH
↪4880440884169110259  4880440884169110259
13    RSLW               Leck_RL       Leck_RL    AGFW Symposium DH
↪5644481773793849108  5644481773793849108
14    RSLW              wDH_RD_A      wDH_RD_A    AGFW Symposium DH
↪4622192786925004485  4622192786925004485
15    RSLW              wDH_MD_A      wDH_MD_A    AGFW Symposium DH
↪5093705160009582980  5093705160009582980
16    RSLW  wDH_BA_A; 1=RD; 0=MD   wDH_BA_A_RD    AGFW Symposium DH
↪5322890886142492590  5322890886142492590
17    RSLW                   dpA           dpA    AGFW Symposium DH
↪4849866990207957614  4849866990207957614
18    RSLW                    qB            qB    AGFW Symposium DH
↪4771725364091629759  4771725364091629759
19    RSLW                    qC            qC    AGFW Symposium DH
↪4978409087288292434  4978409087288292434
20    RSLW                  None             0  Diverse Steuerungen
↪5486870913514090048  5486870913514090048
21    RSLW                  None             1  Diverse Steuerungen
↪5377084992102722959  5377084992102722959
22    RSLW           Analog Dummy          ADum  Diverse Steuerungen
↪5408457159782566744  5408457159782566744
23    RSLW                  None           100  Diverse Steuerungen
↪5055797784689898209  5055797784689898209
24    RSLW                   NaN            cp        Sekundärwerte
↪4838608935279518502  4838608935279518502
```

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```
0     RMES                    NaN          yUWM        Sekundärwerte ␣
↪5008805081156446169  5008805081156446169
1     RMES                    NaN            mP        Sekundärwerte ␣
↪5180980864512333141  5180980864512333141
2     RMES                    NaN          TRSP        Sekundärwerte ␣
↪4964809001779537631  4964809001779537631
3     RMES                    NaN           TVL        Sekundärwerte ␣
↪5137355888694407298  5137355888694407298
4     RMES                    NaN       wLastMW        Sekundärwerte ␣
↪4833634373103605497  4833634373103605497
5     RMES                    NaN       yLastMW        Sekundärwerte ␣
↪4817923247686815456  4817923247686815456
6     RMES                    NaN          yAMW        Sekundärwerte ␣
↪4726758453134789052  4726758453134789052
7     RMES                    NaN          yBMW        Sekundärwerte ␣
↪5528896084200811302  5528896084200811302
8     RMES                    NaN          yCMW        Sekundärwerte ␣
↪5274276049082272588  5274276049082272588
9     RMES                    NaN       dUWMMin        Sekundärwerte ␣
↪5463544828758888616  5463544828758888616
10    RMES                    NaN       dUWMMax        Sekundärwerte ␣
↪4672771372882677276  4672771372882677276
11    RMES                    NaN       KA-0026        Sekundärwerte ␣
↪5714273708462554381  5714273708462554381
12    RMES                    NaN        QDHGes        Sekundärwerte ␣
↪5345716897595312355  5345716897595312355
13    RMES                  dp / 2     yDH_dp2_A        Sekundärwerte ␣
↪5512879293670562022  5512879293670562022
14    RMES                   None     yDH_pRL_A        Sekundärwerte ␣
↪4639451967914783278  4639451967914783278
0     RLVG                    NaN     wNAEin_vO    AGFW Symposium DH ␣
↪4742316320267545359  4742316320267545359
1     RLVG                    NaN     wNBEin_vO    AGFW Symposium DH ␣
↪5013654033692161674  5013654033692161674
2     RLVG                    NaN     wNCEin_vO    AGFW Symposium DH ␣
↪5670691593026035398  5670691593026035398
3     RLVG                    NaN   wDH_BA_A_MD    AGFW Symposium DH ␣
↪4873987359791313088  4873987359791313088
4     RLVG                    NaN     Leck_1_VL  Diverse Steuerungen ␣
↪5669152199869266879  5669152199869266879
5     RLVG                    NaN    nLeck_1_VL  Diverse Steuerungen ␣
↪5517055963660007188  5517055963660007188
6     RLVG                    NaN       KA-0001  Diverse Steuerungen ␣
↪4937005671108174325  4937005671108174325
7     RLVG                    NaN       KA-0002  Diverse Steuerungen ␣
↪5752519230439786595  5752519230439786595
8     RLVG                    NaN       KA-0009  Diverse Steuerungen ␣
↪5660961189098354654  5660961189098354654
9     RLVG                    NaN       KA-0010  Diverse Steuerungen ␣
↪5510085446018401887  5510085446018401887
10    RLVG                    NaN       KA-0011  Diverse Steuerungen ␣
↪4894802981639605379  4894802981639605379
11    RLVG                    NaN       KA-0012  Diverse Steuerungen ␣
↪5310832758005678867  5310832758005678867
12    RLVG                    NaN       KA-0017  Diverse Steuerungen ␣
↪4879781051055847299  4879781051055847299
13    RLVG                    NaN       KA-0018  Diverse Steuerungen ␣
↪4806239740367977881  4806239740367977881
```

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```
14     RLVG                    NaN       KA-0019  Diverse Steuerungen ␣
↪5447964234902471608  5447964234902471608
15     RLVG                    NaN       KA-0020  Diverse Steuerungen ␣
↪4717907439365620025  4717907439365620025
0      RADD                    NaN            dT        Sekundärwerte ␣
↪4654077245127093202  4654077245127093202
1      RADD                    NaN       dLastMW        Sekundärwerte ␣
↪4611793887272861500  4611793887272861500
2      RADD                    NaN      yUWMLast        Sekundärwerte ␣
↪5574611204646558662  5574611204646558662
3      RADD                    NaN     yDH_pMD_A        Sekundärwerte ␣
↪5255402486218254174  5255402486218254174
4      RADD                    NaN  wDH_MD_A_ERO  Diverse Steuerungen ␣
↪5729434727271745948  5729434727271745948
0      RSTN                    NaN      wNA_RSTN                    A ␣
↪5165635044767172069  5165635044767172069
1      RSTN                    NaN       KA-0044                    A ␣
↪5636962607360173089  5636962607360173089
2      RSTN                    NaN       KA-0045                    A ␣
↪5597572325891198144  5597572325891198144
3      RSTN                    NaN       KA-0046                    A ␣
↪5137384799783014264  5137384799783014264
4      RSTN                    NaN      wNB_RSTN                    B ␣
↪5342104608381486733  5342104608381486733
5      RSTN                    NaN       KA-0053                    B ␣
↪5338620382667478180  5338620382667478180
6      RSTN                    NaN       KA-0057                    B ␣
↪5226612456739754122  5226612456739754122
7      RSTN                    NaN       KA-0058                    B ␣
↪5537037692802520861  5537037692802520861
8      RSTN                    NaN      wNC_RSTN                    C ␣
↪5103693862180601916  5103693862180601916
9      RSTN                    NaN       KA-0059                    C ␣
↪4792266770335818241  4792266770335818241
10     RSTN                    NaN       KA-0060                    C ␣
↪5286169822203128424  5286169822203128424
11     RSTN                    NaN       KA-0061                    C ␣
↪4848495011382561496  4848495011382561496
12     RSTN                    NaN       KA-0004  Diverse Steuerungen ␣
↪5625633953643797107  5625633953643797107
13     RSTN                    NaN       KA-0005  Diverse Steuerungen ␣
↪4851348857631426312  4851348857631426312
14     RSTN                    NaN       KA-0006  Diverse Steuerungen ␣
↪5185169121447805605  5185169121447805605
15     RSTN                    NaN       KA-0008  Diverse Steuerungen ␣
↪4760680402451575539  4760680402451575539
16     RSTN                    NaN       KA-0003  Diverse Steuerungen ␣
↪5249070009027066113  5249070009027066113
17     RSTN                    NaN       KA-0007  Diverse Steuerungen ␣
↪5721409231684230901  5721409231684230901
18     RSTN                    NaN       KA-0013  Diverse Steuerungen ␣
↪5075554822852863012  5075554822852863012
19     RSTN                    NaN       KA-0014  Diverse Steuerungen ␣
↪5320878233009751638  5320878233009751638
20     RSTN                    NaN       KA-0015  Diverse Steuerungen ␣
↪5749069735826810904  5749069735826810904
21     RSTN                    NaN       KA-0016  Diverse Steuerungen ␣
↪5704472379299329003  5704472379299329003
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```
22    RSTN                    NaN       KA-0021  Diverse Steuerungen ␣
↪5629658054546932585  5629658054546932585
23    RSTN                    NaN       KA-0022  Diverse Steuerungen ␣
↪5162821695312832398  5162821695312832398
24    RSTN                    NaN       KA-0023  Diverse Steuerungen ␣
↪5357357577779591773  5357357577779591773
25    RSTN                    NaN       KA-0024  Diverse Steuerungen ␣
↪5357348958190741976  5357348958190741976
26    RSTN                    NaN       KA-0025  Diverse Steuerungen ␣
↪4635966862484721732  4635966862484721732
27    RSTN                    NaN       KA-0027  Diverse Steuerungen ␣
↪5700600513951468434  5700600513951468434
28    RSTN                    NaN       KA-0028  Diverse Steuerungen ␣
↪5367185280774605989  5367185280774605989
29    RSTN                    NaN       KA-0029  Diverse Steuerungen ␣
↪5445770133105602710  5445770133105602710
30    RSTN                    NaN       KA-0030  Diverse Steuerungen ␣
↪4885570100974274375  4885570100974274375
31    RSTN                    NaN       KA-0031  Diverse Steuerungen ␣
↪5223383850171539514  5223383850171539514
32    RSTN           NTR_1_RL_Ein       KA-0032  Diverse Steuerungen ␣
↪5333724089944967011  5333724089944967011
33    RSTN           NTR_1_VL_Ein       KA-0033  Diverse Steuerungen ␣
↪4825143842549434339  4825143842549434339
34    RSTN           NTR_1_VL_Ein       KA-0034  Diverse Steuerungen ␣
↪4693599139501858956  4693599139501858956
35    RSTN           NTR_1_RL_Ein       KA-0035  Diverse Steuerungen ␣
↪5693503518255620080  5693503518255620080
36    RSTN           NTR_3_Aus_VL       KA-0036  Diverse Steuerungen ␣
↪4901609871029871596  4901609871029871596
37    RSTN           NTR_3_Aus_RL       KA-0037  Diverse Steuerungen ␣
↪5327767233627106399  5327767233627106399
38    RSTN           NTR_3_Ein_VL       KA-0038  Diverse Steuerungen ␣
↪5760765754619184144  5760765754619184144
39    RSTN           NTR_3_Ein_RL       KA-0039  Diverse Steuerungen ␣
↪4994799657516451637  4994799657516451637
40    RSTN           NTR_2_Aus_VL       KA-0040  Diverse Steuerungen ␣
↪5348583181653286363  5348583181653286363
41    RSTN           NTR_2_Aus_RL       KA-0041  Diverse Steuerungen ␣
↪5499083775210733192  5499083775210733192
42    RSTN           NTR_2_Ein_VL       KA-0042  Diverse Steuerungen ␣
↪4990389026836623226  4990389026836623226
43    RSTN           NTR_2_Ein_RL       KA-0043  Diverse Steuerungen ␣
↪5697786347617919077  5697786347617919077
44    RSTN                    NaN       KA-0054  Diverse Steuerungen ␣
↪5623872434691357889  5623872434691357889
45    RSTN                    NaN       KA-0055  Diverse Steuerungen ␣
↪5283008774827895454  5283008774827895454
46    RSTN         Test ohne Fkt.       KA-0048  Diverse Steuerungen ␣
↪5451626161773298441  5451626161773298441
47    RSTN         Test ohne Fkt.       KA-0049  Diverse Steuerungen ␣
↪4906317765393876716  4906317765393876716
48    RSTN         Test ohne Fkt.       KA-0050  Diverse Steuerungen ␣
↪5153543836152416454  5153543836152416454
```

**_vRXXX_XXXX** (*OBJTYPE=None*)

One row per R-Node of Type OBJTYPE.

**Args:** OBJTYPE: str: i.e. RHYS

**Returns:** R-Node df None is returned if an exception occurs

**columns:**

- OBJTYPE: str: i.e. RADD
- BESCHREIBUNG
- KA
- CONT
- pk
- tk

**MxSync** (*mx=None*, *ForceNoH5ReadForMx=False*)

Xm: NEW 1st Call: vNRCV_Mx1: vNRCV with MX1-Information. Some Xm-Views with MX2-Information (mx2Idx).Mx: Sir3sID Update in Mx-Object.

**Args:**

**mx (default: None): Mx-Object**

- If no Mx-Object is given the Mx-Object corresponding to the Xm-Object is constructed and returned.
- **Notes:**
  - **The Sync-Result in Xm is persisted if xm were read from H5:**
    - xm.ToH5() is called if xm.h5Read is True.
  - **The Sync-Result in Mx is persisted if mx were read from H5:**
    - mx.ToH5() is called (from __Mx1_Sir3sIDUpd) if Sir3sID-Updates occured and mx.h5Read is True.

**ForceNoH5ReadForMx (deafault: False): has an Effect onlx if a new Mx-Object is constructed**

- **ForceNoH5ReadForMx = True:**
  - the new Mx-Object is constructed with NoH5Read=True
- **ForceNoH5ReadForMx = False:**
  - the new Mx-Object is constructed with NoH5Read = not self.h5Read
  - if the Xm was read from H5 the Mx is constructed with NoH5Read=False
  - if the Xm was not read from H5 the Mx is constructed with NoH5Read=True

**Returns:** Mx-Object if no Mx-Object was given; Nothing else

**Raises:** XmError

```
>>> xm=xms['LocalHeatingNetwork']
>>> mx=xm.MxSync()
```

**_vREdges** ()

Returns vREdges.

columns:

OBJTYPE_Ki BESCHREIBUNG_Ki Kn_Ki: Node ID pk_Ki: Node ID

OBJTYPE_Kk BESCHREIBUNG_Kk Kn_Kk pk_Kk CONT

KnExt_Ki: with OBJTYPE extended Node ID KnExt_Kk: with OBJTYPE extended Node ID

```
>>> import pandas as pd
>>> # ---
>>> xm=xms['DHNetwork']
>>> # ---
>>> vREdges=xm.dataFrames['vREdges']
>>> vREdges[[
...  'CONT'
... ,'CONT_PARENT'
... ,'OBJTYPE_Ki'
... ,'OBJTYPE_Kk'
... ,'Kn_Ki'
... ,'Kn_Kk'
... ,'KnExt_Ki'
... ,'KnExt_Kk'
... ]].sort_values(by=['KnExt_Ki','KnExt_Kk','CONT'])
                    CONT          CONT_PARENT OBJTYPE_Ki OBJTYPE_Kk
↪Kn_Ki         Kn_Kk           KnExt_Ki           KnExt_Kk
28   Diverse Steuerungen    AGFW Symposium DH       RSLW       RUES
↪  0             0             0_RSLW             0_RUES
91   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0032             0_RUES       KA-0032_RSTN
92   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0033             0_RUES       KA-0033_RSTN
94   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0034             0_RUES       KA-0034_RSTN
96   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0035             0_RUES       KA-0035_RSTN
98   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0036             0_RUES       KA-0036_RSTN
100  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0037             0_RUES       KA-0037_RSTN
102  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0038             0_RUES       KA-0038_RSTN
104  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0039             0_RUES       KA-0039_RSTN
106  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0040             0_RUES       KA-0040_RSTN
108  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0041             0_RUES       KA-0041_RSTN
110  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0042             0_RUES       KA-0042_RSTN
112  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0043             0_RUES       KA-0043_RSTN
123  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0048             0_RUES       KA-0048_RSTN
125  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0049             0_RUES       KA-0049_RSTN
127  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  0       KA-0050             0_RUES       KA-0050_RSTN
35   Diverse Steuerungen    AGFW Symposium DH       RSLW       RUES
↪100           100           100_RSLW           100_RUES
29   Diverse Steuerungen    AGFW Symposium DH       RSLW       RUES
↪  1             1             1_RSLW             1_RUES
```

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```
32   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  1       KA-0004             1_RUES       KA-0004_RSTN
34   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  1       KA-0005             1_RUES       KA-0005_RSTN
79   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  1       KA-0025             1_RUES       KA-0025_RSTN
81   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  1       KA-0027             1_RUES       KA-0027_RSTN
83   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  1       KA-0028             1_RUES       KA-0028_RSTN
85   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  1       KA-0029             1_RUES       KA-0029_RSTN
87   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  1       KA-0030             1_RUES       KA-0030_RSTN
89   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪  1       KA-0031             1_RUES       KA-0031_RSTN
30   Diverse Steuerungen    AGFW Symposium DH       RSLW       RUES
↪ADum          ADum          ADum_RSLW          ADum_RUES
46   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0003          ADum_RUES       KA-0003_RSTN
39   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0006          ADum_RUES       KA-0006_RSTN
48   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0007          ADum_RUES       KA-0007_RSTN
41   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0008          ADum_RUES       KA-0008_RSTN
55   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0013          ADum_RUES       KA-0013_RSTN
60   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0014          ADum_RUES       KA-0014_RSTN
57   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0015          ADum_RUES       KA-0015_RSTN
62   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0016          ADum_RUES       KA-0016_RSTN
69   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0021          ADum_RUES       KA-0021_RSTN
74   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0022          ADum_RUES       KA-0022_RSTN
71   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0023          ADum_RUES       KA-0023_RSTN
76   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0024          ADum_RUES       KA-0024_RSTN
90   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0032          ADum_RUES       KA-0032_RSTN
93   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0033          ADum_RUES       KA-0033_RSTN
95   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0034          ADum_RUES       KA-0034_RSTN
97   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0035          ADum_RUES       KA-0035_RSTN
99   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0036          ADum_RUES       KA-0036_RSTN
101  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0037          ADum_RUES       KA-0037_RSTN
103  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0038          ADum_RUES       KA-0038_RSTN
105  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0039          ADum_RUES       KA-0039_RSTN
```

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```
107  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0040          ADum_RUES       KA-0040_RSTN
109  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0041          ADum_RUES       KA-0041_RSTN
111  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0042          ADum_RUES       KA-0042_RSTN
113  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0043          ADum_RUES       KA-0043_RSTN
131                    A    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0044          ADum_RUES       KA-0044_RSTN
133                    A    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0045          ADum_RUES       KA-0045_RSTN
122  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0048          ADum_RUES       KA-0048_RSTN
124  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0049          ADum_RUES       KA-0049_RSTN
126  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0050          ADum_RUES       KA-0050_RSTN
139                    B    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0053          ADum_RUES       KA-0053_RSTN
141                    B    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0057          ADum_RUES       KA-0057_RSTN
147                    C    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0059          ADum_RUES       KA-0059_RSTN
148                    C    AGFW Symposium DH       RUES       RSTN
↪ADum       KA-0060          ADum_RUES       KA-0060_RSTN
47   Diverse Steuerungen    AGFW Symposium DH       RLVG       RLVG       KA-
↪0001       KA-0002       KA-0001_RLVG       KA-0002_RLVG
45   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0001       KA-0003       KA-0001_RLVG       KA-0003_RSTN
49   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0002       KA-0007       KA-0002_RLVG       KA-0007_RSTN
56   Diverse Steuerungen    AGFW Symposium DH       RLVG       RLVG       KA-
↪0009       KA-0010       KA-0009_RLVG       KA-0010_RLVG
54   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0009       KA-0013       KA-0009_RLVG       KA-0013_RSTN
58   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0010       KA-0015       KA-0010_RLVG       KA-0015_RSTN
61   Diverse Steuerungen    AGFW Symposium DH       RLVG       RLVG       KA-
↪0011       KA-0012       KA-0011_RLVG       KA-0012_RLVG
59   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0011       KA-0014       KA-0011_RLVG       KA-0014_RSTN
63   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0012       KA-0016       KA-0012_RLVG       KA-0016_RSTN
70   Diverse Steuerungen    AGFW Symposium DH       RLVG       RLVG       KA-
↪0017       KA-0018       KA-0017_RLVG       KA-0018_RLVG
68   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0017       KA-0021       KA-0017_RLVG       KA-0021_RSTN
72   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0018       KA-0023       KA-0018_RLVG       KA-0023_RSTN
75   Diverse Steuerungen    AGFW Symposium DH       RLVG       RLVG       KA-
↪0019       KA-0020       KA-0019_RLVG       KA-0020_RLVG
73   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0019       KA-0022       KA-0019_RLVG       KA-0022_RSTN
77   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN       KA-
↪0020       KA-0024       KA-0020_RLVG       KA-0024_RSTN
166        Sekundärwerte  Diverse Steuerungen       RMES       RADD       KA-
↪0026      yUWMLast       KA-0026_RMES      yUWMLast_RADD
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```
0      AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES    Leck_1_
↪Ein    Leck_1_Ein    Leck_1_Ein_RSLW    Leck_1_Ein_RUES
43   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG    Leck_1_
↪Ein       KA-0001    Leck_1_Ein_RUES       KA-0001_RLVG
36   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG    Leck_1_
↪Ein     Leck_1_VL    Leck_1_Ein_RUES     Leck_1_VL_RLVG
38   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN     Leck_
↪1_VL       KA-0006     Leck_1_VL_RLVG       KA-0006_RSTN
40   Diverse Steuerungen    AGFW Symposium DH       RLVG       RLVG     Leck_
↪1_VL    nLeck_1_VL     Leck_1_VL_RLVG    nLeck_1_VL_RLVG
1      AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES    Leck_2_
↪Ein    Leck_2_Ein    Leck_2_Ein_RSLW    Leck_2_Ein_RUES
52   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG    Leck_2_
↪Ein       KA-0009    Leck_2_Ein_RUES       KA-0009_RLVG
51   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG    Leck_2_
↪Ein       KA-0011    Leck_2_Ein_RUES       KA-0011_RLVG
2      AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES    Leck_3_
↪Ein    Leck_3_Ein    Leck_3_Ein_RSLW    Leck_3_Ein_RUES
66   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG    Leck_3_
↪Ein       KA-0017    Leck_3_Ein_RUES       KA-0017_RLVG
65   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG    Leck_3_
↪Ein       KA-0019    Leck_3_Ein_RUES       KA-0019_RLVG
17     AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES    Leck_
↪Menge    Leck_Menge    Leck_Menge_RSLW    Leck_Menge_RUES
78   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN    Leck_
↪Menge       KA-0025    Leck_Menge_RUES       KA-0025_RSTN
80   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN    Leck_
↪Menge       KA-0027    Leck_Menge_RUES       KA-0027_RSTN
82   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN    Leck_
↪Menge       KA-0028    Leck_Menge_RUES       KA-0028_RSTN
84   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN    Leck_
↪Menge       KA-0029    Leck_Menge_RUES       KA-0029_RSTN
86   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN    Leck_
↪Menge       KA-0030    Leck_Menge_RUES       KA-0030_RSTN
88   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN    Leck_
↪Menge       KA-0031    Leck_Menge_RUES       KA-0031_RSTN
19     AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES       ␣
↪Leck_RL       Leck_RL       Leck_RL_RSLW       Leck_RL_RUES
44   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG       ␣
↪Leck_RL       KA-0001       Leck_RL_RUES       KA-0001_RLVG
50   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG       ␣
↪Leck_RL       KA-0011       Leck_RL_RUES       KA-0011_RLVG
64   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG       ␣
↪Leck_RL       KA-0019       Leck_RL_RUES       KA-0019_RLVG
18     AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES       ␣
↪Leck_VL       Leck_VL       Leck_VL_RSLW       Leck_VL_RUES
53   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG       ␣
↪Leck_VL       KA-0009       Leck_VL_RUES       KA-0009_RLVG
67   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG       ␣
↪Leck_VL       KA-0017       Leck_VL_RUES       KA-0017_RLVG
37   Diverse Steuerungen    AGFW Symposium DH       RUES       RLVG       ␣
↪Leck_VL     Leck_1_VL       Leck_VL_RUES     Leck_1_VL_RLVG
168        Sekundärwerte  Diverse Steuerungen       RMES       RUES        ␣
↪QDHGes        QDHGes        QDHGes_RMES        QDHGes_RUES
152        Sekundärwerte  Diverse Steuerungen       RMES       RADD          ␣
↪TRSP            dT          TRSP_RMES            dT_RADD
153        Sekundärwerte  Diverse Steuerungen       RMES       RADD            ␣
↪TVL             dT           TVL_RMES            dT_RADD
```

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```
159        Sekundärwerte  Diverse Steuerungen       RADD       RUES
↪dLastMW       dLastMW       dLastMW_RADD       dLastMW_RUES
164        Sekundärwerte  Diverse Steuerungen       RMES       RUES
↪dUWMMax       dUWMMax       dUWMMax_RMES       dUWMMax_RUES
163        Sekundärwerte  Diverse Steuerungen       RMES       RUES
↪dUWMMin       dUWMMin       dUWMMin_RMES       dUWMMin_RUES
25     AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪dpA           dpA           dpA_RSLW           dpA_RUES
134                    A    AGFW Symposium DH       RUES       RSTN
↪dpA       KA-0046           dpA_RUES       KA-0046_RSTN
42   Diverse Steuerungen    AGFW Symposium DH       RLVG       RSTN    nLeck_
↪1_VL       KA-0008    nLeck_1_VL_RLVG       KA-0008_RSTN
26     AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪ qB            qB            qB_RSLW            qB_RUES
142                    B    AGFW Symposium DH       RUES       RSTN
↪ qB       KA-0058            qB_RUES       KA-0058_RSTN
27     AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪ qC            qC            qC_RSLW            qC_RUES
150                    C    AGFW Symposium DH       RUES       RSTN
↪ qC       KA-0061            qC_RUES       KA-0061_RSTN
6      AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪vorOrtNA      vorOrtNA      vorOrtNA_RSLW      vorOrtNA_RUES
7      AGFW Symposium DH    AGFW Symposium DH       RUES       RLVG
↪vorOrtNA     wNAEin_vO      vorOrtNA_RUES     wNAEin_vO_RLVG
9      AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪vorOrtNB      vorOrtNB      vorOrtNB_RSLW      vorOrtNB_RUES
140                    B    AGFW Symposium DH       RUES       RSTN
↪vorOrtNB       KA-0057      vorOrtNB_RUES       KA-0057_RSTN
143                    B    AGFW Symposium DH       RUES       RSTN
↪vorOrtNB       KA-0058      vorOrtNB_RUES       KA-0058_RSTN
10     AGFW Symposium DH    AGFW Symposium DH       RUES       RLVG
↪vorOrtNB     wNBEin_vO      vorOrtNB_RUES     wNBEin_vO_RLVG
12     AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪vorOrtNC      vorOrtNC      vorOrtNC_RSLW      vorOrtNC_RUES
132                    A    AGFW Symposium DH       RUES       RSTN
↪vorOrtNC       KA-0045      vorOrtNC_RUES       KA-0045_RSTN
135                    A    AGFW Symposium DH       RUES       RSTN
↪vorOrtNC       KA-0046      vorOrtNC_RUES       KA-0046_RSTN
149                    C    AGFW Symposium DH       RUES       RSTN
↪vorOrtNC       KA-0060      vorOrtNC_RUES       KA-0060_RSTN
151                    C    AGFW Symposium DH       RUES       RSTN
↪vorOrtNC       KA-0061      vorOrtNC_RUES       KA-0061_RSTN
13     AGFW Symposium DH    AGFW Symposium DH       RUES       RLVG
↪vorOrtNC     wNCEin_vO      vorOrtNC_RUES     wNCEin_vO_RLVG
24     AGFW Symposium DH    AGFW Symposium DH       RLVG       RUES   wDH_BA_
↪A_MD   wDH_BA_A_MD   wDH_BA_A_MD_RLVG   wDH_BA_A_MD_RUES
121  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN   wDH_BA_
↪A_MD       KA-0055   wDH_BA_A_MD_RUES       KA-0055_RSTN
22     AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES   wDH_BA_
↪A_RD   wDH_BA_A_RD   wDH_BA_A_RD_RSLW   wDH_BA_A_RD_RUES
119  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN   wDH_BA_
↪A_RD       KA-0054   wDH_BA_A_RD_RUES       KA-0054_RSTN
23     AGFW Symposium DH    AGFW Symposium DH       RUES       RLVG   wDH_BA_
↪A_RD   wDH_BA_A_MD   wDH_BA_A_RD_RUES   wDH_BA_A_MD_RLVG
116  Diverse Steuerungen    AGFW Symposium DH       RADD       RUES  wDH_MD_A_
↪ERO  wDH_MD_A_ERO  wDH_MD_A_ERO_RADD  wDH_MD_A_ERO_RUES
120  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN  wDH_MD_A_
↪ERO       KA-0055  wDH_MD_A_ERO_RUES       KA-0055_RSTN
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21      AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES       wDH_
↪MD_A      wDH_MD_A      wDH_MD_A_RSLW      wDH_MD_A_RUES
115  Diverse Steuerungen    AGFW Symposium DH       RUES       RADD       wDH_
↪MD_A  wDH_MD_A_ERO      wDH_MD_A_RUES  wDH_MD_A_ERO_RADD
118  Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN  wDH_RD_A_
↪ERO       KA-0054  wDH_RD_A_ERO_RUES       KA-0054_RSTN
20      AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES       wDH_
↪RD_A      wDH_RD_A      wDH_RD_A_RSLW      wDH_RD_A_RUES
117  Diverse Steuerungen    AGFW Symposium DH       RUES       RUES       wDH_
↪RD_A  wDH_RD_A_ERO      wDH_RD_A_RUES  wDH_RD_A_ERO_RUES
155        Sekundärwerte  Diverse Steuerungen       RMES       RUES
↪wLastMW       wLastMW       wLastMW_RMES       wLastMW_RUES
158        Sekundärwerte  Diverse Steuerungen       RUES       RADD
↪wLastMW       dLastMW       wLastMW_RUES       dLastMW_RADD
15      AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪wLast         wLast         wLast_RSLW         wLast_RUES
31   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪wLast       KA-0004         wLast_RUES       KA-0004_RSTN
128                    A    AGFW Symposium DH       RUES       RSTN
↪wNAEin      wNA_RSTN        wNAEin_RUES      wNA_RSTN_RSTN
8       AGFW Symposium DH    AGFW Symposium DH       RLVG       RUES
↪wNAEin_vO        wNAEin     wNAEin_vO_RLVG        wNAEin_RUES
3       AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪wNA           wNA           wNA_RSLW           wNA_RUES
129                    A    AGFW Symposium DH       RUES       RSTN
↪wNA      wNA_RSTN           wNA_RUES      wNA_RSTN_RSTN
130                    A    AGFW Symposium DH       RUES       RSTN
↪wNBEin       KA-0044        wNBEin_RUES       KA-0044_RSTN
138                    B    AGFW Symposium DH       RUES       RSTN
↪wNBEin       KA-0053        wNBEin_RUES       KA-0053_RSTN
136                    B    AGFW Symposium DH       RUES       RSTN
↪wNBEin      wNB_RSTN        wNBEin_RUES      wNB_RSTN_RSTN
11      AGFW Symposium DH    AGFW Symposium DH       RLVG       RUES
↪wNBEin_vO        wNBEin     wNBEin_vO_RLVG        wNBEin_RUES
4       AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪wNB           wNB           wNB_RSLW           wNB_RUES
137                    B    AGFW Symposium DH       RUES       RSTN
↪wNB      wNB_RSTN           wNB_RUES      wNB_RSTN_RSTN
146                    C    AGFW Symposium DH       RUES       RSTN
↪wNCEin       KA-0059        wNCEin_RUES       KA-0059_RSTN
144                    C    AGFW Symposium DH       RUES       RSTN
↪wNCEin      wNC_RSTN        wNCEin_RUES      wNC_RSTN_RSTN
14      AGFW Symposium DH    AGFW Symposium DH       RLVG       RUES
↪wNCEin_vO        wNCEin     wNCEin_vO_RLVG        wNCEin_RUES
5       AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪wNC           wNC           wNC_RSLW           wNC_RUES
145                    C    AGFW Symposium DH       RUES       RSTN
↪wNC      wNC_RSTN           wNC_RUES      wNC_RSTN_RSTN
16      AGFW Symposium DH    AGFW Symposium DH       RSLW       RUES
↪wTRST         wTRST         wTRST_RSLW         wTRST_RUES
33   Diverse Steuerungen    AGFW Symposium DH       RUES       RSTN
↪wTRST       KA-0005         wTRST_RUES       KA-0005_RSTN
160        Sekundärwerte  Diverse Steuerungen       RMES       RUES
↪yAMW          yAMW          yAMW_RMES          yAMW_RUES
161        Sekundärwerte  Diverse Steuerungen       RMES       RUES
↪yBMW          yBMW          yBMW_RMES          yBMW_RUES
162        Sekundärwerte  Diverse Steuerungen       RMES       RUES
↪yCMW          yCMW          yCMW_RMES          yCMW_RUES
```

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```
169        Sekundärwerte  Diverse Steuerungen       RMES       RUES     yDH_
↪dp2_A     yDH_dp2_A     yDH_dp2_A_RMES     yDH_dp2_A_RUES
114  Diverse Steuerungen    AGFW Symposium DH       RUES       RADD     yDH_
↪dp2_A  wDH_MD_A_ERO     yDH_dp2_A_RUES  wDH_MD_A_ERO_RADD
170        Sekundärwerte  Diverse Steuerungen       RUES       RADD     yDH_
↪dp2_A     yDH_pMD_A     yDH_dp2_A_RUES     yDH_pMD_A_RADD
172        Sekundärwerte  Diverse Steuerungen       RADD       RUES     yDH_
↪pMD_A     yDH_pMD_A     yDH_pMD_A_RADD     yDH_pMD_A_RUES
173        Sekundärwerte  Diverse Steuerungen       RMES       RUES     yDH_
↪pRL_A     yDH_pRL_A     yDH_pRL_A_RMES     yDH_pRL_A_RUES
171        Sekundärwerte  Diverse Steuerungen       RUES       RADD     yDH_
↪pRL_A     yDH_pMD_A     yDH_pRL_A_RUES     yDH_pMD_A_RADD
156        Sekundärwerte  Diverse Steuerungen       RMES       RUES      ␣
↪yLastMW       yLastMW       yLastMW_RMES       yLastMW_RUES
157        Sekundärwerte  Diverse Steuerungen       RUES       RADD      ␣
↪yLastMW       dLastMW       yLastMW_RUES       dLastMW_RADD
167        Sekundärwerte  Diverse Steuerungen       RADD       RUES      ␣
↪yUWMLast      yUWMLast      yUWMLast_RADD      yUWMLast_RUES
154        Sekundärwerte  Diverse Steuerungen       RMES       RUES         ␣
↪yUWM          yUWM          yUWM_RMES          yUWM_RUES
165        Sekundärwerte  Diverse Steuerungen       RUES       RADD         ␣
↪yUWM      yUWMLast          yUWM_RUES      yUWMLast_RADD
>>> import networkx as nx
>>> G=nx.from_pandas_edgelist(vREdges, source='KnExt_Ki', target='KnExt_Kk',␣
↪edge_attr=True,create_using=nx.DiGraph())
>>> list(G.selfloop_edges())
[]
>>> pathNodes=nx.shortest_path(G,'Leck_1_Ein_RSLW','KA-0008_RSTN')
>>> pathNodes
['Leck_1_Ein_RSLW', 'Leck_1_Ein_RUES', 'Leck_1_VL_RLVG', 'nLeck_1_VL_RLVG',
↪'KA-0008_RSTN']
>>> sink_nodes = [node for node, outdegree in G.out_degree(G.nodes()) if␣
↪outdegree == 0]
>>> source_nodes = [node for node, indegree in G.in_degree(G.nodes()) if␣
↪indegree == 0]
>>> import re
>>> for source, sink in [(source, sink) for sink in sink_nodes for source in␣
↪source_nodes]: # ueber alle Quellen pro Senke ...
...     if re.search('_RSTN$',sink) != None:
...         for path in nx.all_simple_paths(G, source=source, target=sink):
...             if sink=='KA-0008_RSTN':
...                 path
['Leck_1_Ein_RSLW', 'Leck_1_Ein_RUES', 'Leck_1_VL_RLVG', 'nLeck_1_VL_RLVG',
↪'KA-0008_RSTN']
['Leck_VL_RSLW', 'Leck_VL_RUES', 'Leck_1_VL_RLVG', 'nLeck_1_VL_RLVG', 'KA-
↪0008_RSTN']
['ADum_RSLW', 'ADum_RUES', 'KA-0008_RSTN']
>>> #---
>>> # dasselbe mit Knotennamen ohne Postfix ...
>>> G=nx.from_pandas_edgelist(vREdges, source='Kn_Ki', target='Kn_Kk', edge_
↪attr=True,create_using=nx.DiGraph())
>>> # alle RUES Eingänge deren ID mit der des aufnehmenden Signals identisch␣
↪ist führen dann zu Schleifen ...
>>> # ... die entfernt werden muessen wenn Quellen am Indegree erkannt werden␣
↪sollen ...
>>> G.remove_edges_from(list(G.selfloop_edges()))
>>> pathNodes=nx.shortest_path(G,'Leck_1_Ein','KA-0008')
```

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```
>>> pathNodes # (auf die Pfadknotensequenz haben Schleifen keinen Einfluss,␣
↪das Ergebnis waere mit den Schleifen dasselbe ...)
['Leck_1_Ein', 'Leck_1_VL', 'nLeck_1_VL', 'KA-0008']
>>> sink_nodes = [node for node, outdegree in G.out_degree(G.nodes()) if␣
↪outdegree == 0]
>>> source_nodes = [node for node, indegree in G.in_degree(G.nodes()) if␣
↪indegree == 0]
>>> for source, sink in [(source, sink) for sink in sink_nodes for source in␣
↪source_nodes]: # ueber alle Quellen pro Senke ...
...         for path in nx.all_simple_paths(G, source=source, target=sink):
...             if sink=='KA-0008':
...                 path
['Leck_1_Ein', 'Leck_1_VL', 'nLeck_1_VL', 'KA-0008']
['Leck_VL', 'Leck_1_VL', 'nLeck_1_VL', 'KA-0008']
['ADum', 'KA-0008']
```

**_MxSyncAddMx** (*ForceNoH5ReadForMx=False*)

Mx-Object corresponding to the Xm-Object is constructed and returned.

**ForceNoH5ReadForMx (deafault: False)**

- **ForceNoH5ReadForMx = True:**
  - the new Mx-Object is constructed with NoH5Read=True
- **ForceNoH5ReadForMx = False:**
  - **the new Mx-Object is constructed with NoH5Read = not self.h5Read**
    - if the Xm was read from H5 the Mx is constructed with NoH5Read=False
    - if the Xm was not read from H5 the Mx is constructed with NoH5Read=True

**Returns:** Mx-Object

**Raises:** XmError

**MxAdd** (*mx=None*, *ForceNoH5ReadForMx=False*, *timeReq=None*, *aggReq=None*, *timeReq2nd=None*)

Add MX-Resultcolumns to some Xm-Views. NEW 1st Call: vROHRVecResults, vAGSN.

**Args:**

- mx, ForceNoH5ReadForMx: same Args as for MxSync; see description there
- **timeReq:**
  - TIMESTAMP
  - if None 1st TIME in Mx is used
- **aggReq:**
  - 'TIME','TMIN','TMAX' (source: MXS) or 'MIN','MAX', ... (source: mx.getVecAggs())
  - timeReq is considered as TIMESTAMPL
  - timeReq2nd is considered as TIMESTAMPR
- **timeReq2nd:**
  - TIMESTAMP
  - if None last TIME in Mx is used

**Views with MX2-Results added:**

- in the Xm-Views below col mx2Idx must exist (i.e. MxSync mus have been called)
- mx2Idx is considered to be the last of the Model-cols
- right from mx2Idx Result-cols are added if not already existing
- already existing Result-cols are overwritten
- **mx2Idx-Views:**
  - vKNOT (KNOT...)
  - vROHR (ROHR...) - only Non-VEC-Channel-Results are added
  - vFWVB (FWVB...)
  - vVBEL (KNOT..._i and KNOT..._k and Q)
- NEW 1st Call:
  - **vROHRVecResults: VEC-Channel-Results for Pipe-Interior-Pts (IPts):**
    - pk
    - mx2Idx
    - IptIdx: S,0,...,E - Interior Point Index; S=Start EdgeDefNode, E=End EdgeDefNode, 0=1st Ipt in EdgeDefDirection
    - one column per VEC-Channel
  - **vAGSN**
    - from vVBEL: KNOT..._i and KNOT..._k and Q
    - from vROHRVecResults: vecResults
    - **Topology:**
      - nextNODE
      - IptIdx
    - **Geometry:**
      - dx
      - x
      - xVbel
      - Z (the corresponding Z_i, Z_k and ZVEC are droped)
    - **Results:**
      - Q: from Q before and QMVEC for PIPEs; in Schnittrichtung; QMVEC is droped
      - **for available KNOT...#_i, KNOT...#_k and ...#VEC:**

        i.e. KNOT~*~*~*~P_i KNOT~*~*~*~P_k ROHR~*~*~*~PVEC

        P is new column

        the correspondig 3 columns are droped

**Notes:**

- **The Add-Result is persisted if df were read from H5:**

– xm.ToH5() is called if xm.h5Read is True.

**Returns:** Mx-Object if no Mx-Object was given; Nothing else

**Raises:** XmError

```
>>> xm=xms['LocalHeatingNetwork']
>>> mx=xm.MxAdd()
>>> print(xm._getvXXXXAsOneString(vXXXX='vKNOT',filterColList=['BESCHREIBUNG',
↪'IDREFERENZ','NAME','KNOT~*~*~*~PH']))
                      BESCHREIBUNG IDREFERENZ         NAME  KNOT~*~*~*~PH
0                             None         -1       R-K004       2.302971
1                             None         -1       V-K002       3.985846
2                             None         -1       V-K001       4.083384
3                             None         -1       V-K000       4.121495
4                             None         -1       R-K001       2.043288
5                             None         -1       R-K003       2.283565
6                             None         -1       R-K000       2.004937
7                             None         -1       R-K005       2.309655
8                             None         -1          R-L       2.000133
9                             None         -1       R-K002       2.141440
10                            None         -1       V-K004       3.825970
11                            None         -1       V-K005       3.819467
12                            None         -1       R-K007       2.314658
13                            None         -1       V-K006       3.816599
14                            None         -1       R-K006       2.312659
15                            None         -1       V-K003       3.845104
16                            None         -1          V-L       4.125885
17                            None         -1       V-K007       3.814690
18                            None         -1           R2       4.311307
19                            None         -1          V-1       4.126019
20                            None         -1           R3       4.291591
21  Druckhaltung - 2 bar Ruhedruck         -1  PKON-Knoten       2.000000
22          Anbindung Druckhaltung         -1          R-1       2.000000
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHR',filterColList=['BESCHREIBUNG',
↪'IDREFERENZ','NAME_i','NAME_k','ROHR~*~*~*~QMAV']))
   BESCHREIBUNG IDREFERENZ  NAME_i  NAME_k  ROHR~*~*~*~QMAV
0          None         -1  R-K004  R-K005        -8.509475
1          None         -1  V-K002  V-K003        19.059780
2          None         -1  R-K003  R-K004       -15.378901
3          None         -1  V-K004  V-K005         8.509476
4          None         -1  V-K001  V-K002        22.987946
5          None         -1  R-K006  R-K007        -3.928166
6          None         -1  V-K000  V-K001        22.987947
7          None         -1  V-K003  V-K004        15.378901
8          None         -1  V-K005  V-K006         3.928167
9          None         -1  R-K001  R-K002       -22.987946
10         None         -1  R-K002  R-K003       -19.059778
11         None         -1  R-K005  R-K006        -3.928166
12         None         -1  V-K006  V-K007         3.928167
13         None         -1  R-K000  R-K001       -22.987946
14         None         -1     R-L  R-K000       -22.987946
15         None         -1     V-L  V-K000        22.987947
>>> print(xm._getvXXXXAsOneString(vXXXX='vFWVB'))
  BESCHREIBUNG IDREFERENZ   W0  LFK  W0LFK  TVL0  TRS0  LFKT      W  W_min  W_
↪max  INDTR  TRSK  VTYP  DPHAUS  IMBG  IRFV                    pk            ␣
↪      tk  NAME_i KVR_i TM_i   XKOR_i   YKOR_i ZKOR_i  pXCor_i  pYCor_i ␣
↪NAME_k KVR_k TM_k   XKOR_k   YKOR_k ZKOR_k  pXCor_k  pYCor_k                ␣
↪                     CONT CONT_ID CONT_LFDNR                          WBLZ ␣
↪mx2Idx  FWVB~*~*~*~W  FWVB~*~*~*~QM  FWVB~*~*~*~IAKTIV
```

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```
0            1         -1  200  0.8  160.0    90    50  LFKT  160.0  160.0 ␣
↪160.0      1    55    14     0.7     0   0.0  4643800032883366034 ␣
↪4643800032883366034  V-K002     1   90  2541059  5706265     20    319.0   ␣
↪ 56.0  R-K002     2   60  2541059  5706265     20    319.0     56.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1  [BLNZ1,␣
↪BLNZ1u5, BLNZ1u5u7]       0         160.0       3.928166                0.0
1            3         -1  200  1.0  200.0    90    65  LFKT  200.0  200.0 ␣
↪200.0      1    65    14     0.7     0   0.0  4704603947372595298 ␣
↪4704603947372595298  V-K004     1   90  2541539  5706361     20    799.0   ␣
↪152.0  R-K004     2   60  2541539  5706361     20    799.0    152.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1                ␣
↪          []       1         200.0       6.869426                0.0
2            4         -1  200  0.8  160.0    90    60  LFKT  160.0  160.0 ␣
↪160.0      1    60    14     0.7     0   0.0  5121101823283893406 ␣
↪5121101823283893406  V-K005     1   90  2541627  5706363     20    887.0   ␣
↪154.0  R-K005     2   60  2541627  5706363     20    887.0    154.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1  [BLNZ1u5,␣
↪BLNZ1u5u7, BLNZ5]       2         160.0       4.581308                0.0
3            5         -1  200  0.8  160.0    90    55  LFKT  160.0  160.0 ␣
↪160.0      1    55    14     0.7     0   0.0  5400405917816384862 ␣
↪5400405917816384862  V-K007     1   90  2541899  5706325     20   1159.0   ␣
↪116.0  R-K007     2   60  2541899  5706325     20   1159.0    116.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1                ␣
↪ [BLNZ1u5u7]       3         160.0       3.928166                0.0
4            2         -1  200  0.6  120.0    90    60  LFKT  120.0  120.0 ␣
↪120.0      1    62    14     0.7     0   0.0  5695730293103267172 ␣
↪5695730293103267172  V-K003     1   90  2541457  5706345     20    717.0   ␣
↪136.0  R-K003     2   60  2541457  5706345     20    717.0    136.0 ␣
↪Nahwärmenetz mit 1000 kW Anschlussleistu    1001         -1                ␣
↪          []       4         120.0       3.680879                0.0
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['BESCHREIBUNG',
↪'IDREFERENZ','NAME_i','NAME_k','Q']))
                                          BESCHREIBUNG IDREFERENZ       ␣
↪NAME_i  NAME_k            Q
OBJTYPE OBJID                                                           ␣
↪
FWES    5638756766880678918  BHKW - Modul - 1000 kW therm.         -1        ␣
↪  R3     V-1      22.9879
FWVB    4643800032883366034                              1         -1       V-
↪K002  R-K002      3.92817
        4704603947372595298                              3         -1       V-
↪K004  R-K004      6.86943
        5121101823283893406                              4         -1       V-
↪K005  R-K005      4.58131
        5400405917816384862                              5         -1       V-
↪K007  R-K007      3.92817
        5695730293103267172                              2         -1       V-
↪K003  R-K003      3.68088
KLAP    4801110583764519435                           None         -1        ␣
↪  R2      R3      22.9879
PGRP    4986517622672493603                    Pumpengruppe         -1        ␣
↪ R-1      R3         None
PUMP    5481331875203087055                     Umwälzpumpe         -1        ␣
↪ R-1      R2      22.9879
ROHR    4613782368750024999                           None         -1       R-
↪K004  R-K005     -8.50947
        4614949065966596185                           None         -1       V-
↪K002  V-K003      19.0598
```

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```
        4637102239750163477                          None         -1       R-
↪K003  R-K004     -15.3789
        4713733238627697042                          None         -1       V-
↪K004  V-K005      8.50948
        4769996343148550485                          None         -1        ␣
↪ R-L  R-K000     -22.9879
        4789218195240364437                          None         -1       V-
↪K001  V-K002      22.9879
        4939422678063487923                          None         -1        ␣
↪ V-L  V-K000      22.9879
        4945727430885351042                          None         -1       R-
↪K006  R-K007     -3.92817
        4984202422877610920                          None         -1       V-
↪K000  V-K001      22.9879
        5037777106796980248                          None         -1       V-
↪K003  V-K004      15.3789
        5123819811204259837                          None         -1       V-
↪K005  V-K006      3.92817
        5266224553324203132                          None         -1       R-
↪K001  R-K002     -22.9879
        5379365049009065623                          None         -1       R-
↪K002  R-K003     -19.0598
        5611703699850694889                          None         -1       R-
↪K005  R-K006     -3.92817
        5620197984230756681                          None         -1       V-
↪K006  V-K007      3.92817
        5647213228462830353                          None         -1       R-
↪K000  R-K001     -22.9879
VENT    4678923650983295610                          None         -1        ␣
↪ V-1     V-L      22.9879
        4897018421024717974                          None         -1        ␣
↪ R-L     R-1      22.9879
        5525310316015533093                          None         -1  PKON-
↪Knoten     R-1  2.19997e-06
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHRVecResults'))
                    pk  mx2Idx IptIdx  ROHR~*~*~*~SVEC  ROHR~*~*~*~PVEC ␣
↪ROHR~*~*~*~RHOVEC  ROHR~*~*~*~TVEC  ROHR~*~*~*~MVEC  ROHR~*~*~*~ZVEC
0   4613782368750024999       0      S         0.000000         3.302971      ␣
↪   983.700012             60.0        -2.363743             20.0
1   4613782368750024999       0      E        88.019997         3.309655      ␣
↪   983.700012             60.0        -2.363743             20.0
2   4614949065966596185       1      S         0.000000         4.985845      ␣
↪   965.700012             90.0         5.294384             20.0
3   4614949065966596185       1      E       405.959991         4.845104      ␣
↪   965.700012             90.0         5.294384             20.0
4   4637102239750163477       2      S         0.000000         3.283566      ␣
↪   983.700012             60.0        -4.271917             20.0
5   4637102239750163477       2      E        83.550003         3.302971      ␣
↪   983.700012             60.0        -4.271917             20.0
6   4713733238627697042       3      S         0.000000         4.825970      ␣
↪   965.700012             90.0         2.363743             20.0
7   4713733238627697042       3      E        88.019997         4.819468      ␣
↪   965.700012             90.0         2.363743             20.0
8   4789218195240364437       5      S         0.000000         5.083383      ␣
↪   965.700012             90.0         6.385540             20.0
9   4789218195240364437       5      E       195.529999         4.985845      ␣
↪   965.700012             90.0         6.385540             20.0
```

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```
10  4945727430885351042       7      S         0.000000         3.312659 ␣
↪   983.700012             60.0        -1.091157             20.0
11  4945727430885351042       7      E       109.769997         3.314658 ␣
↪   983.700012             60.0        -1.091157             20.0
12  4984202422877610920       8      S         0.000000         5.121495 ␣
↪   965.700012             90.0         6.385540             20.0
13  4984202422877610920       8      E        76.400002         5.083383 ␣
↪   965.700012             90.0         6.385540             20.0
14  5037777106796980248       9      S         0.000000         4.845104 ␣
↪   965.700012             90.0         4.271917             20.0
15  5037777106796980248       9      E        83.550003         4.825970 ␣
↪   965.700012             90.0         4.271917             20.0
16  5123819811204259837      10      S         0.000000         4.819468 ␣
↪   965.700012             90.0         1.091157             20.0
17  5123819811204259837      10      E       164.910004         4.816599 ␣
↪   965.700012             90.0         1.091157             20.0
18  5266224553324203132      11      S         0.000000         3.043288 ␣
↪   983.700012             60.0        -6.385540             20.0
19  5266224553324203132      11      E       195.529999         3.141441 ␣
↪   983.700012             60.0        -6.385540             20.0
20  5379365049009065623      12      S         0.000000         3.141441 ␣
↪   983.700012             60.0        -5.294383             20.0
21  5379365049009065623      12      E       405.959991         3.283566 ␣
↪   983.700012             60.0        -5.294383             20.0
22  5611703699850694889      13      S         0.000000         3.309655 ␣
↪   983.700012             60.0        -1.091157             20.0
23  5611703699850694889      13      E       164.910004         3.312659 ␣
↪   983.700012             60.0        -1.091157             20.0
24  5620197984230756681      14      S         0.000000         4.816599 ␣
↪   965.700012             90.0         1.091157             20.0
25  5620197984230756681      14      E       109.769997         4.814690 ␣
↪   965.700012             90.0         1.091157             20.0
26  5647213228462830353      15      S         0.000000         3.004937 ␣
↪   983.700012             60.0        -6.385540             20.0
27  5647213228462830353      15      E        76.400002         3.043288 ␣
↪   983.700012             60.0        -6.385540             20.0
28  4769996343148550485       4      S         0.000000         3.000133 ␣
↪   983.700012             60.0        -6.385540             20.0
29  4769996343148550485       4      E        73.419998         3.004937 ␣
↪   983.700012             60.0        -6.385540             20.0
30  4939422678063487923       6      S         0.000000         5.125884 ␣
↪   965.700012             90.0         6.385541             20.0
31  4939422678063487923       6      E        68.599998         5.121495 ␣
↪   965.700012             90.0         6.385541             20.0
>>> xm.MxAdd(mx=mx,aggReq='TMAX')
>>> dfTMax=xm.dataFrames['vROHRVecResults'].copy()
>>> xm.MxAdd(mx=mx,aggReq='TMIN')
>>> dfTMin=xm.dataFrames['vROHRVecResults'].copy()
>>> xm.MxAdd(mx=mx,aggReq='MAX')
>>> dfMax=xm.dataFrames['vROHRVecResults'].copy()
>>> xm.MxAdd(mx=mx,aggReq='MIN')
>>> dfMin=xm.dataFrames['vROHRVecResults'].copy()
>>> import pandas as pd
>>> decimals=pd.Series([6],index=['ROHR~*~*~*~PVEC'])
>>> dfTMax.round(decimals=decimals).equals(dfMax.round(decimals=decimals))
True
>>> dfTMin.round(decimals=decimals).equals(dfMin.round(decimals=decimals))
```

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```
True
```

**_MxAddvVBEL** (*dfSource*)

(Re-)constructing vVBEL with MX2-Results.

**Arguments:** dfSource

**Result:**

**View with MX2-Results added:**

- in the Xm-View below col mx2Idx must exist
- mx2Idx is considered to be the last of the Model-cols
- right from mx2Idx Result-cols are added if not already existing
- already existing Result-cols are overwritten
- **mx2Idx-View:**
  - ...
  - vVBEL (KNOT..._i and KNOT..._k and Q)

**Raises:** XmError

```
>>> xm=xms['GPipes']
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['BESCHREIBUNG
↪','IDREFERENZ','tk NAME_i','CONT_i',' CONT_VKNO_i','Z_i','pk_i']))
                            BESCHREIBUNG             IDREFERENZ   CONT_i  Z_
↪i                 pk_i
OBJTYPE OBJID                                                              ␣
↪
ROHR    4979507900871287244  _Split_Split  3S4979507900871287244  M-1-0-1   ␣
↪0  4731210032713520411
        5114681686941855110        _Split  3S5114681686941855110  M-1-0-1   ␣
↪0  5709889458254995435
        5244313507655010738           NaN  3S5244313507655010738  M-1-0-1   ␣
↪0  5256558483525770176
        5694016449043789006           NaN  3S5694016449043789006  M-1-0-1   ␣
↪0  4731210032713520411
VENT    5116489323526156845           NaN  3S5508684139418025293  M-1-0-1 ␣
↪100  4683988347517083361
        5309992331398639768           NaN  3S5309992331398639768  M-1-0-1   ␣
↪0  5046108271210239718
        5508684139418025293           NaN  3S5508684139418025293  M-1-0-1 ␣
↪100  4683988347517083361
        5745097345184516675           NaN  3S5745097345184516675  M-1-0-1   ␣
↪0  5308591811899364960
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['BESCHREIBUNG
↪','IDREFERENZ','tk NAME_k','CONT_k',' CONT_VKNO_k','Z_k','pk_k']))
                            BESCHREIBUNG             IDREFERENZ   CONT_k  Z_
↪k                 pk_k
OBJTYPE OBJID                                                              ␣
↪
ROHR    4979507900871287244  _Split_Split  3S4979507900871287244  M-1-0-1   ␣
↪0  5441322867018839631
```

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```
        5114681686941855110        _Split  3S5114681686941855110  M-1-0-1
↪0  5441322867018839631
        5244313507655010738           NaN  3S5244313507655010738  M-1-0-1
↪100  4683988347517083361
        5694016449043789006           NaN  3S5694016449043789006  M-1-0-1
↪0  5061043246189134395
VENT    5116489323526156845           NaN  3S5508684139418025293  M-1-0-1
↪0  5709889458254995435
        5309992331398639768           NaN  3S5309992331398639768  M-1-0-1
↪0  5256558483525770176
        5508684139418025293           NaN  3S5508684139418025293  M-1-0-1
↪0  5709889458254995435
        5745097345184516675           NaN  3S5745097345184516675  M-1-0-1
↪0  4731210032713520411
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['LAYR','L','D
↪','mx2Idx']))
                            LAYR           L    D  mx2Idx
OBJTYPE OBJID
ROHR    4979507900871287244   []       10000  500       2
        5114681686941855110   []       10000  500       1
        5244313507655010738   []      160000  500       0
        5694016449043789006   []  100.498688  450       3
VENT    5116489323526156845   []           0  666       3
        5309992331398639768   []           0  800       0
        5508684139418025293   []           0  800       2
        5745097345184516675   []           0  800       1
```

```
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['KNOT~*~*~*~
↪RHON_i','KNOT~*~*~*~H_i','KNOT~*~*~*~LFAKTAKT_i','KNOT~*~*~*~P_i','KNOT~*~
↪*~*~PH_i','KNOT~*~*~*~PH_EIN_i','KNOT~*~*~*~QM_i','KNOT~*~*~*~RHO_i','KNOT~
↪*~*~*~T_i','KNOT~*~*~*~EH_i']))
                             KNOT~*~*~*~RHON_i  KNOT~*~*~*~H_i  KNOT~*~*~*~
↪LFAKTAKT_i  KNOT~*~*~*~P_i  KNOT~*~*~*~PH_i  KNOT~*~*~*~PH_EIN_i  KNOT~*~*~
↪*~QM_i  KNOT~*~*~*~RHO_i  KNOT~*~*~*~T_i  KNOT~*~*~*~EH_i
OBJTYPE OBJID
↪
↪
↪
ROHR    4979507900871287244               0.83    10106.165039
↪     1.0       11.106165        10.106165            10.106165          0.
↪00000           7.888546       49.020111     13046.647461
        5114681686941855110               0.83    16375.773438
↪     1.0       17.375772        16.375772            16.375772          0.
↪00000          12.826390       40.874298     13008.437500
        5244313507655010738               0.83    39973.992188
↪     1.0       40.973991        39.973991            39.973991          0.
↪00000          31.891241       39.985443     12777.368164
        5694016449043789006               0.83    10106.165039
↪     1.0       11.106165        10.106165            10.106165          0.
↪00000           7.888546       49.020111     13046.647461
VENT    5116489323526156845               0.83    16440.441406
↪     1.0       17.440441        16.440441            16.440441          0.
↪00000          12.874356       40.910797     13111.193359
        5309992331398639768               0.83    40000.000000
↪     1.0       41.000000        40.000000            40.000000     118257.
↪53125          31.911810       40.000000     12777.444336
        5508684139418025293               0.83    16440.441406
↪     1.0       17.440441        16.440441            16.440441          0.
↪00000          12.874356       40.910797     13111.193359
```

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```
        5745097345184516675                0.83    10000.000000                ␣
↪     1.0       11.000000        10.000000            10.000000    -118257.
↪53125          7.812713       48.967743     13034.717773
>>> print(xm._getvXXXXAsOneString(vXXXX='vVBEL',filterColList=['KNOT~*~*~*~
↪HMAX_INST_i','KNOT~*~*~*~HMIN_INST_i','KNOT~*~*~*~PMAX_INST_i','KNOT~*~*~*~
↪PMIN_INST_i','KNOT~*~*~*~IAKTIV_i','KNOT~*~*~*~PDAMPF_i']))
                              KNOT~*~*~*~HMAX_INST_i  KNOT~*~*~*~HMIN_INST_i ␣
↪KNOT~*~*~*~PMAX_INST_i  KNOT~*~*~*~PMIN_INST_i  KNOT~*~*~*~IAKTIV_i  KNOT~
↪*~*~*~PDAMPF_i
OBJTYPE OBJID                                                                ␣
↪                                                                            ␣
↪
ROHR    4979507900871287244            10106.165039            10106.165039 ␣
↪             11.106165               11.106165                  0.0       ␣
↪          0.0
        5114681686941855110            16375.773438            16375.773438 ␣
↪             17.375772               17.375772                  0.0       ␣
↪          0.0
        5244313507655010738            39973.992188            39973.992188 ␣
↪             40.973991               40.973991                  0.0       ␣
↪          0.0
        5694016449043789006            10106.165039            10106.165039 ␣
↪             11.106165               11.106165                  0.0       ␣
↪          0.0
VENT    5116489323526156845            16440.441406            16440.441406 ␣
↪             17.440441               17.440441                  0.0       ␣
↪          0.0
        5309992331398639768            40000.000000            40000.000000 ␣
↪             41.000000               41.000000                  0.0       ␣
↪          0.0
        5508684139418025293            16440.441406            16440.441406 ␣
↪             17.440441               17.440441                  0.0       ␣
↪          0.0
        5745097345184516675            10000.000000            10000.000000 ␣
↪             11.000000               11.000000                  0.0       ␣
↪          0.0
```

**_MxAddvROHRVecResults** (*dfSource*)

(Re-)constructing vROHRVecResults.

**Arguments:**

- dfSource

**Result:**

- **vROHRVecResults: VEC-Channel-Results for Pipe-Interior-Pts (IPts):**
  - pk
  - mx2Idx
  - IptIdx: S,0,...,E - Interior Point Index; S=Start EdgeDefNode, E=End EdgeDefNode, 0=1st Ipt in EdgeDefDirection
  - one column per VEC-Channel

**Raises:** XmError

```
>>> xm=xms['GPipes']
>>> mx=xm.MxSync()
>>> mxVecsFileDataLst=mx.getMxsVecsFileData()
>>> mxVecsFileData=mxVecsFileDataLst[0]
>>> # construct MultiIndex Start ... ---
>>> colsToBeUnpacked=mxVecsFileData.columns.tolist() # all columns
>>> arrays=[[mxVecsFileData.index[0]]*len(colsToBeUnpacked),colsToBeUnpacked]
>>> tuples = list(zip(*arrays))
>>> import pandas as pd
>>> mIndex = pd.MultiIndex.from_tuples(tuples, names=['Timestamp', 'Sir3sID
↪'])
>>> # construct MultiIndex End ... ---
>>> dfUnpacked=mx.unPackMxsVecsFileDataDf(mxVecsFileData,mIndex,
↪returnMultiIndex=False)
>>> xm._MxAddvROHRVecResults(dfSource=dfUnpacked)
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHRVecResults',sortList=['ROHR~*~
↪*~*~PHVEC','ROHR~*~*~*~SVEC'],ascending=False))
                     pk  mx2Idx IptIdx  ROHR~*~*~*~SVEC  ROHR~*~*~*~TVEC ␣
↪ROHR~*~*~*~ZVEC  ROHR~*~*~*~PVEC  ROHR~*~*~*~MVEC  ROHR~*~*~*~RHOVEC  ROHR~
↪*~*~*~PHVEC  ROHR~*~*~*~QMVEC
8   5244313507655010738       0      S         0.000000        39.985443    ␣
↪       0.000        40.973988        27.264917          31.891241        ␣
↪39.973988       3077.763672
9   5244313507655010738       0      0      5000.000000        40.407806    ␣
↪       3.125        40.453373        27.264917          31.443642        ␣
↪39.453373       3121.575439
10  5244313507655010738       0      1     10000.000000        40.397430    ␣
↪       6.250        39.925545        27.264917          30.999212        ␣
↪38.925545       3166.328857
11  5244313507655010738       0      2     15000.000000        40.387299    ␣
↪       9.375        39.390327        27.264917          30.549555        ␣
↪38.390327       3212.934082
12  5244313507655010738       0      3     20000.000000        40.377533    ␣
↪      12.500        38.847401        27.264917          30.094410        ␣
↪37.847401       3261.526123
13  5244313507655010738       0      4     25000.000000        40.368195    ␣
↪      15.625        38.296429        27.264917          29.633537        ␣
↪37.296429       3312.250732
14  5244313507655010738       0      5     30000.000000        40.359253    ␣
↪      18.750        37.737045        27.264917          29.166674        ␣
↪36.737045       3365.269043
15  5244313507655010738       0      6     35000.000000        40.350830    ␣
↪      21.875        37.168861        27.264917          28.693537        ␣
↪36.168861       3420.760010
16  5244313507655010738       0      7     40000.000000        40.342926    ␣
↪      25.000        36.591457        27.264917          28.213825        ␣
↪35.591457       3478.922119
17  5244313507655010738       0      8     45000.000000        40.335602    ␣
↪      28.125        36.004372        27.264917          27.727211        ␣
↪35.004372       3539.977539
18  5244313507655010738       0      9     50000.000000        40.328918    ␣
↪      31.250        35.407112        27.264917          27.233337        ␣
↪34.407112       3604.174561
19  5244313507655010738       0     10     55000.000000        40.322968    ␣
↪      34.375        34.799137        27.264917          26.731815        ␣
↪33.799137       3671.793457
20  5244313507655010738       0     11     60000.000000        40.317780    ␣
↪      37.500        34.179863        27.264917          26.222223        ␣
↪33.179863       3743.149414
```

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```
21  5244313507655010738       0     12     65000.000000        40.313477
↪      40.625        33.548645        27.264917          25.704098
↪32.548645       3818.601318
22  5244313507655010738       0     13     70000.000000        40.310181
↪      43.750        32.904778        27.264917          25.176931
↪31.904778       3898.557129
23  5244313507655010738       0     14     75000.000000        40.308014
↪      46.875        32.247482        27.264917          24.640163
↪31.247482       3983.484131
24  5244313507655010738       0     15     80000.000000        40.307098
↪      50.000        31.575897        27.264917          24.093176
↪30.575897       4073.921387
25  5244313507655010738       0     16     85000.000000        40.307587
↪      53.125        30.889069        27.264917          23.535278
↪29.889069       4170.492676
26  5244313507655010738       0     17     90000.000000        40.309753
↪      56.250        30.185926        27.264917          22.965700
↪29.185926       4273.925781
27  5244313507655010738       0     18     95000.000000        40.313782
↪      59.375        29.465275        27.264917          22.383583
↪28.465275       4385.075684
28  5244313507655010738       0     19    100000.000000        40.320038
↪      62.500        28.725761        27.264917          21.787949
↪27.725761       4504.953613
29  5244313507655010738       0     20    105000.000000        40.328827
↪      65.625        27.965855        27.264917          21.177694
↪26.965855       4634.768066
30  5244313507655010738       0     21    110000.000000        40.340607
↪      68.750        27.183802        27.264917          20.551556
↪26.183802       4775.974121
31  5244313507655010738       0     22    115000.000000        40.355988
↪      71.875        26.377584        27.264917          19.908077
↪25.377584       4930.345703
32  5244313507655010738       0     23    120000.000000        40.375671
↪      75.000        25.544865        27.264917          19.245565
↪24.544865       5100.068359
33  5244313507655010738       0     24    125000.000000        40.400543
↪      78.125        24.682890        27.264917          18.562040
↪23.682890       5287.872559
34  5244313507655010738       0     25    130000.000000        40.431854
↪      81.250        23.788412        27.264917          17.855148
↪22.788412       5497.221191
35  5244313507655010738       0     26    135000.000000        40.471252
↪      84.375        22.857525        27.264917          17.122063
↪21.857525       5732.586426
36  5244313507655010738       0     27    140000.000000        40.520874
↪      87.500        21.885471        27.264917          16.359346
↪20.885471       5999.854980
37  5244313507655010738       0     28    145000.000000        40.583862
↪      90.625        20.866375        27.264917          15.562735
↪19.866375       6306.970215
38  5244313507655010738       0     29    150000.000000        40.664673
↪      93.750        19.792791        27.264917          14.726833
↪18.792791       6664.956543
39  5244313507655010738       0     30    155000.000000        40.770111
↪      96.875        18.655102        27.264917          13.844651
↪17.655102       7089.647949
```

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```
40   5244313507655010738       0      E    160000.000000        40.910797
↪     100.000        17.440462        27.264917          12.906846
↪16.440462       7604.778320
5    5114681686941855110       1      S         0.000000        40.874298
↪       0.000        17.375792        27.264917          12.826405
↪16.375792       7652.472168
6    5114681686941855110       1      0      5000.000000        43.653412
↪       0.000        16.058306        27.264917          11.750417
↪15.058306       8353.209961
7    5114681686941855110       1      E     10000.000000        43.972473
↪       0.000        14.612197        27.264917          10.651937
↪13.612197       9214.634766
4    4979507900871287244       2      E     10000.000000        43.972504
↪       0.000        14.612185       -27.264917          10.620180
↪13.612185      -9242.188477
3    4979507900871287244       2      0      5000.000000        48.149719
↪       0.000        12.987305       -27.264917           9.317566
↪11.987305     -10534.264648
2    4979507900871287244       2      S         0.000000        49.020142
↪       0.000        11.106176       -27.264917           7.919193
↪10.106176     -12394.407227
0    5694016449043789006       3      S         0.000000        10.000000
↪       0.000        11.106160         0.000000           6.933107
↪10.106160          0.000000
1    5694016449043789006       3      E       100.498688        10.000000
↪       0.000         1.000000         0.000000           0.791800
↪0.000000          0.000000
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHRVecResults'))
                     pk  mx2Idx IptIdx  ROHR~*~*~*~SVEC  ROHR~*~*~*~TVEC
↪ROHR~*~*~*~ZVEC  ROHR~*~*~*~PVEC  ROHR~*~*~*~MVEC  ROHR~*~*~*~RHOVEC  ROHR~
↪*~*~*~PHVEC  ROHR~*~*~*~QMVEC
0    5694016449043789006       3      S         0.000000        10.000000
↪       0.000        11.106160         0.000000           6.933107
↪10.106160          0.000000
1    5694016449043789006       3      E       100.498688        10.000000
↪       0.000         1.000000         0.000000           0.791800
↪0.000000          0.000000
2    4979507900871287244       2      S         0.000000        49.020142
↪       0.000        11.106176       -27.264917           7.919193
↪10.106176     -12394.407227
3    4979507900871287244       2      0      5000.000000        48.149719
↪       0.000        12.987305       -27.264917           9.317566
↪11.987305     -10534.264648
4    4979507900871287244       2      E     10000.000000        43.972504
↪       0.000        14.612185       -27.264917          10.620180
↪13.612185      -9242.188477
5    5114681686941855110       1      S         0.000000        40.874298
↪       0.000        17.375792        27.264917          12.826405
↪16.375792       7652.472168
6    5114681686941855110       1      0      5000.000000        43.653412
↪       0.000        16.058306        27.264917          11.750417
↪15.058306       8353.209961
7    5114681686941855110       1      E     10000.000000        43.972473
↪       0.000        14.612197        27.264917          10.651937
↪13.612197       9214.634766
8    5244313507655010738       0      S         0.000000        39.985443
↪       0.000        40.973988        27.264917          31.891241
↪39.973988       3077.763672
```

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```
9   5244313507655010738       0      0      5000.000000        40.407806
↪        3.125        40.453373        27.264917          31.443642
↪39.453373       3121.575439
10  5244313507655010738       0      1     10000.000000        40.397430
↪        6.250        39.925545        27.264917          30.999212
↪38.925545       3166.328857
11  5244313507655010738       0      2     15000.000000        40.387299
↪        9.375        39.390327        27.264917          30.549555
↪38.390327       3212.934082
12  5244313507655010738       0      3     20000.000000        40.377533
↪       12.500        38.847401        27.264917          30.094410
↪37.847401       3261.526123
13  5244313507655010738       0      4     25000.000000        40.368195
↪       15.625        38.296429        27.264917          29.633537
↪37.296429       3312.250732
14  5244313507655010738       0      5     30000.000000        40.359253
↪       18.750        37.737045        27.264917          29.166674
↪36.737045       3365.269043
15  5244313507655010738       0      6     35000.000000        40.350830
↪       21.875        37.168861        27.264917          28.693537
↪36.168861       3420.760010
16  5244313507655010738       0      7     40000.000000        40.342926
↪       25.000        36.591457        27.264917          28.213825
↪35.591457       3478.922119
17  5244313507655010738       0      8     45000.000000        40.335602
↪       28.125        36.004372        27.264917          27.727211
↪35.004372       3539.977539
18  5244313507655010738       0      9     50000.000000        40.328918
↪       31.250        35.407112        27.264917          27.233337
↪34.407112       3604.174561
19  5244313507655010738       0     10     55000.000000        40.322968
↪       34.375        34.799137        27.264917          26.731815
↪33.799137       3671.793457
20  5244313507655010738       0     11     60000.000000        40.317780
↪       37.500        34.179863        27.264917          26.222223
↪33.179863       3743.149414
21  5244313507655010738       0     12     65000.000000        40.313477
↪       40.625        33.548645        27.264917          25.704098
↪32.548645       3818.601318
22  5244313507655010738       0     13     70000.000000        40.310181
↪       43.750        32.904778        27.264917          25.176931
↪31.904778       3898.557129
23  5244313507655010738       0     14     75000.000000        40.308014
↪       46.875        32.247482        27.264917          24.640163
↪31.247482       3983.484131
24  5244313507655010738       0     15     80000.000000        40.307098
↪       50.000        31.575897        27.264917          24.093176
↪30.575897       4073.921387
25  5244313507655010738       0     16     85000.000000        40.307587
↪       53.125        30.889069        27.264917          23.535278
↪29.889069       4170.492676
26  5244313507655010738       0     17     90000.000000        40.309753
↪       56.250        30.185926        27.264917          22.965700
↪29.185926       4273.925781
27  5244313507655010738       0     18     95000.000000        40.313782
↪       59.375        29.465275        27.264917          22.383583
↪28.465275       4385.075684
```

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```
28  5244313507655010738       0     19    100000.000000        40.320038
↪       62.500        28.725761        27.264917          21.787949
↪27.725761       4504.953613
29  5244313507655010738       0     20    105000.000000        40.328827
↪       65.625        27.965855        27.264917          21.177694
↪26.965855       4634.768066
30  5244313507655010738       0     21    110000.000000        40.340607
↪       68.750        27.183802        27.264917          20.551556
↪26.183802       4775.974121
31  5244313507655010738       0     22    115000.000000        40.355988
↪       71.875        26.377584        27.264917          19.908077
↪25.377584       4930.345703
32  5244313507655010738       0     23    120000.000000        40.375671
↪       75.000        25.544865        27.264917          19.245565
↪24.544865       5100.068359
33  5244313507655010738       0     24    125000.000000        40.400543
↪       78.125        24.682890        27.264917          18.562040
↪23.682890       5287.872559
34  5244313507655010738       0     25    130000.000000        40.431854
↪       81.250        23.788412        27.264917          17.855148
↪22.788412       5497.221191
35  5244313507655010738       0     26    135000.000000        40.471252
↪       84.375        22.857525        27.264917          17.122063
↪21.857525       5732.586426
36  5244313507655010738       0     27    140000.000000        40.520874
↪       87.500        21.885471        27.264917          16.359346
↪20.885471       5999.854980
37  5244313507655010738       0     28    145000.000000        40.583862
↪       90.625        20.866375        27.264917          15.562735
↪19.866375       6306.970215
38  5244313507655010738       0     29    150000.000000        40.664673
↪       93.750        19.792791        27.264917          14.726833
↪18.792791       6664.956543
39  5244313507655010738       0     30    155000.000000        40.770111
↪       96.875        18.655102        27.264917          13.844651
↪17.655102       7089.647949
40  5244313507655010738       0      E    160000.000000        40.910797
↪      100.000        17.440462        27.264917          12.906846
↪16.440462       7604.778320
>>> xm=xms['LocalHeatingNetwork']
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> print(xm._getvXXXXAsOneString(vXXXX='vROHRVecResults',filterColList=[
↪'mx2Idx','IptIdx','ROHR~*~*~*~SVEC'],index=True))
    mx2Idx IptIdx  ROHR~*~*~*~SVEC
0        0      S         0.000000
1        0      E        88.019997
2        1      S         0.000000
3        1      E       405.959991
4        2      S         0.000000
5        2      E        83.550003
6        3      S         0.000000
7        3      E        88.019997
8        5      S         0.000000
9        5      E       195.529999
10       7      S         0.000000
11       7      E       109.769997
```

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```
12       8      S         0.000000
13       8      E        76.400002
14       9      S         0.000000
15       9      E        83.550003
16      10      S         0.000000
17      10      E       164.910004
18      11      S         0.000000
19      11      E       195.529999
20      12      S         0.000000
21      12      E       405.959991
22      13      S         0.000000
23      13      E       164.910004
24      14      S         0.000000
25      14      E       109.769997
26      15      S         0.000000
27      15      E        76.400002
28       4      S         0.000000
29       4      E        73.419998
30       6      S         0.000000
31       6      E        68.599998
```

**_MxAddvAGSN** ( )

(Re-)constructing vAGSN.

Result:

- **vAGSN**
  - from vVBEL: KNOT…_i and KNOT…_k and Q
  - from vROHRVecResults: vecResults
  - **Topology:**
    - nextNODE
    - IptIdx
  - **Geometry:**
    - dx
    - x
    - xVbel
    - Z (the corresponding Z_i, Z_k and ZVEC are droped)
  - **Results:**
    - Q: from Q before and QMVEC for PIPEs; in Schnittrichtung; QMVEC is droped
    - **for available KNOT…#_i, KNOT…#_k and …#VEC:**
      - i.e. KNOT~*~*~*~P_i KNOT~*~*~*~P_k ROHR~*~*~*~PVEC
      - P is new column
      - the correspondig 3 source-columns are droped

**Raises:** XmError

```
>>> xm=xms['GPipes']
>>> mx=xm.MxSync()
>>> xm.MxAdd(mx=mx)
>>> vAGSN=xm.dataFrames['vAGSN']
>>> schnitt=vAGSN[vAGSN['NAME']=='LR']
>>> xm.dataFrames['schnitt']=schnitt
>>> print(xm._getvXXXXAsOneString(vXXXX='schnitt',filterColList=['OBJTYPE',
↪'NAME_i','NAME_k','IptIdx','nextNODE','x','PH','Q'],index=True))
    OBJTYPE NAME_i NAME_k IptIdx nextNODE         x        PH        Q
79     VENT     GL     G1      S       G1       0.0        40   118258
80     VENT     GL     G1      E       G1       0.0    39.974   118258
81     ROHR     G1    GKS      S      GKS       0.0    39.974  3077.76
82     ROHR     G1    GKS      0      GKS    5000.0   39.4534  3121.58
83     ROHR     G1    GKS      1      GKS   10000.0   38.9255  3166.33
84     ROHR     G1    GKS      2      GKS   15000.0   38.3903  3212.93
85     ROHR     G1    GKS      3      GKS   20000.0   37.8474  3261.53
86     ROHR     G1    GKS      4      GKS   25000.0   37.2964  3312.25
87     ROHR     G1    GKS      5      GKS   30000.0    36.737  3365.27
88     ROHR     G1    GKS      6      GKS   35000.0   36.1689  3420.76
89     ROHR     G1    GKS      7      GKS   40000.0   35.5915  3478.92
90     ROHR     G1    GKS      8      GKS   45000.0   35.0044  3539.98
91     ROHR     G1    GKS      9      GKS   50000.0   34.4071  3604.17
92     ROHR     G1    GKS     10      GKS   55000.0   33.7991  3671.79
93     ROHR     G1    GKS     11      GKS   60000.0   33.1799  3743.15
94     ROHR     G1    GKS     12      GKS   65000.0   32.5486   3818.6
95     ROHR     G1    GKS     13      GKS   70000.0   31.9048  3898.56
96     ROHR     G1    GKS     14      GKS   75000.0   31.2475  3983.48
97     ROHR     G1    GKS     15      GKS   80000.0   30.5759  4073.92
98     ROHR     G1    GKS     16      GKS   85000.0   29.8891  4170.49
99     ROHR     G1    GKS     17      GKS   90000.0   29.1859  4273.93
100    ROHR     G1    GKS     18      GKS   95000.0   28.4653  4385.08
101    ROHR     G1    GKS     19      GKS  100000.0   27.7258  4504.95
102    ROHR     G1    GKS     20      GKS  105000.0   26.9659  4634.77
103    ROHR     G1    GKS     21      GKS  110000.0   26.1838  4775.97
104    ROHR     G1    GKS     22      GKS  115000.0   25.3776  4930.35
105    ROHR     G1    GKS     23      GKS  120000.0   24.5449  5100.07
106    ROHR     G1    GKS     24      GKS  125000.0   23.6829  5287.87
107    ROHR     G1    GKS     25      GKS  130000.0   22.7884  5497.22
108    ROHR     G1    GKS     26      GKS  135000.0   21.8575  5732.59
109    ROHR     G1    GKS     27      GKS  140000.0   20.8855  5999.85
110    ROHR     G1    GKS     28      GKS  145000.0   19.8664  6306.97
111    ROHR     G1    GKS     29      GKS  150000.0   18.7928  6664.96
112    ROHR     G1    GKS     30      GKS  155000.0   17.6551  7089.65
113    ROHR     G1    GKS      E      GKS  160000.0   16.4405  7604.78
114    VENT    GKS    GKD      S      GKD  160000.0   16.4404   118258
115    VENT    GKS    GKD      E      GKD  160000.0   16.3758   118258
116    ROHR    GKD     G3      S       G3  160000.0   16.3758  7652.47
117    ROHR    GKD     G3      0       G3  165000.0   15.0583  8353.21
118    ROHR    GKD     G3      E       G3  170000.0   13.6122  9214.63
119    ROHR     G4     G3      S       G4  170000.0   13.6122  9242.19
120    ROHR     G4     G3      0       G4  175000.0   11.9873  10534.3
121    ROHR     G4     G3      E       G4  180000.0   10.1062  12394.4
122    VENT     GR     G4      S       GR  180000.0   10.1062   118258
123    VENT     GR     G4      E       GR  180000.0        10   118258
```

**_Xm__Mx1_Sir3sIDUpd**(*mx*, *checkAllChannels=True*)

Update NAME1,2 and Sir3sID in mx.mx1Df and mx.df.

**Args:** mx: Mx-Object checkAllChannels: if False only Channels with empty NAME1 are updated; default: True: all Channels are checked and updated if necessary

**Notes:**

**The following OBJTYPEs are covered:**

- KNOT
- WBLZ
- RXXXX
- all Channels in vVBEL

mx.ToH5() is called if Sir3sID-Updates occured and mx.h5Read is True.

**Raises:** XmError

**_Xm__Mx1_Sir3sIDUpd_ObjTypeNode**(*mx=None*, *dfUpd=None*, *dfNAME1=None*, *NAME1Col='NAME'*)

Update Sir3sID and NAME1 in mx.mx1Df and mx.df for Channels in dfUpd.

**Args:** mx: Mx-Object dfUpd: df with OBJTYPE,NAME1,NAME2,OBJTYPE_PK,ATTRTYPE,Sir3sID to be updated dfNAME1: df with NAME1-Information NAME1Col: col in dfNAME1 with NAME1-Information

**Returns:** nOfSir3sIDsUpdated

**Note:** only wrong Channels are updated

**Raises:** XmError

**_Xm__Mx1_vNRCV**(*mx*)

vNRCV_Mx1 (vNRCV with Mx1-Information) is added to dataFrames.

**Args:** mx: Mx-Object

**self.dataFrames['vNRCV_Mx1']**

**index**

- reindex

**FILTERed**

- existing MX-Channels only
- cRefLfdNr: 1st references only

**SORTed**

- Sir3sID

**columns NEW**

- Sir3sID

**Raises:** XmError

**_Xm__Mx2_vFWVB**(*mx*)

Mx2-Information into vFWVB.

**Args:** mx: Mx-Object

**self.dataFrames['vFWVB']**

**columns NEW**

- mx2Idx

**Raises:** XmError

**_Xm__Mx2_vKNOT**(*mx*)

Mx2-Information into vKNOT.

**Args:** mx: Mx-Object

**self.dataFrames['vKNOT']**

**columns NEW**

- mx2Idx

**Raises:** XmError

**_Xm__Mx2_vROHR**(*mx*)

Mx2-Information into vROHR.

**Args:** mx: Mx-Object

**self.dataFrames['vROHR']**

**columns NEW**

- mx2NoPts
- mx2Idx

**Raises:** XmError

**_Xm__Mx2_vVBEL**(*mx, edges=['ROHR', 'VENT', 'FWVB', 'FWES', 'PUMP', 'KLAP', 'REGV', 'PREG', 'MREG', 'DPRG', 'PGRP']*)

Mx2-Information into vVBEL.

**Args:** mx: Mx-Object

**self.dataFrames['vVBEL']:**

**columns NEW**

- mx2Idx
- **Notes:**
  - for all edges mx2Idx is taken from mx.mx2Df

**Raises:** XmError

**_Xm__MxAddForOneDf**(*dfTarget=None*, *dfSource=None*, *multiIndexKey=None*, *testStr='testStr'*)

Add MX2-Resultdata from dfSource as cols to returned dfTarget.

**Args:** dfTarget: df with col mx2Idx dfSource: df with mx2Idx-corresponding index and cols (containing MX2-Resultdata) to be added multiIndexKey: value for 1st Index if dfTarget is Multiindexed - i.e. 'XXXX'

**Notes:**

- all cols from dfSource are added at the end of dfTarget in dfSource-sequence
- the cols can already exist in dfTarget
- if so, _all cols must already exist ...
- ... the dfTarget-sequence should but must be not necessary the dfSource-sequence

**Returns:** dfTarget

**Raises:** XmError

CHAPTER

THREE

# RM

```
>>> # ---
>>> # SETUP
>>> # ---
>>> import os
>>> import logging
>>> logger = logging.getLogger('PT3S.Rm')
>>> # ---
>>> # path
>>> # ---
>>> if __name__ == "__main__":
...   try:
...      dummy=__file__
...      logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ','path =␣
↪os.path.dirname(__file__)'," ."))
...      path = os.path.dirname(__file__)
...   except NameError:
...      logger.debug("{0:s}{1:s}{2:s}".format('DOCTEST: __main__ Context: ","path =
↪'.' because __file__ not defined and: "," from Rm import Rm"))
...      path = '.'
...      from Rm import Rm
... else:
...    path = '.'
...    logger.debug("{0:s}{1:s}".format('Not __main__ Context: ","path = '.' ."))
>>> try:
...    from PT3S import Mx
... except ImportError:
...    logger.debug("{0:s}{1:s}".format("DOCTEST: from PT3S import Mx: ImportError: ",
↪"trying import Mx instead ... maybe pip install -e . is active ..."))
...    import Mx
>>> try:
...    from PT3S import Xm
... except ImportError:
...    logger.debug("{0:s}{1:s}".format("DOCTEST: from PT3S import Xm: ImportError: ",
↪"trying import Xm instead ... maybe pip install -e . is active ..."))
...    import Xm
>>> # ---
>>> # testDir
>>> # ---
>>> # globs={'testDir':'testdata'}
>>> try:
...    dummy= testDir
... except NameError:
...    testDir='testdata'
>>> # ---
```

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```
>>> # dotResolution
>>> # ---
>>> # globs={'dotResolution':''}
>>> try:
...     dummy= dotResolution
... except NameError:
...     dotResolution=''
>>> import pandas as pd
>>> import matplotlib.pyplot as plt
>>> pd.set_option('display.max_columns',None)
>>> pd.set_option('display.width',666666666)
>>> # ---
>>> # LocalHeatingNetwork SETUP
>>> # ---
>>> xmlFile=os.path.join(os.path.join(path,testDir),'LocalHeatingNetwork.XML')
>>> xm=Xm.Xm(xmlFile=xmlFile)
>>> mx1File=os.path.join(path,os.path.join(testDir,'WDLocalHeatingNetwork\B1\V0\BZ1\M-
↪1-0-1'+dotResolution+'.MX1'))
>>> mx=Mx.Mx(mx1File=mx1File,NoH5Read=True,NoMxsRead=True)
>>> mx.setResultsToMxsFile(NewH5Vec=True)
5
>>> xm.MxSync(mx=mx)
>>> rm=Rm(xm=xm,mx=mx)
>>> # ---
>>> # Plot 3Classes False
>>> # ---
>>> plt.close('all')
>>> ppi=72 # matplotlib default
>>> dpi_screen=2*ppi
>>> fig=plt.figure(dpi=dpi_screen,linewidth=1.)
>>> timeDeltaToT=mx.df.index[2]-mx.df.index[0]
>>> # 3Classes und FixedLimits sind standardmaessig Falsch; RefPerc ist␣
↪standardmaessig Wahr
>>> # die Belegung von MCategory gemaess FixedLimitsHigh/Low erfolgt immer ...
>>> pFWVB=rm.pltNetDHUS(timeDeltaToT=timeDeltaToT,pFWVBMeasureCBFixedLimitHigh=0.80,
↪pFWVBMeasureCBFixedLimitLow=0.66,pFWVBGCategory=['BLNZ1u5u7'],pVICsDf=pd.DataFrame({
↪'Kundenname': ['VIC1'],'Knotenname': ['V-K007']}))
>>> # ---
>>> # Check pFWVB Return
>>> # ---
>>> f=lambda x: "{0:8.5f}".format(x)
>>> print(pFWVB[['Measure','MCategory','GCategory','VIC']].to_string(formatters={
↪'Measure':f}))
   Measure MCategory  GCategory   VIC
0  0.80971       Top  BLNZ1u5u7   NaN
1  0.66692    Middle              NaN
2  0.66043    Middle  BLNZ1u5u7   NaN
3  0.65551    Bottom  BLNZ1u5u7  VIC1
4  0.68548    Middle              NaN
>>> # ---
>>> # Print
>>> # ---
>>> (wD,fileName)=os.path.split(xm.xmlFile)
>>> (base,ext)=os.path.splitext(fileName)
>>> plotFileName=wD+os.path.sep+base+'.'+'pdf'
>>> if os.path.exists(plotFileName):
...     os.remove(plotFileName)
```

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```
>>> plt.savefig(plotFileName,dpi=2*dpi_screen)
>>> os.path.exists(plotFileName)
True
>>> # ---
>>> # Plot 3Classes True
>>> # ---
>>> plt.close('all')
>>> # FixedLimits wird automatisch auf Wahr gesetzt wenn 3Classes Wahr ...
>>> pFWVB=rm.pltNetDHUS(timeDeltaToT=timeDeltaToT,pFWVBMeasure3Classes=True,
↪pFWVBMeasureCBFixedLimitHigh=0.80,pFWVBMeasureCBFixedLimitLow=0.66)
>>> # ---
>>> # LocalHeatingNetwork Clean Up
>>> # ---
>>> if os.path.exists(mx.h5File):
...     os.remove(mx.h5File)
>>> if os.path.exists(mx.mxsZipFile):
...     os.remove(mx.mxsZipFile)
>>> if os.path.exists(mx.h5FileVecs):
...     os.remove(mx.h5FileVecs)
>>> if os.path.exists(plotFileName):
...     os.remove(plotFileName)
```

`Rm.`**`pltNetFigAx`**(*pDf*, ***kwds*)

Erzeugt eine für die Netzdarstellung verzerrungsfreie Axes-Instanz.

- verwendet gcf() (will return an existing figure if one is open, or it will make a new one if there is no active figure)
- an already existing figure might be created this way: fig=plt.figure(dpi=2*72,linewidth=1.)
- errechnet die verzerrungsfreie Darstellung unter Berücksichtigung einer zukünftigen horizontalen Farblegende
- erzeugt eine Axes-Instanz
- setzt Attribute der Axes-Instanz
- setzt Attribute der Figure-Instanz

**Args:** pDf: dataFrame

**Coordinates:**

- pXCor_i: colName in pDf (default: 'pXCor_i'): x-Start Coordinate of all Edges to be plotted
- pYCor_i: colName in pDf (default: 'pYCor_i'): y-Start Coordinate of all Edges to be plotted
- pXCor_k: colName in pDf (default: 'pXCor_k'): x-End Coordinate of all Edges to be plotted
- pYCor_k: colName in pDf (default: 'pYCor_k'): y-End Coordinate of all Edges to be plotted

**Colorlegend:**

- CBFraction: fraction of original axes to use for colorbar (default: 0.05)
- CBHpad: fraction of original axes between colorbar and new image axes (default: 0.0275)

**Figure:**

- pltTitle: title [not suptitle] (default: 'pltNetFigAx')
- figFrameon: figure frame (background): displayed or invisible (default: True)

- figEdgecolor: edge color of the Figure rectangle (default: 'black')
- figFacecolor: face color of the Figure rectangle (default: 'white')

`Rm.pltNetNodes`(*pDf*, ***kwds*)

Scatters NODEs on gca().

**Args:** pDf: dataFrame

**NODE: Size (Attribute)**

- pAttribute: colName (default: 'Attribute') in pDf
- pSizeFactor: (deafault: 1.)
- scatter Sy-Area in pts^2 = pSizeFactor * Attribute

**NODE: Color (Measure)**

- pMeasure: colName (default: 'Measure') in pDf
- pMeasureColorMap (default: plt.cm.autumn)
- pMeasureAlpha (default: 0.9)
- pMeasureClip (default: False)
- CBFixedLimits (default: True)
- CBFixedLimitLow (default: 0.)
- CBFixedLimitHigh (default: 1.)

**NODE: 3Classes**

- pMeasure3Classes (default: True)
- pMCategory: colName (default: 'MCategory') in pDf
- pMCatTopTxt (default: 'Top')
- pMCatMidTxt (default: 'Middle')
- pMCatBotTxt (default: 'Bottom')
- pMCatTopColor (default: 'palegreen')
- pMCatTopAlpha (default: 0.9)
- pMCatTopClip (default: False)
- pMCatMidColorMap (default: plt.cm.autumn)
- pMCatMidAlpha (default: 0.9)
- pMCatMidClip (default: False)
- pMCatBotColor (default: 'violet')
- pMCatBotAlpha (default: 0.9)
- pMCatBotClip (default: False)

**NODE:**

- pXCor: colName (default: 'pXCor_i') in pDf
- pYCor: colName (default: 'pYCor_i') in pDf

**Returns:** (pcN, vmin, vmax)

- pcN: die mit Farbskala gezeichneten Symbole
- vmin/vmax: die für die Farbskala verwendeten Extremalwerte

`Rm.pltNetPipes`(*pDf*, ***kwds*)

Plots Lines with Marker on gca().

**Args:** pDf: dataFrame

**PIPE-Line:**

- pAttribute: column in pDf (default: 'Attribute')
- pAttributeLs (default: '-')
- pAttributeSizeFactor: plot linewidth in pts = pAttributeSizeFactor (default: 1.0) * Attribute
- pAttributeColorMap (default: plt.cm.binary)
- pAttributeColorMapUsageStart (default: 1./3; Wertebereich: [0,1])
  - Farbskala nach vorh. min./max. Wert
  - die Farbskala wird nur ab UsageStart genutzt
  - d.h. Werte die eine "kleinere" Farbe hätten, bekommen die Farbe von UsageStart

**PIPE-Marker:**

- pMeasure: column in pDf (default: 'Measure')
- pMeasureMarker (default: '.')
- pMeasureSizeFactor: plot markersize in pts = pMeasureSizeFactor (default: 1.0) * Measure
- pMeasureColorMap (default: plt.cm.cool)
- pMeasureColorMapUsageStart (default: 0.; Wertebereich: [0,1])
  - Farbskala nach vorh. min./max. Wert
  - die Farbskala wird nur ab UsageStart genutzt
  - d.h. Werte die eine "kleinere" Farbe hätten, bekommen die Farbe von UsageStart

**PIPE:**

- pWAYPXCors: column in pDf (default: 'pWAYPXCors')
- pWAYPYCors: column in pDf (default: 'pWAYPYCors')
- pClip (default: False)

`Rm.pltNetLegendColorbar`(*pc*, *pDf*, ***kwds*)

Erzeugt eine Axes cax für den Legendenbereich aus ax (=gca()) und zeichnet auf cax die Farblegende (die Farbskala mit allen Eigenschaften).

**Args:** pc: (eingefaerbte) PathCollection (aus pltNetNodes); wird für die Erzeugung der Farbskala zwingend benoetigt pDf: dataFrame (default: None)

**Measure:**

- pMeasure: colName in pDf (default: 'Measure')
- pMeasureInPerc: Measure wird interpretiert in Prozent [0-1] (default: True)
- pMeasure3Classes (default: False d.h. Measure wird nicht in 3 Klassen dargestellt)

**CBFixedLimits (Ticks):**

- CBFixedLimits (default: False d.h. Farbskala nach vorh. min./max. Wert)
- CBFixedLimitLow (default: .10)
- CBFixedLimitHigh (default: .95)

**Label:**

- pMeasureUNIT (default: '[]')
- pMeasureTYPE (default: '')

**CB**

- CBFraction: fraction of original axes to use for colorbar (default: 0.05)
- CBHpad: fraction of original axes between colorbar and new image axes (default: 0.0275)
- CBLabelPad (default: -50)
- CBTicklabelsHPad (default: 0.)
- CBAspect: ratio of long to short dimension (default: 10.)
- CBShrink: fraction by which to shrink the colorbar (default: 0.3)
- CBAnchorHorizontal: horizontaler Fußpunkt der colorbar in Plot-% (default: 0.)
- CBAnchorVertical: vertikaler Fußpunkt der colorbar in Plot-% (default: 0.2)

**Return:** cax

`Rm.`**`pltNetLegendColorbar3Classes`**(*pDf*, ***kwds*)

Zeichnet auf gca() die ergaenzenden Legendeninformationen bei 3 Klassen.

- scatters the Top-Symbol
- scatters the Bot-Symbol
- the "Mid-Symbol" is the (already existing) colorbar with (already existing) ticks and ticklabels

Args:

pDf: dataFrame

**Category:**

- pMCategory: colName in pDf (default: 'MCategory')
- pMCatTopText
- pMCatMidText
- pMCatBotText

**CBLegend (3Classes) - Parameterization of the representative Symbols**

- CBLe3cTopVPad (default: 1+1*1/4)
- CBLe3cMidVPad (default: .5)
- CBLe3cBotVPad (default: 0-1*1/4)
  - 1 is the height of the Colorbar
  - the VPads (the vertical Sy-Positions) are defined in cax.transAxes Coordinates
  - cax is the Colorbar Axes
- CBLe3cSySize=10**2 (Sy-Area in pts^2)

- CBLe3cSyType='o'

**Color:**

- pMCatBotColor='violet'
- pMCatTopColor='palegreen'

**Returns:** (bbTop, bbMid, bbBot): the boundingBoxes of the 3Classes-Symbols

**class** Rm.**Rm**(*xm=None*, *mx=None*)

**classmethod pltNetPipes**(*pDf*, ***kwds*)

Plots colored PIPES.

**Args:**

**DATA:**

**pDf: dataFrame**

- query: query to filter pDf; default: None; Exp.: ="CONT_ID == '1001'"
- fmask: function to filter pDf; default: None; Exp.: =lambda row: True if row.KVR_i=='2' and row.KVR_k=='2' else False
- query and fmask are used both (query 1st) if not None
- sort_values_by: list of colNames defining the plot order; default: None
- sort_values_ascending; default: False (d.h. kleine zuletzt und damit i.d.R. auch dünne über dicke)

**AXES:** pAx: Axes to be plotted on; if not specified: gca() is used

**Colorlegend:**

- CBFraction in % (default: 5)
- CBHpad (default: 0.05)
- CBLabel (default: pAttribute/pAttributeFunc)

**PIPE-Attribute:**

- pAttribute: column in pDf (default: 'Attribute')
- **pAttributeFunc:**
  - function to be used to construct a new col to be plotted
  - if pAttributeFunc is not None pAttribute is not used: pAttribute is set to 'pAttributeFunc'
  - the new constructed col is named 'pAttributeFunc'; this name can be used in sort_values_by

**PIPE-Color:**

- pAttributeColorMap (default: plt.cm.cool)
- Farbskalamapping:
- 
- pAttributeColorMapMin (default: pAttribute.min()); ordnet der kleinsten Farbe einen Wert zu

- pAttributeColorMapMax (default: pAttribute.max()); ordnet der größten Farbe einen Wert zu
- Standard: Farbskala wird (wenn die Werte die vorstehend parametrierten Min/Max erreichen) voll ausgenutzt
- **pAttributeColorMapUsageStart (default: 0.; Wertebereich: [0,1[)**
  - hier: die Farbskala wird unten nur ab UsageStart genutzt …
  - … d.h. Werte die eine "kleinere" Farbe hätten, bekommen die Farbe von UsageStart
- **pAttributeColorMapUsageEnd (default: 1.; Wertebereich: ]0,1])**
  - hier: die Farbskala wird oben nur bis UsageEnd genutzt …
  - … d.h. Werte die eine "größere" Farbe hätten, bekommen die Farbe von UsageEnd

**PIPE-Color 2nd:**

- um "unwichtige" Bereiche zu "dimmen":
- Beispiele:
- räumlich: nicht-Schnitt Bereiche; Bestand (2nd) vs. Ausbau; Zonen unwichtig (2nd) vs. Zonen wichtig
- "Ampel": "uninteressante" Wertebereiche
- es wird dieselbe Spalte pAttribute/pAttributeFunc für die 2. Farbskala verwendet
- es wird derselbe Linienstil (pAttributeLs) für die 2. Farbskala verwendet
- es wird dieselbe Dicke pAttrLineSize (pAttribute/pAttributeFunc) für die 2. Farbskala verwendet
- nur die Farbskala ist anders sowie ggf. das Farbskalamapping
- pAttributeColorMapFmask: function to filter pDf to decide to plot with colorMap; default: =lambda row: True
- pAttributeColorMap2ndFmask: function to filter pDf to decide to plot with colorMap2nd; default: =lambda row: False
- mit den beiden Funktionsmasken kann eine Filterung zusätzlich zu query und fmask realisiert werden
- die Funktionsmasken sollten schnittmengenfrei sein - wenn nicht: 2nd überschreibt
- pAttributeColorMap2nd (default: plt.cm.binary)
- Farbskalamapping:
-
- pAttributeColorMap2ndMin (default: pAttributeColorMapMin)
- pAttributeColorMap2ndMax (default: pAttributeColorMapMax)
- pAttributeColorMap2ndUsageStart (default: 0.; Wertebereich: [0,1[)
- pAttributeColorMap2ndUsageEnd (default: 1.; Wertebereich: ]0,1])

**PIPE-Linestyle:**

- pAttributeLs (default: '-')
- same for all colors if mutliple colors are specified

**PIPE-Linesize:**

- pAttrLineSize: column in pDf; if not specified: pAttribute/pAttributeFunc
- pAttrLineSizeFactor (>0): plot linewidth in pts = pAttrLineSizeFactor (default: ...) * fabs(pAttrLineSize)
- ...: 1./(pDf[pAttrLineSize].std()*2.)
- same for all colors if mutliple colors are specified

**PIPE-Geometry:**

- pWAYPXCors: column in pDf (default: 'pWAYPXCors')
- pWAYPYCors: column in pDf (default: 'pWAYPYCors')
- pClip (default: True)

```
>>> import pandas as pd
>>> import matplotlib.pyplot as plt
>>> import matplotlib.gridspec as gridspec
>>> import math
>>> # ---
>>> xm=xms['DHNetwork']
>>> #mx=mxs['DHNetwork']
>>> # ---
>>> plt.close()
>>> size_DINA3quer=(16.5, 11.7)
>>> plt.rc('figure',figsize=size_DINA3quer)
>>> plt.rc('figure',dpi=72)
>>> plt.rc('savefig',dpi=72*2)
>>> fig=plt.figure()
>>> gs = gridspec.GridSpec(4, 2)
>>> # ---
>>> vROHR=xm.dataFrames['vROHR']
>>> # ---
>>> import Rm
>>> # ---
>>> # Attribute with neg. Values
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[0])
>>> Rm.Rm.pltNetPipes(vROHR
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttribute='ROHR~*~*~*~QMAV'
...     )
>>> # ---
>>> # Function as Attribute
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[1])
>>> Rm.Rm.pltNetPipes(vROHR
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     )
>>> # --------------------------
```

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```
>>> # ---
>>> # Farbskalamapping u. -eigenschaften u. Plotorder groß auf klein
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[2])
>>> Rm.Rm.pltNetPipes(vROHR
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     ,pAttributeColorMapMin=0.
...     ,pAttributeColorMapMax=1600.
...     ,CBLabel='Q [t/h]'
...     ,sort_values_by=['pAttributeFunc']
...     ,sort_values_ascending=True
...     )
>>> # --------------------------
>>> # ---
>>> # Farbskalamapping u. -eigenschaften u. Plotorder klein auf groß
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[3])
>>> Rm.Rm.pltNetPipes(vROHR
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     ,pAttributeColorMapMin=0.
...     ,pAttributeColorMapMax=1600.
...     ,CBLabel='Q [t/h]'
...     ,sort_values_by=['pAttributeFunc']
...     )
>>> # --------------------------
>>> # ---
>>> # Farbskalamapping u. -eigenschaften u. Plotorder  groß auf klein /␣
↪2nd Color
>>> # --------------------------
>>> vAGSN=xm.dataFrames['vAGSN']
>>> hpRL=vAGSN[(vAGSN['LFDNR']=='1') & (vAGSN['Layer']==2)]
>>> pDf=pd.merge(vROHR
...     ,hpRL[hpRL.IptIdx=='S']
...     ,how='left'
...     ,left_on='pk'
...     ,right_on='OBJID'
...     ,suffixes=('','_AGSN')).filter(items=vROHR.columns.tolist()+[
↪'OBJID'])
>>> axNfd = fig.add_subplot(gs[4])
>>> Rm.Rm.pltNetPipes(pDf
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     ,pAttributeColorMapMin=0.
...     ,pAttributeColorMapMax=1600.
...     ,CBLabel='Q [t/h]'
...     ,sort_values_by=['pAttributeFunc']
```

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```
...     ,sort_values_ascending=True
...     ,pAttributeColorMapFmask=lambda row: True if not pd.isnull(row.
↪OBJID) else False
...     ,pAttributeColorMap2ndFmask=lambda row: True if pd.isnull(row.
↪OBJID) else False
...     )
>>> # --------------------------
>>> # ---
>>> # Farbskalamapping u. -eigenschaften u. Plotorder  groß auf klein /␣
↪2nd Color / UsageLimits
>>> # --------------------------
>>> axNfd = fig.add_subplot(gs[5])
>>> Rm.Rm.pltNetPipes(pDf
...     ,query="CONT_ID == '1001'"
...     ,fmask=lambda row: True if row.KVR_i=='2' and row.KVR_k=='2'␣
↪else False
...     ,pAx=axNfd
...     ,pAttributeFunc=lambda row: math.fabs(row['ROHR~*~*~*~QMAV'])
...     ,pAttributeColorMapMin=0.
...     ,pAttributeColorMapMax=1600.
...     ,CBLabel='Q [t/h]'
...     ,sort_values_by=['pAttributeFunc']
...     ,sort_values_ascending=True
...     ,pAttributeColorMapFmask=lambda row: True if not pd.isnull(row.
↪OBJID) else False
...     ,pAttributeColorMap2ndFmask=lambda row: True if pd.isnull(row.
↪OBJID) else False
...     ,pAttributeColorMap2ndUsageStart=.5/5. # nicht zu weiß
...     ,pAttributeColorMap2ndUsageEnd=2.5/5. # nicht zu schwarz
...     ,pAttributeColorMapUsageStart=5/16.
...     ,pAttributeColorMapUsageEnd=11/16.
...     )
>>> # --------------------------
>>> gs.tight_layout(fig)
>>> plt.savefig('pltNetPipes.pdf',format='pdf')
```

**pltNetDHUS**(***kwds*)

Plot: Net: DistrictHeatingUnderSupply.

Args (optional):

**TIMEs (als TIMEDELTA zu Szenariumbeginn):**

- timeDeltaToRef: Reference Scenariotime (for MeasureInRefPerc-Calculations) (default: pd.to_timedelta('0 seconds'))
- timeDeltaToT: Scenariotime (default: pd.to_timedelta('0 seconds'))

**FWVB**

- **pFWVBFilterFunction: Filterfunction to be applied to FWVB to determine the FWVB to be plotted**
  - **default: lambda df: (df.CONT_ID.astype(int).isin([1001])) & (df.W0LFK>0)**
    - **CONT_IDisIn: [1001]**
      - um zu vermeiden, dass FWVB aus Bloecken gezeichnet werden (unwahrscheinlich, dass es solche gibt)

        * **W0LFK>0:**
            · um zu vermeiden, dass versucht wird, FWVB mit der Soll-Leistung 0 zu zeichnen (pFWVBAttribute default is 'W0LFK')

**FWVB Attribute (Size, z-Order)**

- pFWVBAttribute: columnName (default: 'W0LFK')
  - the column must be able to be converted to a float
  - the conversion is done before FilterFunction
  - **see ApplyFunction and NaNValue for conversion details:**
    * **pFWVBAttributeApplyFunction: Function to be applied to column pFWVBAttribute**
      · default: lambda x: pd.to_numeric(x,errors='coerce')
    * **pFWVBAttributeApplyFunctionNaNValue: Value for NaN-Values produced by pFWVBAttribu**
      · default: 0
      · .fillna(pFWVBAttributeApplyFunktionNaNValue).astype(float) is called after ApplyFunction
- pFWVBAttributeAsc: z-Order (default: False d.h. "kleine auf große")
- pFWVBAttributeRefSize: scatter Sy-Area in pts^2 of for RefSizeValue (default: 10**2)
  - corresponding RefSizeValue is Attribute.std() or Attribute.mean() if Attribute.std() is < 1

**FWVB (plot only large (small, medium) FWVB . . . )**

- quantil_pFWVBAttributeHigh <= (default: 1.)
- quantil_pFWVBAttributeLow >= (default: .0)
- default: all FWVB are plotted
- note that Attribute >0 is a precondition

**FWVB Measure (Color)**

- pFWVBMeasure (default: 'FWVB~*~*~*~W')
  - float() must be possible
- pFWVBMeasureInRefPerc (default: True d.h. Measure wird verarbeitet in Prozent T zu Ref)
  - 0-1
  - if refValue is 0 than refPerc-Result is set to 1
- pFWVBMeasureAlpha/Colormap/Clip
- 3Classes
  - **pFWVBMeasure3Classes (default: False)**
    * **False:**
      · Measure wird nicht in 3 Klassen dargestellt

        - die Belegung von MCategory gemaess FixedLimitsHigh/Low erfolgt dennoch
    - CatTexts (werden verwendet wenn 3Classes Wahr gesetzt ist)
        - für CBLegend (3Classes) als _zusätzliche Beschriftung rechts
        - als Texte für die Spalte MCategory in return pFWVB
        - pMCatTopText
        - pMCatMidText
        - pMCatBotText
    - CatAttribs (werden verwendet wenn 3Classes Wahr gesetzt ist)
        - für die Knotendarstellung
        - pMCatTopAlpha/Color/Clip
        - pMCatMidAlpha/Colormap/Clip
        - pMCatBotAlpha/Color/Clip
- CBFixedLimits
    - pFWVBMeasureCBFixedLimits (default: False d.h. Farbskala nach vorh. min./max. Wert)
        - wird Wahr gesetzt sein, wenn 3Classes Wahr gesetzt ist
        - damit die mittlere Farbskala den Klassengrenzen "gehorcht"
    - pFWVBMeasureCBFixedLimitLow (default: .10)
    - pFWVBMeasureCBFixedLimitHigh (default: .95)

**CB**

- CBFraction: fraction of original axes to use for colorbar (default: 0.05)
- CBHpad: fraction of original axes between colorbar and new image axes (default: 0.0275)
- CBLabelPad (default: -50)
- CBTicklabelsHPad (default: 0.)
- CBAspect: ratio of long to short dimension (default: 10.)
- CBShrink: fraction by which to shrink the colorbar (default: .3)
- CBAnchorHorizontal: horizontaler Fußpunkt der colorbar in Plot-% (default: 0.)
- CBAnchorVertical: vertikaler Fußpunkt der colorbar in Plot-% (default: 0.2)

**CBLegend (3Classes) - Parameterization of the representative Symbols**

- CBLe3cTopVPad (default: 1+1*1/4)
- CBLe3cMidVPad (default: .5)
- CBLe3cBotVPad (default: 0-1*1/4)
    - 1 is the height of the Colorbar
    - the VPads (the vertical Sy-Positions) are defined in cax.transAxes Coordinates
    - cax is the Colorbar Axes

- CBLe3cSySize=10**2 (Sy-Area in pts^2)
- CBLe3cSyType='o'

**ROHR**

- **pROHRFilterFunction: Filterfunction to be applied to PIPEs to determine the PIPEs to be plotted**
  - **default: lambda df: (df.KVR.astype(int).isin([2])) & (df.CONT_ID.astype(int).isin([1001])) & (df.**
    - **KVRisIn: [2]**
      - 1: supply-line
      - 2: return-line
    - **CONT_IDisIn: [1001]**
      - um zu vermeiden, dass Rohre aus Bloecken gezeichnet werden (deren Koordinaten nicht zu den Koordinaten von Rohren aus dem Ansichtsblock passen)
    - **DI>0:**
      - um zu vermeiden, dass versucht wird, Rohre mit dem Innendurchmesser 0 zu zeichnen (pROHRAttribute default is 'DI')

**ROHR (PIPE-Line: Size and Color, z-Order)**

- **pROHRAttribute: columnName (default: 'DI')**
  - the column must be able to be converted to a float
  - the conversion is done before FilterFunction
  - **see ApplyFunction and NaNValue for conversion details:**
    - **pROHRAttributeApplyFunction: Function to be applied to column pROHRAttribute**
      - default: lambda x: pd.to_numeric(x,errors='coerce')
    - **pROHRAttributeApplyFunctionNaNValue: Value for NaN-Values produced by pROHRAttri**
      - default: 0
      - .fillna(pROHRAttributeApplyFunktionNaNValue).astype(float) is called after ApplyFunction
- pROHRAttributeAsc: z-Order (default: False d.h. "kleine auf grosse")
- pROHRAttributeLs (default: '-')
- pROHRAttributeRefSize: plot linewidth in pts for RefSizeValue (default: 1.0)
  - corresponding RefSizeValue is Attribute.std() or Attribute.mean() if Attribute.std() is < 1
- pROHRAttributeColorMap (default: plt.cm.binary)
- pROHRAttributeColorMapUsageStart (default: 1./3; Wertebereich: [0,1])
  - Farbskala nach vorh. min./max. Wert
  - die Farbskala wird nur ab UsageStart genutzt

  - d.h. Werte die eine "kleinere" Farbe haetten, bekommen die Farbe von UsageStart

**ROHR (plot only large (small, medium) pipes . . . )**

- quantil_pROHRAttributeHigh <= (default: 1.)
- quantil_pROHRAttributeLow >= (default: .75)
- default: only the largest 25% are plotted
- note that Attribute >0 is a precondition

**ROHR (PIPE-Marker: Size and Color)**

- pROHRMeasure columnName (default: 'ROHR~*~*~*~QMAV')
- pROHRMeasureApplyFunction: Function to be applied to column pROHRMeasure (default: lambda x: math.fabs(x))
- pROHRMeasureMarker (default: '.')
- pROHRMeasureRefSize: plot markersize for RefSizeValue in pts (default: 1.0)
  - corresponding RefSizeValue is Measure.std() or Measure.mean() if Measure.std() is < 1
  - if pROHRMeasureRefSize is None: plot markersize will be plot linewidth
- pROHRMeasureColorMap (default: plt.cm.cool)
- pROHRMeasureColorMapUsageStart (default: 0.; Wertebereich: [0,1])
  - Farbskala nach vorh. min./max. Wert
  - die Farbskala wird nur ab UsageStart genutzt
  - d.h. Werte die eine "kleinere" Farbe hätten, bekommen die Farbe von UsageStart

**NRCVs - NumeRiCal Values to be displayed**

- **pFIGNrcv: List of Sir3sID RegExps to be displayed (i.e. ['KNOT~PKON-Knoten~S*~S+~QM']) defau**
  the 1st Match is used if a RegExp matches more than 1 Channel

  **further Examples for RegExps (and corresponding Texts):**

  - WBLZ~WärmeblnzGes~S*~S+~WES (Generation)
  - WBLZ~WärmeblnzGes~S*~S+~WVB (Load)
  - WBLZ~WärmeblnzGes~S*~S+~WVERL (Loss)

  WBLZ~[S ]+~S*~S+~S+: Example for a RegExp matching all Channels with OBJTYPE WBLZ
- pFIGNrcvTxt: corresponding (same length required!) List of Texts (i.e. ['Kontrolle DH']) default: None
- **pFIGNrcvFmt (i.e. '{:12s}: {:8.2f} {:6s}')**
  - Text (from pFIGNrcvTxt)
  - Value
  - UNIT (determined from Channel-Data)
- **pFIGNrcvPercFmt (i.e. ' {:6.1f}%')**
  - ValueInRefPercent
  - if refValue==0: 100%

- pFIGNrcvXStart (.5 default)
- pFIGNrcvYStart (.5 default)

**Category - User Heat Balances to be displayed**

- pFWVBGCategory: List of Heat Balances to be displayed (i.e. ['BLNZ1u5u7']) default: None
- pFWVBGCategoryUnit: Unit of all these Balances (default: '[kW]'])
- pFWVBGCategoryXStart (.1 default)
- pFWVBGCategoryYStart (.9 default)
- **pFWVBGCategoryCatFmt (i.e. '{:12s}: {:6.1f} {:4s}')**
  - Category NAME
  - Category Load
  - pFWVBGCategoryUnit
- **pFWVBGCategoryPercFmt (i.e. ' {:6.1f}%')**
  - Last Ist/Soll
- **pFWVBGCategory3cFmt (i.e. ' {:5d}/{:5d}/{:5d}')**
  - NOfTops
  - NOfMids
  - NOfBots

**VICs - VeryImportantCustomers whose Values to be displayed**

- **pVICsDf: DataFrame with VeryImportantCustomers (Text & Specification)**

  **columns expected:**

  - Kundenname (i.e. 'VIC1') - Text
  - Knotenname (i.e. 'V-K007') - Specification by Supply-Node

  i.e.: pd.DataFrame({'Kundenname': ['VIC1'],'Knotenname': ['V-K007']})

- **pVICsPercFmt (i.e. '{:12s}: {:6.1f}%')**
  - Kundenname
  - Load in Percent to Reference
- **pVICsFmt (i.e. '{:12s}: {:6.1f} {:6s}')**
  - Kundenname
  - Load
  - pFWVBGCategoryUnit
- pVICsXStart (.5 default)
- pVICsYStart (.1 default)

**Figure:**

- pltTitle: title [not suptitle] (default: 'pltNetFigAx')

- figFrameon: figure frame (background): displayed or invisible (default: True)
- figEdgecolor: edge color of the Figure rectangle (default: 'black')
- figFacecolor: face color of the Figure rectangle (default: 'white')

**Returns:**

**pFWVB**

- **columns changed (compared to vFWVB):**
  - pFWVBAttribute (wg. z.B. pFWVBAttributeApplyFunction und .astype(float))
- **columns added (compared to vFWVB):**
  - Measure (in % zu Ref wenn pFWVBMeasureInRefPer=True)
  - MeasureRef (Wert von Measure im Referenzzustand)
  - MeasureOrig (Wert von Measure)
  - **MCategory: str (Kategorisierung von Measure mit FixedLimitHigh/Low-Werten):**
    - TopText or
    - MidText or
    - BotText
  - GCategory: list (non-empty only if req. GCategories are a subset of the available Categories and object belongs to a req. Category)
  - VIC (filled with Kundenname from pVICsDf)
- **rows (compared to vFWVB):**
  - pFWVB enthaelt dieselben Objekte wie vFWVB
  - aber: die geplotteten Objekte sind ggf. nur eine Teilmenge (wg. z.B. pFWVB-FilterFunction)

**exception** Rm.**RmError**(*value*)

CHAPTER

FOUR

# INDICES AND TABLES

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- modindex
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